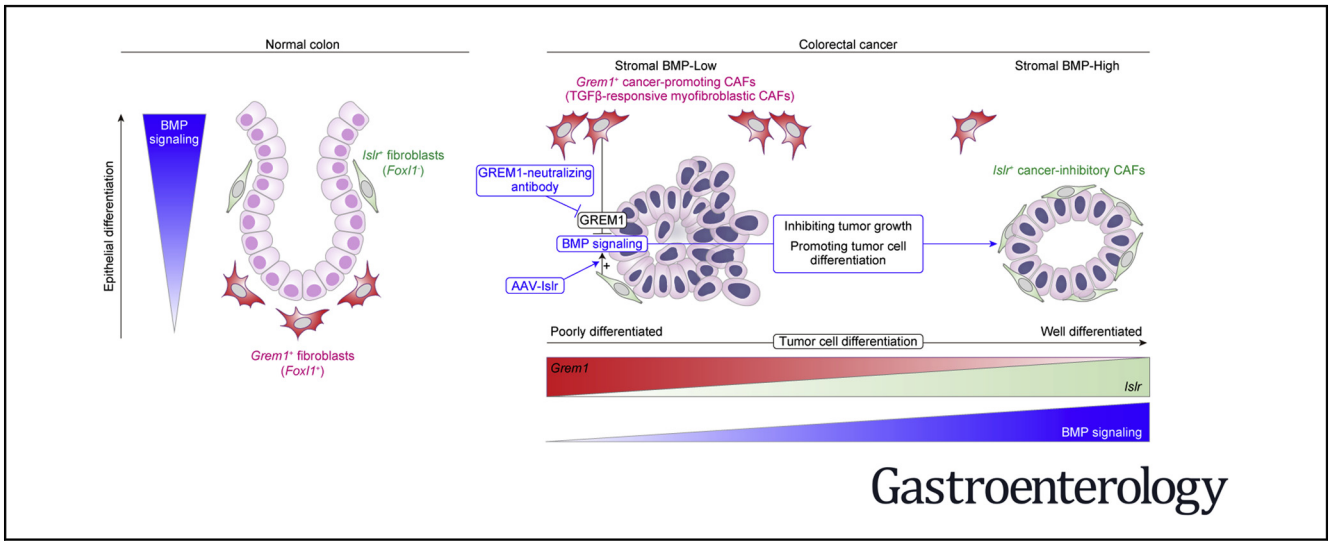
i An update to this article is included at the end

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The Balance of Stromal BMP Signaling Mediated by *GREM1* and *ISLR* Drives Colorectal Carcinogenesis

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See Covering the Cover synopsis on page 977.

BACKGROUND & AIMS: Cancer-associated fibroblasts (CAFs), key constituents of the tumor microenvironment, either promote or restrain tumor growth. Attempts to therapeutically target CAFs have been hampered by our incomplete understanding of these functionally heterogeneous cells. Key growth factors in the intestinal epithelial niche, bone morphogenetic proteins (BMPs), also play a critical role in colorectal cancer (CRC) progression. However, the crucial proteins regulating stromal BMP balance and the potential application of BMP signaling to manage CRC remain

largely unexplored. **METHODS:** Using human CRC RNA expression data, we identified CAF-specific factors involved in BMP signaling, then verified and characterized their expression in the CRC stroma by in situ hybridization. CRC tumoroids and a mouse model of CRC hepatic metastasis were used to test approaches to modify BMP signaling and treat CRC. **RESULTS:** We identified *Grem1* and *Islr* as CAF-specific genes involved in BMP signaling. Functionally, GREM1 and ISLR acted to inhibit and promote BMP signaling, respectively. *Grem1* and *Islr* marked distinct fibroblast subpopulations and were differentially regulated by transforming growth factor β and FOXL1, providing an underlying mechanism to explain fibroblast biological dichotomy. In patients with CRC, high *GREM1* and *ISLR* expression levels were associated with poor and favorable survival, respectively. A GREM1-neutralizing antibody or fibroblast *Islr* overexpression reduced CRC tumoroid growth and promoted *Lgr5*$^{+}$ intestinal stem cell differentiation. Finally, adeno-associated virus 8 (AAV8)–mediated delivery of *Islr* to hepatocytes increased BMP signaling and improved survival in our mouse model of hepatic metastasis. **CONCLUSIONS:** Stromal BMP signaling predicts and modifies CRC progression and survival, and it can be therapeutically targeted by novel AAV-directed gene delivery to the liver.

Keywords: Colorectal Cancer; Cancer-Associated Fibroblasts; Tumor Microenvironment; Bone Morphogenetic Protein.

WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

Colorectal cancer (CRC) progression is promoted or restrained by cancer-associated fibroblasts (CAFs). However, the biological mechanisms underlying CAF heterogeneity and therapeutic strategies to target the CRC stroma remain largely unexplored.

NEW FINDINGS

Mesenchymal BMP signaling increased by ISLR and decreased by GREM1 regulates CRC progression. Augmenting BMP signaling by adeno-associated virus (AAV)-mediated delivery of *Islr* to hepatocytes inhibited CRC hepatic metastasis growth.

LIMITATIONS

This study was performed using mouse models and human tissue samples. Future studies are necessary to determine the therapeutic efficacy of modulating stromal BMP signaling in patients with CRC.

IMPACT

Targeting BMP signaling in the mesenchyme by novel approaches such as AAV-directed gene delivery to the liver, could prove effective adjuvant approaches to treat CRC.

Colorectal cancer (CRC) is a major cause of cancer mortality.[1] Despite advances in surgical techniques and medical therapies targeting tumor cells, endothelial cells, and immune cells, the majority of patients with metastatic CRC still die of their disease.[1]

Cancer-associated fibroblasts (CAFs), a key constituent of the tumor microenvironment, influence CRC initiation, progression, and dissemination and can promote drug resistance via secretion of growth factors, chemokines, extracellular matrix, and proangiogenic factors.[2] CAFs are not a uniformly protumorigenic entity; rather, CAFs are composed of functionally heterogeneous subpopulations, including tumor-promoting CAFs and tumor-retarding CAFs.[2] However, the markers and mechanisms underlying this CAF biological dichotomy are largely unknown, which has hampered therapeutic attempts to exploit these differences.[2]

One key family of growth factors secreted by CAFs, as well as cancer cells, are the bone morphogenetic proteins (BMPs).[3] BMPs belong to the transforming growth factor β (TGF-β) superfamily. Binding of BMP ligands, such as BMP2, 4, and 7, to type I and II BMP receptors induces phosphorylation of SMAD1/5/8, which in turn binds SMAD4 to increase target gene expression such as *ID1*, *2*, *3*, and *4*.[3] BMP gradients partly define the intestinal epithelial stem cell niche in the normal colon and serve to promote or retard cancer progression in a context-dependent manner.[3] In the normal colon, the epithelial stem cell niche is maintained by low BMP and high Wnt at the crypt base, whereas epithelial cell differentiation is driven by increasing BMP and low Wnt toward the luminal surface.[4] The BMP gradient is finely tuned by BMP inhibitors, such as GREM1 and NOGGIN, which are secreted by fibroblasts near the crypt base.[4–6] In CRC, inactivation of BMP signaling through germline or sporadic mutations in BMP receptors and *SMAD4* contributes to CRC predisposition and progression.[4] Numerous studies have shown the tumor-retarding role of BMP signaling in CRC cells themselves.[3,7,8] BMP signaling has been shown to reduce stemness of intestinal stem cells such as *Lgr5*$^{+}$ intestinal stem cells, leading to epithelial differentiation.[7,8] Most CRC studies, however, have failed to address the function of stromal BMP signaling in CRC progression.

Here, we identified gremlin 1 (*GREM1*) and immunoglobulin superfamily containing leucine-rich repeat (*ISLR*), specifically and distinctly expressed by different types of CRC CAFs, as important regulators of BMP signaling within the tumor microenvironment.

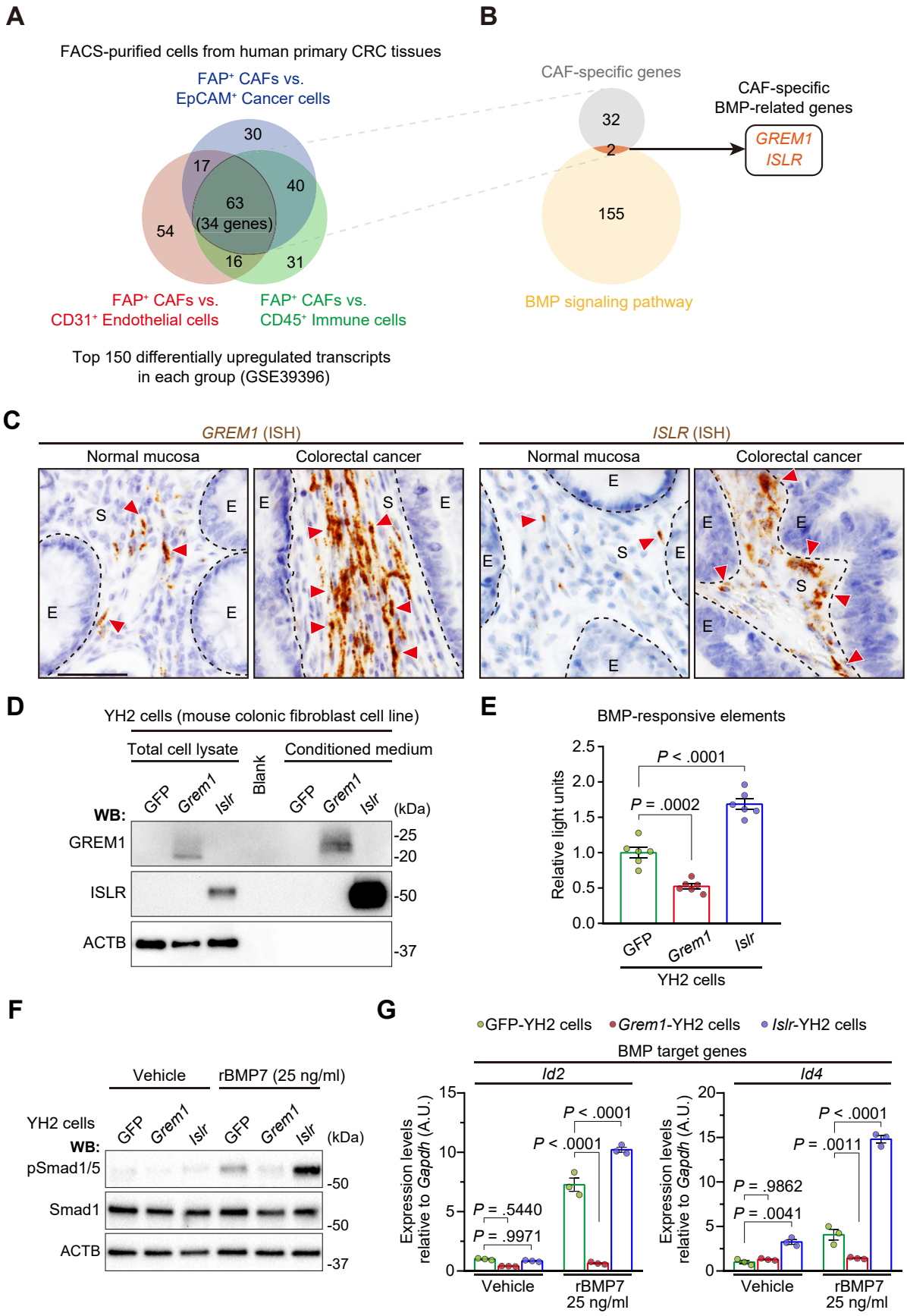
GREM1, a ligand-sequestering antagonist for BMP2, 4, and 7, is expressed by mesenchymal stem/stromal cells in the bone marrow and intestinal fibroblasts.[6] *GREM1* expressed by CAFs can accelerate tumor cell proliferation via inhibition of BMP

§ Authors share co-senior authorship.

Abbreviations used in this paper: αSMA, α smooth muscle actin; AAV, adeno-associated virus; AP, *Apc*$^{\Delta/\Delta}$ and *Trp53* $^{\Delta/\Delta}$; APS, *Apc*$^{\Delta/\Delta}$, *Trp53* $^{\Delta/\Delta}$, and *Smad4* $^{\Delta/\Delta}$; BMP, bone morphogenetic protein; CAF, cancer-associated fibroblast; ChIP, chromatin immunoprecipitation; CRISPR, clustered regularly interspaced short palindromic repeats; CM, conditioned medium; CRC, colorectal cancer; EpCAM, epithelial cell adhesion molecule; FACS, fluorescence-activated cell sorting; FAP, fibroblast activation protein alpha; GFP, green fluorescent protein; GO, Gene Ontology; HA, hemagglutinin; IF, immunofluorescence; iRSC, intestinal reticular stem cell; ISH, in situ hybridization; Meflin, mesenchymal stromal cell- and fibroblast-expressing Linx paralogue; PCR, polymerase chain reaction; qRT-PCR, quantitative reverse-transcription polymerase chain reaction; scRNA-seq, single-cell RNA sequencing; SEM, standard error of the mean; smFISH, single-molecule fluorescent in situ hybridization; TGF-β, transforming growth factor β.

Most current article

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A
FACS-purified cells from human primary CRC tissues
FAP⁺ CAFs vs. EpCAM⁺ Cancer cells
30
17
40
63 (34 genes)
54
16
31
FAP⁺ CAFs vs. CD31⁺ Endothelial cells
FAP⁺ CAFs vs. CD45⁺ Immune cells
Top 150 differentially upregulated transcripts in each group (GSE39396)
B
CAF-specific genes
32
2
155
CAF-specific BMP-related genes
GREM1
ISLR
BMP signaling pathway
C
GREM1 (ISH)
ISLR (ISH)
Normal mucosa
Colorectal cancer
Normal mucosa
Colorectal cancer
S
E
D
YH2 cells (mouse colonic fibroblast cell line)
Total cell lysate
Blank
Conditioned medium
WB:
GFP
Grem1
Islr
(kDa)
GREM1
ISLR
ACTB
-25
-20
-50
-37
E
BMP-responsive elements
Relative light units
2.5
2.0
1.5
1.0
0.5
0.0
P < .0001
P = .0002
GFP
Grem1
Islr
YH2 cells
F
Vehicle
rBMP7 (25 ng/ml)
YH2 cells
WB:
pSmad1/5
Smad1
ACTB
-50
-50
-37
G
GFP-YH2 cells
Grem1-YH2 cells
Islr-YH2 cells
BMP target genes
Id2
Id4
Expression levels relative to Gapdh (A.U.)
P < .0001
P < .0001
P = .5440
P = .9971
P < .0001
P = .0011
P = .9862
P = .0041
Vehicle
rBMP7 25 ng/ml

signaling.[9] In patients with CRC or breast cancer, high *GREM1* expression is associated with poor prognosis.[10–12] GREM1 secreted by glioma cancer stem cells blocks BMP2-induced differentiation of glioma cells, thereby maintaining their proliferation and stemness.[13] Furthermore, overexpression of *GREM1* in intestinal epithelial cells initiates colonic tumorigenesis, supporting the protumorigenic role of *GREM1*.[14,15]

ISLR (also known as Meflin), a glycosylphosphatidylinositol-anchored membrane protein, which is also secreted, was recently identified as a specific marker for mesenchymal stem/stromal cells and fibroblasts in various organs including the bone marrow, heart, and pancreas.[16,17] In contrast to the tumor-promoting role of *GREM1*, our recent study found that *ISLR* defines a subset of tumor-retarding CAFs that are distinct from α-smooth muscle actin (αSMA)$^+$ CAFs in pancreatic cancer.[16] ISLR interacts with BMP7 to augment BMP7-Smad1/5 signaling.[17] Although a recent report has indicated that *ISLR* is highly expressed by fibroblasts in the inflamed colon and CRC,[18] the biological role of ISLR in CRC related to BMP signaling remains unknown.

In this study, after the identification and validation of *GREM1* and *ISLR* as 2 functionally opposing BMP-related genes specifically expressed by CRC CAFs, we examined the prognostic significance of *GREM1* and *ISLR* expression levels in human CRC. Then, we characterized distinct expression patterns of *GREM1* and *ISLR* in both normal colon and CRC and the potential mechanism by which this CAF polarization occurs. Next, we explored whether a GREM1-neutralizing antibody or conditioned medium transfer from *Islr*-overexpressing colonic fibroblasts could restrain CRC organoid growth. Finally, we investigated whether adeno-associated virus (AAV) 8–mediated ectopic overexpression of *Islr* in hepatocytes could retard CRC liver metastasis progression in mice as a novel therapeutic approach to restrain CRC metastasis progression.

Materials and Methods

Statistical Analysis

A comparison of 2 groups was performed by using 2-tailed unpaired *t* tests or Mann-Whitney *U* tests. For multiple comparisons, we used analysis of variance (ANOVA) with subsequent Tukey or Sidak post hoc analysis (for parametric tests) or Kruskal-Wallis test followed by Dunn post-hoc multiple comparisons (for nonparametric tests). For survival analyses, Kaplan-Meier survival estimation with a log rank (Mantel-Cox) test was performed. Statistical analyses were conducted using GraphPad Prism 8.00 (GraphPad) or SPSS Statistics, version 25 (IBM). *P* values of less than .05 were considered statistically significant.

For all other Materials and Methods, see the Supplementary Materials.

Results

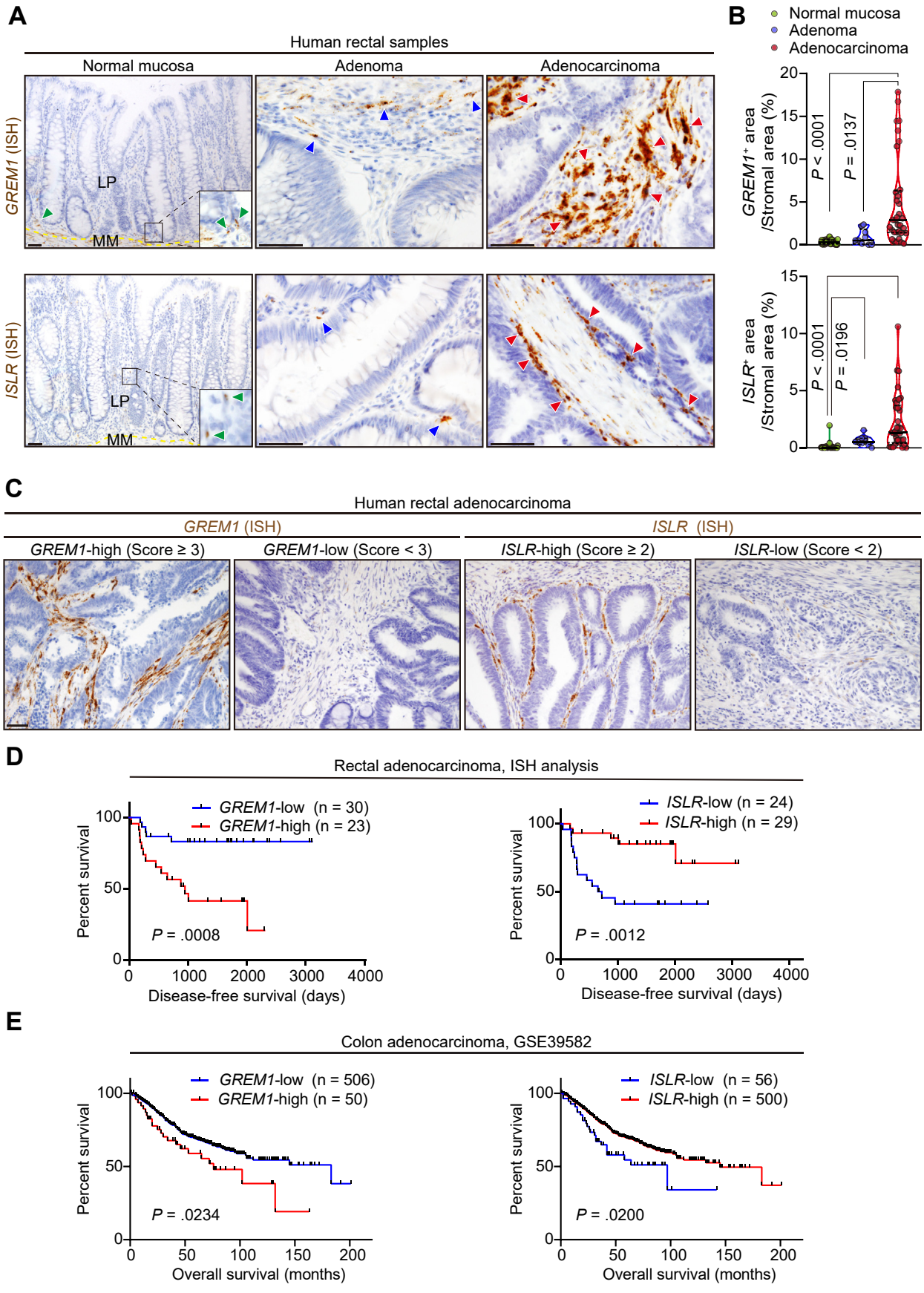
Identification of Cancer-Associated Fibroblast–Specific Expression of the Bone Morphogenetic Protein Antagonist GREM1 and the Bone Morphogenetic Protein Potentiator ISLR in Colorectal Cancer

To identify which BMP-related genes are specifically expressed by CAFs, we first analyzed expression microarray data from a study of fluorescence-activated cell sorting (FACS)–purified cells from human primary CRC tissues.[19] The top 150 differentially expressed gene probes up-regulated in CAFs in each group (FAP$^+$ CAFs vs EpCAM$^+$ cancer cells, FAP$^+$ CAFs vs CD31$^+$ endothelial cells, and FAP$^+$ CAFs vs CD45$^+$ immune cells) were selected for our analysis, resulting in the identification of 34 genes specifically expressed in human CRC CAFs (Figure 1*A*, Supplementary Figure 1*A* and *B*, and Supplementary Table 1). Next, to examine for genes involved in BMP signaling, we compared the 34 CAF-specific genes with BMP-relevant genes listed in Gene Ontology (GO) and identified by literature review (BMP signaling pathway; GO0030509)[17] (Figure 1*B*). This analysis identified 2 genes, *GREM1* and *ISLR*, as human CRC CAF-specific genes relevant to BMP signaling.

To validate *GREM1* and *ISLR* expression in human CRC CAFs, we performed RNA in situ hybridization (ISH) on CRC patient samples and confirmed that both *GREM1* and *ISLR* are highly expressed by fibroblastic cells in the human CRC stroma compared to the normal colorectal stroma (Figure 1*C* and Supplementary Figure 2). Consistent with the microarray data, single-molecule fluorescent ISH (smFISH) for *GREM1* and *ISLR* followed by FAP immunofluorescence showed that *GREM1* and *ISLR* were expressed by FAP$^+$ CAFs in human CRC sections (Supplementary Figure 3). Moreover, the fibroblast-specific expression pattern of *GREM1* and *ISLR* was corroborated by analyses of publicly available single-cell RNA sequencing (scRNA-seq) data from human normal colon mucosa and primary CRC[20] (Supplementary Figure 4*A* and *B*).

We next sought to verify the functional roles of GREM1 and ISLR in the regulation of BMP signaling, using lentivirus-mediated overexpression of *Grem1* or *Islr* in a mouse colonic

Figure 1. Identification of *GREM1* and *ISLR* as a BMP antagonist and potentiator, respectively, specifically expressed by CRC CAFs. (*A*, *B*) Analysis of expression microarray data from FACS-purified cells from human primary CRC tissues. (*A*) Venn diagram depicting the overlap of the top 150 differentially up-regulated transcripts in the 3 groups as indicated. (*B*) Venn diagram showing the overlap of 34 CAF-specific genes and 157 BMP-related genes identified by GO of the BMP signaling pathway (GO: 0030509) and Hara et al.[17] (*C*) ISH for *GREM1* and *ISLR* in the human normal colorectal mucosa and CRC. Dotted lines indicate the borders between epithelial cells (E) and the stroma (S). Red arrowheads denote *GREM1* or *ISLR* expression. Scale bar, 50 μm. (*D–G*) Lentivirus-mediated overexpression of *Grem1* and *Islr* in a mouse colonic fibroblast cell line, YH2 cells, represses and augments BMP signaling, respectively. (*D*) Western blotting (WB) showing *Grem1* and *Islr* overexpression in the total cell lysates and CM. (*E*) Luciferase assays of BMP-responsive elements; n = 6. (*F*, *G*) YH2 cells were stimulated with recombinant BMP7, followed by (*F*) WB and (*G*) qRT-PCR; n = 3. Mean ± standard error of the mean (SEM). One-way ANOVA (*E*) or 2-way ANOVA (*G*) with post hoc Tukey multiple comparisons. A.U., arbitrary unit.


A
Human rectal samples
Normal mucosa
Adenoma
Adenocarcinoma
GREM1 (ISH)
ISLR (ISH)
LP
MM
B
Normal mucosa
Adenoma
Adenocarcinoma
GREM1+ area /Stromal area (%)
P < .0001
P = .0137
ISLR+ area /Stromal area (%)
P < .0001
P = .0196
C
Human rectal adenocarcinoma
GREM1 (ISH)
ISLR (ISH)
GREM1-high (Score ≥ 3)
GREM1-low (Score < 3)
ISLR-high (Score ≥ 2)
ISLR-low (Score < 2)
D
Rectal adenocarcinoma, ISH analysis
GREM1-low (n = 30)
GREM1-high (n = 23)
P = .0008
ISLR-low (n = 24)
ISLR-high (n = 29)
P = .0012
Percent survival
Disease-free survival (days)
E
Colon adenocarcinoma, GSE39582
GREM1-low (n = 506)
GREM1-high (n = 50)
P = .0234
ISLR-low (n = 56)
ISLR-high (n = 500)
P = .0200
Overall survival (months)

fibroblast cell line, YH2 cells. GREM1 and ISLR overexpression was detected in the conditioned medium from *Grem1*-overexpressing and *Islr*-overexpressing YH2 cells, respectively, suggesting that GREM1 and ISLR were secreted into the medium (Figure 1*D*). Luciferase assays of BMP-responsive elements showed that GREM1 overexpression suppressed, whereas ISLR overexpression augmented, BMP signaling (Figure 1*E*). Furthermore, GREM1 overexpression inhibited BMP7-mediated phosphorylation of Smad1/5, a downstream effector of BMP signaling, thereby preventing the BMP7-induced increase in the expression of BMP target genes *Id2* and *Id4* (Figure 1*F* and *G*). Conversely, these surrogates for the BMP signaling pathway were increased by ISLR overexpression. When *Grem1*-overexpressing YH2 cells were admixed with *Islr*-overexpressing YH2 cells, GREM1 and ISLR counteracted each other's effect on BMP7 signaling (Supplementary Figure 5). Similar to the GREM1 antagonism of BMP7 signaling, GREM1 overexpression also prevented the BMP2-induced increase in *Id2* and *Id4*. In contrast, ISLR overexpression promoted the BMP2-mediated increase in *Id4*, but not *Id2* (Supplementary Figure 6). Collectively, these data indicate that 2 functionally opposing stromal regulators of BMP signaling, *GREM1* and *ISLR*, are up-regulated in CRC CAFs and may contribute to fine-tuning of BMP signaling within the CRC stroma.

GREM1 *and* ISLR *Expression Are Up-regulated During Colorectal Carcinogenesis in Humans*

Next, we investigated whether expression of *GREM1* and *ISLR* is up-regulated during CRC progression. ISH for *GREM1* and *ISLR*, as well as scRNA-seq and expression microarray analyses, showed that *GREM1* and *ISLR* expression were increased during human colorectal carcinogenesis (Figure 2*A* and *B* and Supplementary Figures 4*A* and 7*A*–*C*). *GREM1* and *ISLR* up-regulation was also observed in the stroma of liver metastases of human CRC compared to the normal liver tissues (Supplementary Figure 8*A*–*C*). Furthermore, in line with the fibroblast-specific expression of *GREM1* and *ISLR*, analyses of The Cancer Genome Atlas and expression microarray data showed that the highest expression levels of *GREM1* and *ISLR* were observed in a stroma-rich molecular subtype of CRC (consensus molecular subtype 4 [CMS4]) (Supplementary Figure 9*A* and *B*). This is consistent with a recent article showing that CMS4 tumors displayed the highest *GREM1* transcript levels.[10] Overall, these data suggest that *GREM1* and *ISLR* expression are up-regulated in the CRC stroma during colorectal carcinogenesis.

GREM1 *and* ISLR *Expression Levels Are Associated With Poor and Favorable Clinical Outcomes in Patients With Colorectal Cancer, Respectively*

To investigate the clinical significance of *GREM1* and *ISLR* expression in CRC CAFs, we evaluated *GREM1* and *ISLR* expression by ISH in 53 rectal cancer surgical samples (Figure 2*C* and Supplementary Table 2). Survival analyses showed that high *GREM1* expression (score of $\geq$3) and high *ISLR* expression (score of $\geq$2) were independent prognostic factors for poor and favorable disease-free survival, respectively, in patients with rectal cancer (Figure 2*D* and Supplementary Tables 2 and 3).

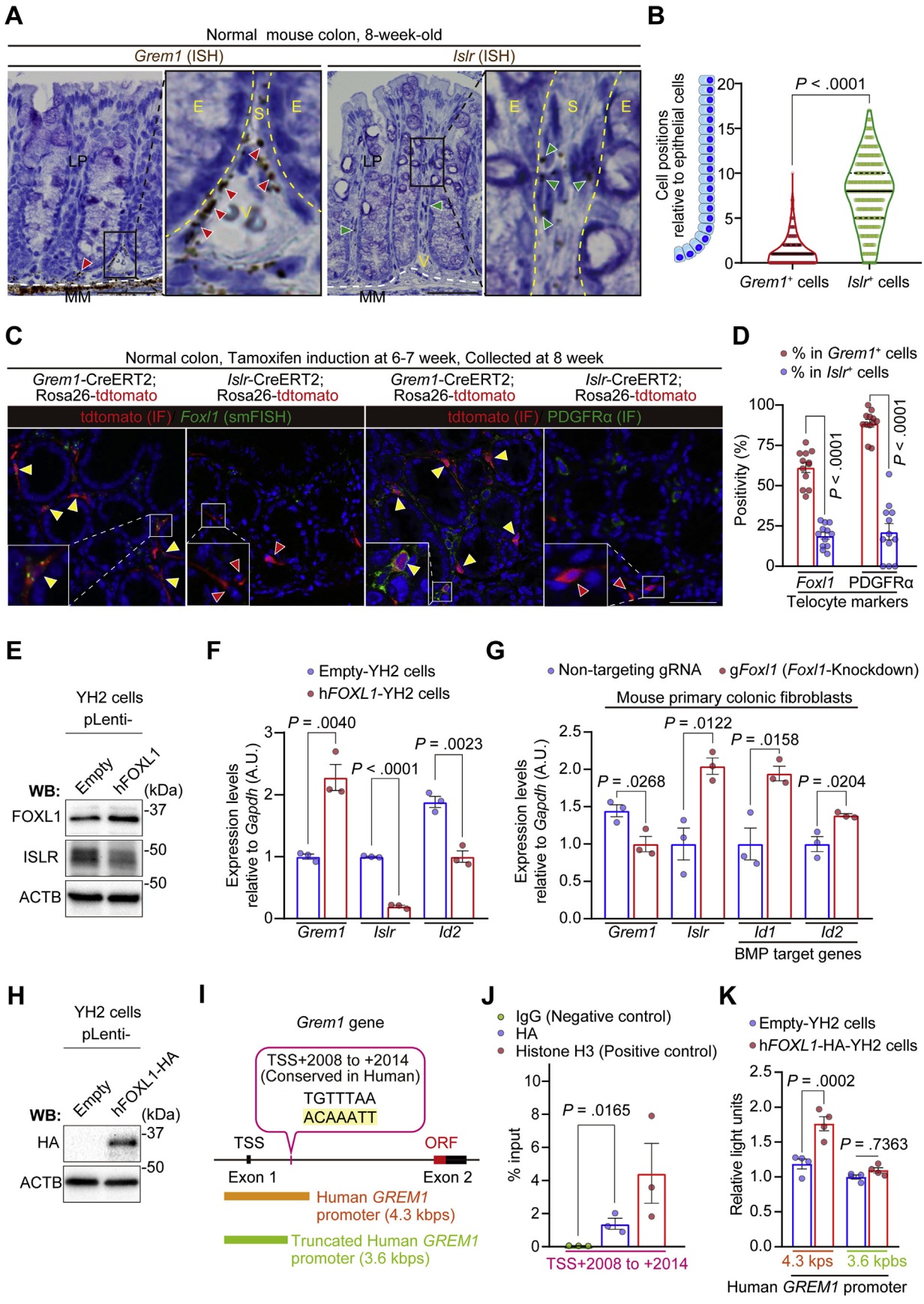
Furthermore, analysis of expression microarray data from 556 patients with primary colon cancer confirmed that the *GREM1*-high and *ISLR*-low groups each independently exhibited poor overall survival (Figure 2*E* and Supplementary Table 4). No patients in this cohort had both *GREM1*-high and *ISLR*-low, suggesting that *GREM1*-high patients and *ISLR*-low patients were 2 separate patient subgroups (Supplementary Table 5). Together, these data indicate that *GREM1* and *ISLR* expression levels may serve as prognostic biomarkers in human CRC, with *GREM1* expression associated with poorer and *ISLR* expression associated with improved survival.

Grem1$^+$ *Fibroblasts Are Distinct From* Islr$^+$ *Fibroblasts in the Normal Mouse Colon, With the Majority of* Grem1$^+$ *Fibroblasts Marked by* Foxl1

We next sought to characterize the specific stromal cell types expressing *Grem1* and *Islr* in the normal colon. To this end, we performed *Grem1* and *Islr* ISH using normal mouse colons (Figure 3*A* and *B*). As shown elsewhere,[14] our ISH data confirmed that *Grem1* expression was observed in fibroblastic cells near the base of the colonic crypts in the lamina propria, as well as in muscularis mucosae cells. Interestingly, however, *Islr*$^+$ fibroblasts were located near the middle of the colonic crypts, suggesting that *Grem1*$^+$ intestinal fibroblasts were topographically distinct from *Islr*$^+$ intestinal fibroblasts in the normal colonic crypts.

To further define the fibroblast subpopulations expressing *Grem1* and *Islr*, we studied *Grem1*-CreERT2[6];Rosa26-LSL-tdtomato mice and *Islr*-CreERT2[17];Rosa26-LSL-tdtomato mice and costained with intestinal fibroblast markers (Figure 3*C* and *D*). A recent report has illustrated that subepithelial telocytes identified by expression of *Foxl1* and platelet-derived growth factor receptor alpha (*Pdgfra*) provide key intestinal stem cell

Figure 2. *GREM1* and *ISLR* expression levels are associated with poor and favorable prognosis in patients with CRC, respectively. (*A*, *B*) ISH for *GREM1* and *ISLR* using human rectal samples. (*A*) Representative images. Yellow dotted lines indicate the borders between the lamina propria (LP) and muscularis mucosa (MM). Green, blue, and red arrowheads denote *GREM1* or *ISLR* expression in the normal mucosa, adenoma, and adenocarcinoma, respectively. (*B*) Violin plots depicting *GREM1* and *ISLR* ISH signal$^+$ areas in the stroma. Three high-power fields (400×) per patient: 11 patients with normal mucosa, 3 with adenoma, and 11 with adenocarcinoma. Solid black lines indicate the median; dotted black lines indicate quartiles. (*C*, *D*) ISH analysis of 53 human primary rectal cancer surgical samples. (*C*) Representative images. Cases with a score of $\geq$3 and a score of $\geq$2 were defined as *GREM1*-high and *ISLR*-high, respectively. (*D*) Kaplan-Meier survival curves. (*E*) Kaplan-Meier survival curves in expression microarray data from 556 patients with primary colon cancer. Kruskal-Wallis test followed by Dunn post hoc multiple comparisons (*B*) and log rank test (*D* and *E*). Scale bars, 50 μm.


A
Normal mouse colon, 8-week-old
Grem1 (ISH)
Islr (ISH)
LP
E
S
V
MM
B
Cell positions relative to epithelial cells
P < .0001
Grem1+ cells
Islr+ cells
C
Normal colon, Tamoxifen induction at 6-7 week, Collected at 8 week
Grem1-CreERT2; Rosa26-tdtomato
Islr-CreERT2; Rosa26-tdtomato
tdtomato (IF) Foxl1 (smFISH)
tdtomato (IF) PDGFRα (IF)
D
% in Grem1+ cells
% in Islr+ cells
Positivity (%)
P < .0001
Foxl1
PDGFRα
Telocyte markers
E
YH2 cells pLenti-
Empty
hFOXL1
WB:
(kDa)
FOXL1
ISLR
ACTB
F
Empty-YH2 cells
hFOXL1-YH2 cells
Expression levels relative to Gapdh (A.U.)
P = .0040
P < .0001
P = .0023
Grem1
Islr
Id2
G
Non-targeting gRNA
gFoxl1 (Foxl1-Knockdown)
Mouse primary colonic fibroblasts
P = .0268
P = .0122
P = .0158
P = .0204
Id1
BMP target genes
H
YH2 cells pLenti-
Empty
hFOXL1-HA
HA
ACTB
I
Grem1 gene
TSS+2008 to +2014 (Conserved in Human)
TGTTTAA
ACAAATT
TSS
ORF
Exon 1
Exon 2
Human GREM1 promoter (4.3 kbps)
Truncated Human GREM1 promoter (3.6 kbps)
J
IgG (Negative control)
HA
Histone H3 (Positive control)
% input
P = .0165
TSS+2008 to +2014
K
Empty-YH2 cells
hFOXL1-HA-YH2 cells
Relative light units
P = .0002
P = .7363
4.3 kps
3.6 kpbs
Human GREM1 promoter

niche signaling molecules such as Wnts and *Grem1*.[21] Indeed, analysis of RNA-seq data from FACS-purified *Foxl1*-lineage intestinal telocytes and non–*Foxl1*-lineage intestinal mesenchymal cells[21] suggested that *Grem1* expression was observed in *Foxl1*-lineage$^+$ telocytes, but not in *Foxl1*-lineage$^-$ cells (Supplementary Figure 10). Consistent with this, *Foxl1* smFISH or PDGFRα immunofluorescence (IF) using *Grem1*-CreERT2 mice showed that the majority of the *Grem1*$^+$ fibroblasts, also known as *Grem1*$^+$ intestinal reticular stem cells (iRSCs),[6] expressed the telocyte markers *Foxl1* and PDGFRα in the lamina propria of the normal colon (Figure 3*C* and *D*). In contrast, *Islr*$^+$ fibroblasts exhibited lower positivity for *Foxl1* and PDGFRα than *Grem1*$^+$ iRSCs. These data implied a high degree of overlap between *Grem1*$^+$ iRSCs and *Foxl1*$^+$ telocytes. This prompted us to investigate whether *Foxl1* might drive *Grem1* expression at the expense of *Islr* expression.

FOXL1 *Directly Up-regulates* Grem1 *Transcription While Repressing* Islr *Expression in Mouse Colonic Fibroblasts*

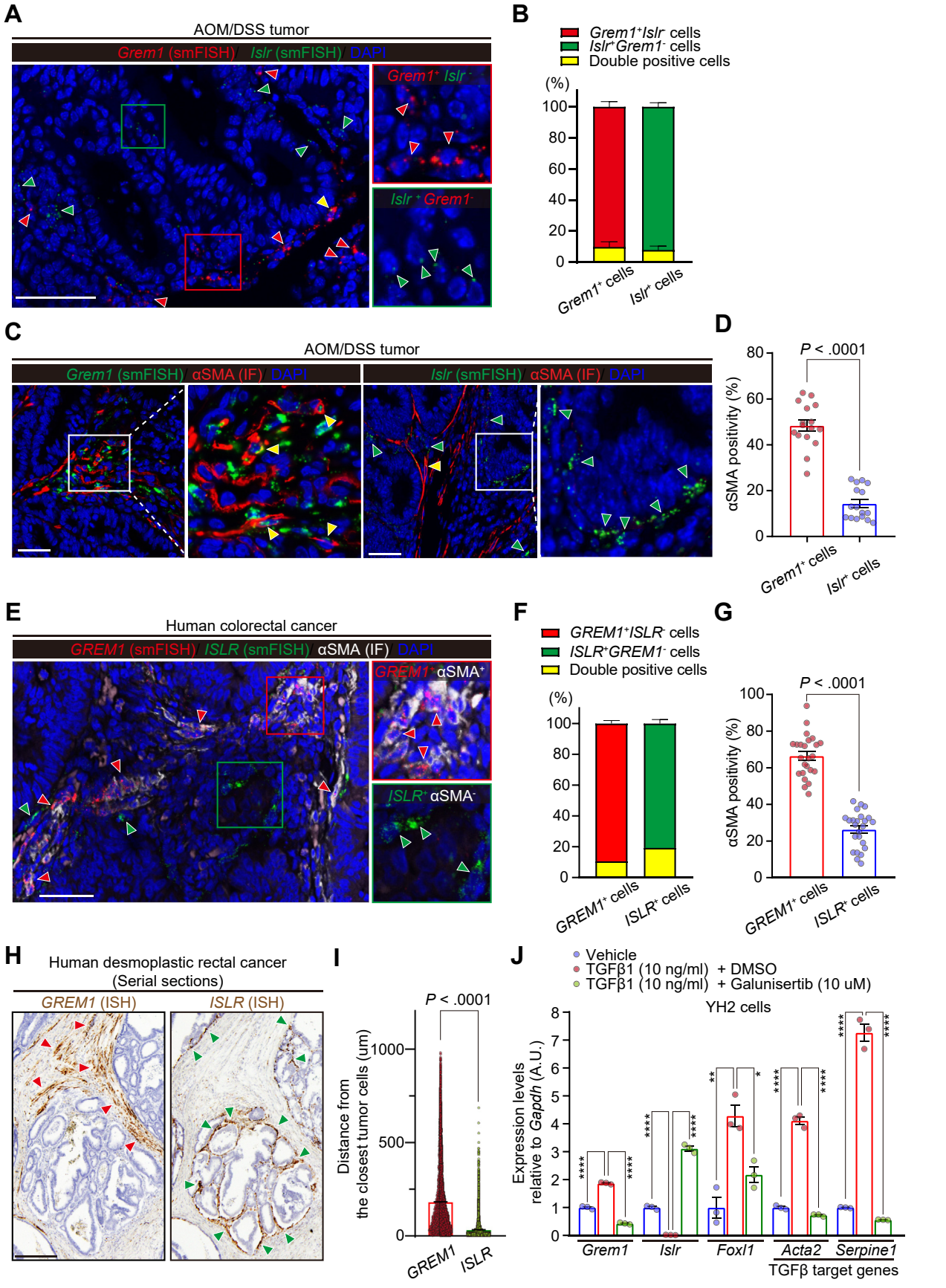
To assess the effect of FOXL1 on the regulation of *Grem1* and *Islr* expression, human *FOXL1*-overexpressing YH2 cells were generated by lentiviral transduction (Figure 3*E*). Consistent with our earlier colocalization analyses, FOXL1 overexpression in YH2 cells induced *Grem1* up-regulation at the expense of *Islr* expression, accompanied by decreased *Id2* expression (Figure 3*F*). Similarly, luciferase reporter assays showed that FOXL1 overexpression increased the activity of the *GREM1*-promoter reporter and reduced the activity of the *ISLR*-promoter reporter in comparison to control empty YH2 lines (Supplementary Figure 11). Conversely, clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9–mediated *Foxl1*-knockdown in mouse primary colonic fibroblasts attenuated *Grem1* expression while inducing up-regulation of *Islr* and BMP target genes (Figure 3*G* and Supplementary Figure 12).

To explore whether FOXL1 is directly involved in regulating *Grem1* expression, we generated *FOXL1*-hemagglutinin (HA) tag–overexpressing YH2 cells (Figure 3*H*) and performed chromatin immunoprecipitation (ChIP) using an anti-HA antibody. ChIP–quantitative polymerase chain reaction (PCR) showed enrichment of FOXL1-HA binding to the *Grem1* promoter (transcriptional start site, +2008 to +2014 bps; an intronic region) compared to immunoglobulin controls (Figure 3*I* and *J*). Consistent herewith, a luciferase assay using the human *GREM1* promoter region in YH2 cells confirmed that FOXL1-mediated augmentation of *GREM1* expression was abrogated by truncation of the *GREM1* promoter to remove the FOXL1-binding region (Figure 3*K*). Collectively, these data indicate that FOXL1 is recruited to the *Grem1/GREM1* promoter to drive *Grem1* expression, providing mechanistic insight into the overlap between *Grem1*$^+$ iRSCs and *Foxl1*$^+$ telocytes.

GREM1$^+$ *Cancer-Associated Fibroblasts Are Myofibroblastic Cancer-Associated Fibroblasts, Which Are Distinct From* ISLR$^+$ *Cancer-Associated Fibroblasts, in Mouse and Human Colorectal Cancer*

Next, we sought to characterize *GREM1*$^+$ CAFs and *ISLR*$^+$ CAFs in the CRC mesenchyme. Using azoxymethane/dextran sulfate sodium mouse CRC and human CRC samples, we carried out *Grem1/GREM1* and *Islr /ISLR* smFISH as well as IF for αSMA, a well-established marker of myofibroblastic CAFs. We found that, both in mouse and human CRC, *Grem1/GREM1*$^+$ CAFs were distinct from *Islr/ISLR*$^+$ CAFs and that αSMA positivity was higher in *Grem1/GREM1*$^+$ CAFs than *Islr/ISLR*$^+$ CAFs (Figure 4*A*–*G*). A collagen gel contraction assay also showed that *Grem1*-overexpressing YH2 cells exhibited increased contraction, a hallmark of activated myofibroblasts[2] (Supplementary Figure 13*A* and *B*). Consistent with our smFISH data, scRNA-seq data from human CRC tissues[20] confirmed that *GREM1* expression levels were inversely correlated with *ISLR* expression (Supplementary Figure 14*A*) and that *GREM1*, but not *ISLR*, transcripts were positively correlated with *ACTA2* expression in CRC CAFs (Supplementary Figure 14*B*). Furthermore, the scRNA-seq data set showed that *GREM1* was predominantly expressed by myofibroblasts. High-*ISLR* transcripts were observed not only in myofibroblasts but also in Stromal 2 fibroblasts that are characterized by spatial proximity to epithelial cells and high

Figure 3. *Grem1* and *Islr* identify distinct subpopulations of intestinal fibroblasts in the normal mouse colon and are differentially regulated by FOXL1. (*A*, *B*) ISH for *Grem1* and *Islr* in the adult normal mouse colon. (*A*) Representative images. Red and green arrowheads denote *Grem1*$^+$ cells and *Islr*$^+$ cells, respectively. Yellow dotted lines delineate the boundaries between epithelial cells (E) and stromal cells (S). White dotted lines indicate the borders between the lamina propria (LP) and muscularis mucosa (MM). V indicates blood vessels. (*B*) Violin plots depicting the positions of mesenchymal cells expressing *Grem1* or *Islr* relative to the adjacent epithelial position; 344 *Grem1*$^+$ cells and 512 *Islr*$^+$ cells from 20 well-oriented crypts/mouse, 4 mice each. Black solid lines indicate the median; black dotted lines indicate quartiles. (*C*, *D*) smFISH for *Foxl1* and IF for PDGFRα in *Grem1*-CreERT2 mice and *Islr*-CreERT2 mice. (*C*) Representative pictures. Yellow arrowheads indicate double-positive cells (*Grem1*$^+$*Foxl1*$^+$ cells or *Grem1*$^+$PDGFRα^+ cells). Red arrowheads denote *Islr* single-positive cells. (*D*) *Foxl1* positivity and PDGFRα positivity in the *Grem1*$^+$ cells and *Islr*$^+$ cells. Four high-power fields (400×)/mouse, 3 mice each. (*E*, *F*) Lentivirus-mediated human *FOXL1* (h*FOXL1*) overexpression in YH2 cells induces *Grem1* up-regulation and decreases *Islr* expression. (*E*) Western blot. (*F*) qRT-PCR (n = 3). (*G*) CRISPR/Cas9-mediated knockdown of *Foxl1* reduces *Grem1* expression while up-regulating *Islr* expression in primary mouse colonic fibroblasts as assessed by qRT-PCR (n = 3 mice each). (*H–K*) FOXL1 interacts with a *Grem1* intron region. (*H*) Western blot showing hFOXL1-HA overexpression in YH2 cells. (*I*) Schematic representation of a FOXL1-binding site in the mouse *Grem1* intron (*highlighted with yellow*) and corresponding human *GREM1* promoter regions used in luciferase assays. (*J*) ChIP–quantitative PCR in YH2 cells (n = 3). (*K*) Luciferase assays of a human *GREM1* promoter (4.3 kbps) and a truncated human *GREM1* promoter (3.6 kbps) that lacks the FOXL1-binding site (n = 4). Mean ± SEM. Mann-Whitney *U* test (*B*), 2-tailed unpaired Student *t* test (*D*, *F*, *G*, and *J*), and 2-way ANOVA with Tukey post hoc multiple comparisons (*K*). Scale bars, 50 μm. ORF, open reading frame; TSS, transcriptional start site.


A
AOM/DSS tumor
Grem1 (smFISH) Islr (smFISH) DAPI
Grem1+ Islr-
Islr+ Grem1-
B
Grem1+Islr- cells
Islr+Grem1- cells
Double positive cells
(%)
100
80
60
40
20
0
Grem1+ cells
Islr+ cells
C
AOM/DSS tumor
Grem1 (smFISH) αSMA (IF) DAPI
Islr (smFISH) αSMA (IF) DAPI
D
P < .0001
αSMA positivity (%)
80
60
40
20
0
Grem1+ cells
Islr+ cells
E
Human colorectal cancer
GREM1 (smFISH) ISLR (smFISH) αSMA (IF) DAPI
GREM1+ αSMA+
ISLR+ αSMA-
F
GREM1+ISLR- cells
ISLR+GREM1- cells
Double positive cells
(%)
100
80
60
40
20
0
GREM1+ cells
ISLR+ cells
G
P < .0001
αSMA positivity (%)
100
80
60
40
20
0
GREM1+ cells
ISLR+ cells
H
Human desmoplastic rectal cancer
(Serial sections)
GREM1 (ISH)
ISLR (ISH)
I
P < .0001
Distance from the closest tumor cells (um)
1000
500
0
GREM1
ISLR
J
Vehicle
TGFβ1 (10 ng/ml) + DMSO
TGFβ1 (10 ng/ml) + Galunisertib (10 uM)
YH2 cells
Expression levels relative to Gapdh (A.U.)
8
7
6
5
4
3
2
1
0
****
**
*
Grem1
Islr
Foxl1
Acta2
Serpine1
TGFβ target genes

expression of BMP ligands, including BMP7[20] (Supplementary Figure 14*C* and *D*). Interestingly, *GREM1*$^+$ CAFs were spatially distinct from *ISLR*$^+$ CAFs in desmoplastic human CRC, with *ISLR*$^+$ CAFs located in closer proximity to cancer cells than *GREM1*$^+$ CAFs (Figure 4*H* and *I*).

As TGF-β1 has well-characterized functions in inducing myofibroblastic differentiation of CAFs,[2] we examined whether TGF-β1 is involved in controlling the differential expression of *Grem1* and *Islr*. Stimulation of YH2 cells with recombinant TGF-β1 increased transcript levels of TGF-β target genes *Serpine1* and *Acta2*, but also *Foxl1* and *Grem*1, while decreasing *Islr* expression. This was rescued by cotreatment with Galunisertib, a specific inhibitor for TGF-β receptor 1 (Figure 4*J*). In keeping with the TGF-β–induced up-regulation of *Foxl1* and *Grem1* in vitro, *GREM1*$^+$ CAFs showed a higher degree of colocalization with *FOXL1* in human CRC sections than *ISLR*$^+$ CAFs did (Supplementary Figure 15). Together, our data suggest that TGF-β1 up-regulates *Foxl1* in fibroblasts that, in turn, bind to the *Grem1* promoter to up-regulate *Grem1* expression. TGF-β1 and Foxl1 also reduce the expression of *Islr*. This signaling pathway may be involved in the selective development or differentiation of αSMA$^+$*Grem1*$^+$ myofibroblastic CAFs, which are distinct from *Islr*$^+$ αSMA$^-$ CAFs, in mouse and human CRC.

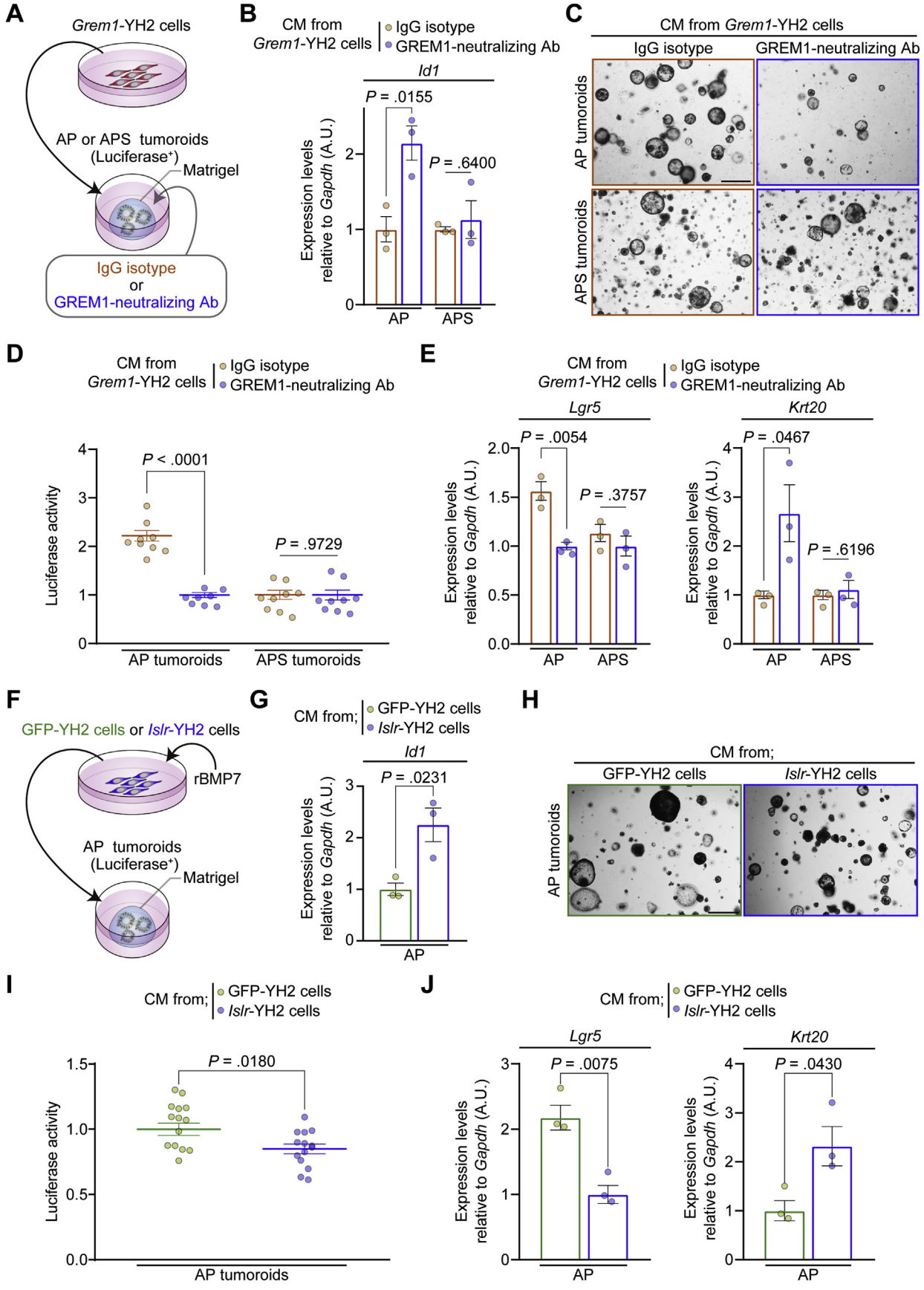
Blocking Bone Morphogenetic Protein Antagonism by Using a GREM1*-Neutralizing Antibody Promotes Colorectal Cancer Organoid Differentiation and Restrains Growth*

What relevance do these stromal changes have on the cancer? We investigated whether augmenting BMP signaling either by *Grem1* inhibition or *Islr* overexpression could retard CRC progression. To this end, we took advantage of CRISPR/Cas9 genome engineering and generated luciferase-expressing *Apc*$^{\Delta/\Delta}$, *Trp53* $^{\Delta/\Delta}$ mouse CRC organoids (henceforth referred to as *AP tumoroids*) (Supplementary Figure 16*A* and *B*). To disrupt BMP signaling, we sequentially mutated *Smad4*, a downstream effector of BMP signaling, to generate *Smad4*-mutant AP tumoroids (hereafter termed *APS tumoroids*) (Supplementary Figure 16*C* and *D*).

First, to test the role of GREM1 in CRC organoid growth, conditioned medium (CM) from *Grem1*-YH2 cells or control green fluorescent protein (GFP)–YH2 cells was transferred to either AP or APS tumoroids (Supplementary Figure 17*A*). As expected, the expression of BMP target genes was repressed by CM transfer from *Grem1*-YH2 cells only in AP tumoroids but not in APS tumoroids (Supplementary Figure 17*B*). Luciferase activity was used to assess an effect of the treatment on viable cell number in luciferase-expressing AP or APS tumoroid cultures. CM from *Grem1*-YH2 cells increased the tumoroid-derived luciferase signals in AP tumoroids (Supplementary Figure 17*C*). This was ameliorated by the loss of *Smad4* in APS tumoroids, indicating that the pro-proliferation effect of GREM1 occurred via antagonism of BMP signaling. Consistent with the role of BMP signaling in promoting epithelial cell differentiation,[4,22] CM from *Grem1*-YH2 cells decreased the expression of *Krt20*, a marker for differentiated CRC cells, in AP tumoroids but not in APS tumoroids (Supplementary Figure 17*D*).

We next examined whether restoring BMP signaling with a GREM1-neutralizing antibody could repress CRC organoid growth. For this purpose, either a GREM1-neutralizing antibody or an IgG isotype was added to the AP and APS tumoroids incubated in CM from *Grem1*-YH2 cells (Figure 5*A*). The effect of the GREM1-neutralizing antibody to abolish GREM1-mediated BMP antagonism was validated by quantitative reverse-transcription PCR (qRT-PCR) of *Id2* in YH2 cells (Supplementary Figure 18*A* and *B*). Blocking the antagonism of BMP signaling by GREM1 using the GREM1-neutralizing antibody restored BMP target gene expression (Figure 5*B*) and reduced tumoroid-derived luciferase signals and tumoroid size in AP tumoroids but not in APS tumoroids (Figure 5*C* and *D* and Supplementary Figure 19*A*). Moreover, treatment with the GREM1-neutralizing antibody decreased *Lgr5* expression with a concomitant increase in *Krt20* expression in AP tumoroids. This effect was abrogated in APS tumoroids (Figure 5*E*). Collectively, our in vitro data suggest that restoring BMP signaling with the GREM1-neutralizing antibody promoted differentiation of *Lgr5*$^+$ intestinal stem cells and attenuated tumoroid growth of *Smad4*–wild-type, but not *Smad4*-mutant, CRC.

Figure 4. *GREM1*$^+$ CAFs are myofibroblastic CAFs, which are distinct from *ISLR*$^+$ CAFs in human and mouse CRC. (*A*, *B*) Dual smFISH for *Grem1* and *Islr* in an azoxymethane/dextran sodium sulfate mouse model of CRC. (*A*) Representative pictures. Red, green, and yellow arrowheads denote *Grem1*$^+$*Islr*$^-$, *Grem1*$^-$*Islr*$^+$, and *Grem1*$^+$*Islr*$^+$ CAFs, respectively. (*B*) Semi-quantification of the ratio of double-positive (*Grem1*$^+$*Islr*$^+$) cells in *Grem1*$^+$ cells and *Islr*$^+$ cells. Four high-power fields (400×)/mouse, 4 mice. (*C*, *D*) *Grem1* smFISH or *Islr* smFISH followed by αSMA IF in azoxymethane/dextran sodium sulfate tumors. (*C*) Representative pictures. Yellow and green arrowheads indicate double-positive cells (*Grem1*$^+$$\alpha$SMA$^+$ cells or *Islr*$^+$$\alpha$SMA$^+$ cells) and *Islr*$^+$$\alpha$SMA$^-$ cells, respectively. (*D*) αSMA positivity in *Grem1*$^+$cells and *Islr*$^+$ cells. Four high-power fields/mouse, 4 mice each. (*E*–*G*) Dual smFISH for *GREM1* and *ISLR* followed by αSMA IF in human CRC. (*E*) Representative pictures. Red and green arrowheads denote *GREM1*$^+$$\alpha$SMA$^+$ cells and *ISLR*$^+$$\alpha$SMA$^-$ cells, respectively. (*F*) Semiquantification of the ratio of double-positive (*GREM1*$^+$*ISLR*$^+$) cells in *GREM1*$^+$ cells and *ISLR*$^+$ cells. (*G*) αSMA positivity in *GREM1*$^+$cells and *ISLR*$^+$ cells; 4–6 high-power fields/patient, 5 patients. (*H*, *I*) *GREM1*$^+$ CAFs are spatially distinct from *ISLR*$^+$ CAFs in human desmoplastic rectal cancer. (*H*) Representative pictures of *GREM1* and *ISLR* ISH on human desmoplastic rectal cancer samples. Red and green arrowheads denote *GREM1* and *ISLR* expression, respectively. (*I*) Quantification of the minimum distance between *GREM1* or *ISLR* ISH signals and the closest tumor cells; n = 38,396 (*GREM1*) and 18,028 DAB$^+$ signals (*ISLR*) from 3 low-power fields (100×)/patient, 7 patients each. (*J*) YH2 cells were stimulated with a vehicle, recombinant TGF-β1, or recombinant TGFβ1 + Galunisertib for 24 hours, followed by qRT-PCR (n = 3). Mean ± SEM. Mann-Whitney *U* test (*D*, *G*, and *I*) and 1-way ANOVA with Tukey post hoc multiple comparisons (*J*). The boxed areas are magnified in the adjacent panels (*A*, *C*, and *E*). Scale bars, 50 μm (*A*, *C*, and *E*) and 250 μm (*H*). ****$P < .0001$, **$P = .0013$, *$P = .0122$. AOM, azoxymethane; DMSO, dimethyl sulfoxide; DSS, dextran sodium sulfate; M, mol/L.


A
Grem1-YH2 cells
AP or APS tumoroids
(Luciferase+)
Matrigel
IgG isotype
or
GREM1-neutralizing Ab
B
CM from
Grem1-YH2 cells
IgG isotype
GREM1-neutralizing Ab
Id1
P = .0155
P = .6400
Expression levels
relative to Gapdh (A.U.)
AP
APS
C
CM from Grem1-YH2 cells
IgG isotype
GREM1-neutralizing Ab
AP tumoroids
APS tumoroids
D
CM from
Grem1-YH2 cells
IgG isotype
GREM1-neutralizing Ab
P < .0001
P = .9729
Luciferase activity
AP tumoroids
APS tumoroids
E
CM from
Grem1-YH2 cells
IgG isotype
GREM1-neutralizing Ab
Lgr5
Krt20
P = .0054
P = .3757
P = .0467
P = .6196
Expression levels
relative to Gapdh (A.U.)
AP
APS
F
GFP-YH2 cells or Islr-YH2 cells
rBMP7
AP tumoroids
(Luciferase+)
Matrigel
G
CM from;
GFP-YH2 cells
Islr-YH2 cells
Id1
P = .0231
Expression levels
relative to Gapdh (A.U.)
AP
H
CM from;
GFP-YH2 cells
Islr-YH2 cells
AP tumoroids
I
CM from;
GFP-YH2 cells
Islr-YH2 cells
P = .0180
Luciferase activity
AP tumoroids
J
CM from;
GFP-YH2 cells
Islr-YH2 cells
Lgr5
Krt20
P = .0075
P = .0430
Expression levels
relative to Gapdh (A.U.)
AP

Conditioned Medium From Islr*-Overexpressing Intestinal Fibroblasts Increases Bone Morphogenetic Protein Signaling, Facilitates Colorectal Cancer Organoid Differentiation, and Attenuates Growth*

Next, we tested whether CM from *Islr*-overexpressing YH2 cells could augment BMP signaling in CRC tumoroids and thus inhibit CRC tumoroid growth. Given that ISLR overexpression promoted BMP signaling in the presence of recombinant BMP7 (Figure 1*F* and *G*), we collected CM from *Islr*- or GFP-overexpressing YH2 cells incubated with recombinant BMP7 and transferred the CM to AP tumoroids (Figure 5*F*). In keeping with our hypothesis of the opposing roles of GREM1 and ISLR, CM from *Islr*-YH2 cells increased *Id1* expression in AP tumoroids, suggesting that ISLR overexpression in fibroblasts enhanced BMP signaling in CRC tumoroids in a paracrine manner (Figure 5*G*). Moreover, medium conditioned by *Islr*-YH2 cells decreased AP tumoroid–derived luciferase signals and tumoroid size (Figure 5*H* and *I* and Supplementary Figure 19*B*). Similar to the GREM1-neutralizing antibody, this tumoroid growth inhibition by *Islr*-YH2 cell media was accompanied by a reduction in *Lgr5* transcripts and increased *Krt20* expression in AP tumoroids (Figure 5*J*). Taken together, our in vitro studies indicate that modulating stromal, secreted BMP regulators of BMP signaling through either GREM1-neutralizing antibody or *Islr*-overexpression facilitates *Lgr5*$^{+}$ intestinal stem cell differentiation and diminishes CRC tumoroid growth.

Modifying the Metastatic Niche by Adeno-associated Virus 8–Mediated In Vivo Overexpression of Islr *in Hepatocytes Retards Colorectal Cancer Hepatic Metastasis*

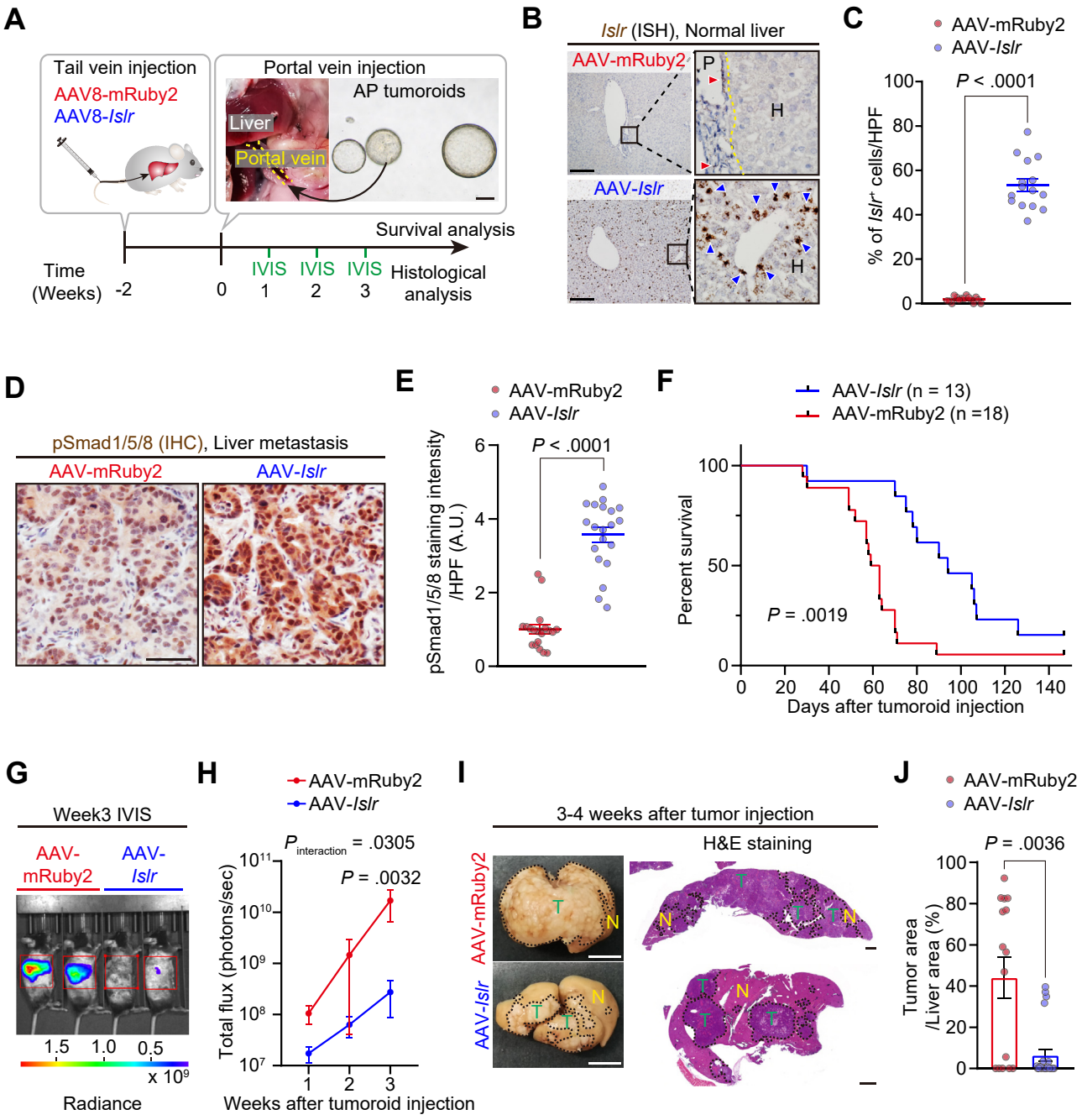
Hepatic metastasis is the major cause of CRC death,[1] and, through portal vein dissemination, complicates most advanced gastrointestinal adenocarcinomas. Therefore, we explored whether enhancing BMP signaling, either by the GREM1-neutralizing antibody or *Islr* overexpression, could impair hepatic metastagenesis of CRC in vivo. For this purpose, we generated a mouse model of CRC hepatic metastasis using intraportal injection of AP tumoroids (Figure 6*A*).

Initially, we examined whether treatment with the GREM1-neutralizing antibody could retard CRC hepatic metastasis and improve survival in comparison to an IgG isotype–treated control group (Supplementary Figure 20*A*). Consistent with our earlier observation in human CRC liver metastases (Supplementary Figure 8*A* and *B*), ISH for *Grem1* confirmed that *Grem1* was expressed by fibroblastic cells in the stroma of mouse CRC hepatic metastases (Supplementary Figure 20*B*). Immunohistochemistry for pSmad1/5/8 showed that the treatment with GREM1-neutralizing antibody restored BMP signaling in metastatic CRC (Supplementary Figure 20*C* and *D*). The GREM1-neutralizing antibody–treated group showed a trend toward prolonged overall survival and a trend toward decreased tumor growth, although the groups, as dosed in this study, did not show significant differences in survival (Supplementary Figure 20*E* and *F*).

Thus, we then focused on the other key stromal BMP signaling regulator, *Islr*. Inspired by recent advances in AAV-mediated gene therapy in human diseases,[23,24] we reasoned that augmenting BMP signaling, via ectopic overexpression of *Islr* in hepatocytes, a liver cell type shown to contribute to a metastatic niche,[25] could potentially ameliorate the progression of CRC hepatic metastasis. We injected AAV8 (an AAV serotype with tropism for murine hepatocytes[26]) encoding either *Islr* or, as a control, a red fluorescent protein (mRuby2) via mouse tail vein (Figure 6*A*) to generate ectopic *Islr* overexpression in hepatocytes in vivo (Figure 6*B* and *C*). Two weeks after the tail vein injection, AP tumoroid cells were injected directly into the portal vein to generate CRC hepatic metastases. The liver-directed delivery of *Islr* enhanced BMP signaling in CRC hepatic metastasis as well as in normal hepatocytes (Figure 6*D* and *E* and Supplementary Figure 21*A*–*C*) and significantly prolonged mouse survival compared to the AAV8-mRuby2 control group (Figure 6*F*). Notably, there was no histologic evidence of liver injury induced by AAV8-*Islr* (Supplementary Figure 22*A* and *B*).

To evaluate alterations in growth kinetics and histopathology by AAV8-*Islr*, we next monitored the growth of CRC hepatic metastases with an in vivo imaging system and harvested all mice 3 to 4 weeks after tumor injection (Figure 6*A*). In line with improved survival, the AAV8-*Islr* group showed reduced tumor-derived luminescence signal, histologic tumor area, and Ki-67 cell proliferation index (Figure 6*G*–*L*). Consistent with our earlier in vitro data showing that stromal *Islr* overexpression promoted CRC tumoroid differentiation, *Islr* overexpression in hepatocytes yielded more differentiated CRC histology (Figure 6*M* and *N*). Furthermore, in agreement

Figure 5. A GREM1-neutralizing antibody or CM from *Islr*-overexpressing intestinal fibroblasts restrains CRC tumoroid growth and promotes *Lgr5*$^{+}$ stem cell differentiation via increased BMP signaling in tumoroids. (*A*) Experimental schematic depicting CM transfer from *Grem1*-overexpressing YH2 cells to AP (*Apc*$^{\Delta/\Delta}$ and *Trp53* $^{\Delta/\Delta}$) tumoroids or APS (*Apc*$^{\Delta/\Delta}$, *Trp53* $^{\Delta/\Delta}$, and *Smad4* $^{\Delta/\Delta}$) tumoroids. Either an IgG isotype or a GREM1-neutralizing antibody was added to the tumoroids. (*B*) qRT-PCR for *Id1* in AP and APS tumoroids (n = 3). (*C*) Representative pictures of AP tumoroids and APS tumoroids. (*D*) Luciferase signals from AP tumoroids and APS tumoroids (n ≥ 8). (*E*) qRT-PCR for *Lgr5* and *Krt20* in AP and APS tumoroids (n = 3). (*F*) Experimental schematic depicting CM transfer from *Islr*-overexpressing YH2 cells to AP tumoroids. CM was collected from *Islr*- or GFP-overexpressing YH2 cells incubated with 10 ng/mL of recombinant BMP7 (rBMP7). (*G*) qRT-PCR for *Id1* in AP tumoroids (n = 3). (*H*) Representative pictures of AP tumoroids. (*I*) Luciferase signals from AP tumoroids (n = 14). (*J*) qRT-PCR for *Lgr5* and *Krt20* in AP tumoroids (n = 3). Scale bars, 500 μm. Mean ± SEM. Two-tailed unpaired Student *t* test (*B*, *D*, *E*, *G*, *I*, and *J*). Note that data normalization was performed within the AP and APS tumoroid groups separately (*B*, *D*, and *E*). Ab, antibody.


A
Tail vein injection
AAV8-mRuby2
AAV8-Islr
Portal vein injection
Liver
Portal vein
AP tumoroids
Survival analysis
Time (Weeks)
-2
0
IVIS 1
IVIS 2
IVIS 3
Histological analysis
B
Islr (ISH), Normal liver
AAV-mRuby2
AAV-Islr
P
H
C
AAV-mRuby2
AAV-Islr
P < .0001
% of Islr+ cells/HPF
D
pSmad1/5/8 (IHC), Liver metastasis
AAV-mRuby2
AAV-Islr
E
AAV-mRuby2
AAV-Islr
P < .0001
pSmad1/5/8 staining intensity /HPF (A.U.)
F
AAV-Islr (n = 13)
AAV-mRuby2 (n =18)
Percent survival
P = .0019
Days after tumoroid injection
G
Week3 IVIS
AAV-mRuby2
AAV-Islr
1.5 1.0 0.5
x 10^9
Radiance
H
AAV-mRuby2
AAV-Islr
P interaction = .0305
P = .0032
Total flux (photons/sec)
Weeks after tumoroid injection
I
3-4 weeks after tumor injection
H&E staining
AAV-mRuby2
AAV-Islr
T
N
J
AAV-mRuby2
AAV-Islr
P = .0036
Tumor area /Liver area (%)
K
Ki-67 (IHC), Liver metastasis
AAV-mRuby2
AAV-Islr
L
AAV-mRuby2
AAV-Islr
P < .0001
Ki67 labeling index (%)
M
EpCAM (IHC), Liver metastasis
AAV-mRuby2
AAV-Islr
Poorly diff.
Moderately diff.
N
AAV-mRuby2
AAV-Islr
P = .0309
Poorly diff tumor area /Tumor area (%)

with antifibrotic roles of BMP7 signaling,[17] αSMA immunostaining and Picro-Sirius red staining showed that fibrosis was reduced in the AAV8-*Islr* group (Supplementary Figure 23*A*–*D*). Taken together, these data suggest that BMP modulation could be an attractive target in CRC metastasis and that leveraging hepatocytes to augment BMP signaling by AAV8-*Islr* could represent an exciting, novel therapeutic opportunity in metastatic CRC.

Discussion

Initially identified more than 50 years ago,[27] BMP is now known to be important in regulating intestinal epithelial homeostasis and cancer cell proliferation.[3,4] The regulation of BMP in the tumor microenvironment and the role of BMP in tumor management, however, are still largely unknown. In the present study, we have shown that *GREM1* and *ISLR* are CAF-specific factors that exert opposing effects on BMP signaling in colonic fibroblasts and define distinct subpopulations of fibroblasts in the normal colon and CRC. FOXL1 and TGF-β may explain, at least in part, the polarization of CAFs into tumor-promoting $GREM1^{+}$ CAFs and tumor-retarding $ISLR^{+}$ CAFs. Moreover, *GREM1* and *ISLR* expression levels were associated with poor and favorable outcomes in patients with CRC. Using organoid culture and a preclinical mouse model, our data support that BMP signaling imbalance, regulated by *Grem1* and *Islr*, drives CRC progression and is a key target for cancer treatment. We provided the therapeutic proof of principle that augmenting BMP signaling, either by using a GREM1-neutralizing antibody or AAV8-*Islr*, represents an attractive future approach to treat CRC.

Previous studies have shown that stromal deletion of BMPR2 could facilitate carcinogenesis in CRC and breast cancer, suggesting tumor-suppressive functions of mesenchymal BMP signaling.[28,29] Moreover, augmentation of BMP signaling mediated either by increased BMP ligand or decreased BMP antagonist secretion from the tumor mesenchyme, which occurs as a consequence of stromal Hedgehog signaling activation, restrains bladder cancer or CRC progression.[22,30] Our work supports these findings and provides novel insights into regulatory mechanisms of BMP signaling in the CRC tumor microenvironment. Given that loss of stromal BMPR2 expression increases cytokine production in a mouse model of breast cancer,[28] further studies are warranted to investigate whether mesenchymal BMP signaling, modulated by *GREM1* and *ISLR*, could also be involved in shaping the immunosuppressive tumor microenvironment.

High stromal TGF-β signaling is associated with worse outcomes for patients with CRC.[19] Here, we provide a potential mechanism underlying CAF heterogeneity, initiated by TGF-β and FOXL1, that polarizes fibroblasts toward $Grem1^{high}Islr^{low}$ CAFs or $Islr^{high}Grem1^{low}$ CAFs. TGF-β drives *Foxl1* expression in CAFs (Figure 4*J*). FOXL1, in turn, directly up-regulates *Grem1* expression to antagonize BMP signaling (Figure 3*E*–*K*) and promote cancer progression by suppressing differentiation of $Lgr5^{+}$ stem cells while inducing epithelial proliferation. In contrast, relatively low levels of TGF-β in the microenvironment of some tumors result in lower levels of FOXL1, which permits higher *Islr* expression and a relative tumor-suppressive and higher-BMP signaling milieu (Figures 3*E*–*G* and 4*J*). This potential TGF-β–FOXL1–*Grem1/Islr* axis that modulates BMP signaling in the colon provides a novel mechanism, to our knowledge, to help understand the polarization of CAFs within the tumor microenvironment and presents a promising target for future cancer treatment.

By using scRNA-seq from normal mouse small intestines, one recent article suggested that *Grem1* is expressed mainly by $Pdgfra^{low+}Cd81^{+}$ trophocytes that are distinct from $Foxl1^{+}$ telocytes.[5] Consistent with this, our *Foxl1* smFISH in *Grem1*-CreERT2 mice showed that there were fewer *Foxl1* and *Grem1* double-positive fibroblasts in the small intestine than in the colon (Supplementary Figure 24). This observation raises the possibility of organ-dependent *Foxl1* expression patterns within the context of gastrointestinal fibroblast heterogeneity, which warrants further research.

One limitation of the present study is that we have not unraveled the origins and lineage hierarchy of $Grem1^{+}$ iRSCs and $Islr^{+}$ fibroblasts despite presenting a potential mechanism for fibroblast polarization. Whether $Grem1^{+}$ CAFs and $Islr^{+}$ CAFs arise from their local progenitors, are recruited from the bone marrow, or are simply new expression profiles within existing cells requires further investigation.

Figure 6. AAV8-mediated *Islr* overexpression in hepatocytes augments BMP signaling and retards CRC hepatic metastasis growth. (*A*) Experimental scheme. Yellow dotted lines outline the portal vein. (*B*, *C*) ISH for *Islr* in the liver 2 weeks after tail vein injection of AAV8-*Islr* or AAV8-mRuby2. (*B*) Representative images. Red and blue arrowheads denote the endogenous expression of *Islr* in fibroblastic cells in the portal area and ectopic overexpression of *Islr* in hepatocytes, respectively. The yellow dotted line indicates the border between the portal area (P) and hepatocytes (H) (*C*) Semiquantification; 5 high-power fields (400×)/mouse, 3 mice each. (*D*, *E*) Immunohistochemistry for pSmad1/5/8 in liver metastases. (*D*) Representative pictures. (*E*) Quantification of 3,3′-diaminobenzidine (DAB) intensity: 5 high-power fields/mouse, 4 mice each. (*F*) Kaplan-Meier survival curves. (*G*, *H*) Luciferase signals from AP tumoroids were assessed by an in vivo imaging system (IVIS). (*G*) Representative images. (*H*) Growth kinetics. Signals within red rectangles in *F* were quantified; n = 5 (AAV-mRuby2) and 8 (AAV-*Islr*) mice. (*I*) Representative macroscopic pictures and H&E-stained sections of liver metastases. Dotted lines indicate borders between tumors (T) and the normal liver (N). (*J*) Quantification of tumor areas using H&E stained sections; 3 liver pieces/mouse, 5 (AAV-mRuby2) and 8 (AAV-*Islr*) mice. (*K*, *L*) Immunohistochemistry for Ki-67: (*K*) representative pictures and (*L*) Ki-67 positivity in total epithelial cells. Four high-power fields/mouse, 4 mice each. (*M*, *N*) Evaluation of tumor cell differentiation status. (*M*) Representative pictures of immunohistochemistry for EpCAM. Green dotted lines indicate tumor budding. (*N*) The ratio of poorly differentiated tumor areas in the total tumor areas; 11 liver pieces each group from 5 (AAV-mRuby2) and 8 (AAV-*Islr*) mice. Scale bars represent 200 μm (*A*, *B*), 1 cm (macroscopic pictures in *I*), 1 mm (H&E staining in *I*), and 50 μm (*D*, *K*, and *M*). Mean ± SEM. Mann-Whitney *U* test (*C*, *E*, *J*, *L*, and *N*), log rank test (*F*), and 2-way repeated-measures ANOVA with post hoc Sidak multiple comparison test at week 3 (*H*). diff, differentiated; IHC, immunohistochemistry; IVIS, in vivo imaging system.

Considering the possible plasticity of CAFs,[2] it is plausible that *Grem1*$^+$ CAFs and *Islr*$^+$ CAFs could undergo phenotypic interconversion during tumor development, a state of dynamic flux between a relatively polarized cancer-retarding or cancer-promoting microenvironment.

We also provided the first experimental evidence that the GREM1-neutralizing antibody promoted differentiation of *Lgr5*$^+$ intestinal stem cells and retarded CRC tumoroid growth only in a *Smad4*–wild-type setting, but not in *Smad4*-mutant tumoroids. Our data reinforce the importance of stratifying patients who may benefit from the GREM1-neutralizing antibody according to the mutation status of BMP-related genes such as *SMAD4* and BMP receptors in future preclinical and clinical trials. Our preliminary data with an in vivo experimental model of CRC hepatic metastases implied that the GREM1-neutralizing antibody still requires further optimization of the therapeutic setting (metastatic prevention vs treatment), dosage regimens, route of administration, and in combination with other agents. However, the combined in vitro and in vivo findings are encouraging.

To our knowledge, our study is the first to use AAV8 to target hepatocytes to treat hepatic metastasis.[23,31] Clinical trials have shown that in vivo gene delivery by AAV holds great promise in patients with nonneoplastic diseases such as inherited genetic diseases and degenerative neuromuscular disorders.[23] Excitingly, recent clinical trials have shown that hepatocyte-directed gene transfer of coagulation factors by AAV substantially improved clinical symptoms in patients with hemophilia.[23,24] In light of new human liver-tropic bioengineered AAVs,[26] our work suggests that AAV-mediated hepatocyte-directed therapy could, in the future, serve as a novel and well-tolerated cancer therapy. Furthermore, our findings pave the way for AAV-mediated delivery of payloads to modulate not only BMP signaling but any other number of the relevant biological hallmarks of cancer.[32]

In conclusion, our data show that stromal BMP signaling, inhibited by *GREM1* and promoted by *ISLR,* is biologically relevant in CRC growth, spread, and survival. By targeting the upstream determinants of mesenchymal expression, such as TGF-β and FOXL1 or by targeting the downstream drivers of BMP signaling such as GREM1 and ISLR, one may identify new approaches to prevent and treat cancer.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at https://doi.org/10.1053/j.gastro.2020.11.011.

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CRediT Authorship Contributions

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Conflicts of interest

These authors disclose the following: Gareth Davies is a researcher at UCB Pharma (Slough, Berkshire, UK); A GREM1-neutralizing antibody was generated by UCB Pharma and provided to researchers at the Gut Cancer Group, the University of Adelaide (Australia), at no cost. Leszek Lisowski has consulted on technologies addressed in this paper and has stock and/or equity in companies with technology broadly related to this paper. The remaining authors disclose no conflicts.

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Supplementary Materials and Methods

Human and Animal Ethics

Human CRC samples were obtained at the time of surgery from patients who had provided informed consent. This study was approved by the Ethics Committee of Nagoya University Graduate School of Medicine (2017-0127). All animal protocols were approved by the Animal Ethics committees of SAHMRI and Nagoya University Graduate School of Medicine (SAM322, SAM189, and 31434).

Antibodies

The following antibodies were used in this study. Goat polyclonal anti-GREM1 antibody (1:1000 for Western blotting [WB]; AF956, R&D Systems), rabbit polyclonal anti-ISLR antibody (1:1000 for WB; HPA050811, Atlas Antibodies), mouse monoclonal anti-ACTB (1:5000 for WB; sc-47778, Santa Cruz Biotechnology), rabbit polyclonal anti-RFP (1:500 for IF; 600-401-379, Rockland), rabbit monoclonal anti–phospho-Smad1/5 (1:1000 for WB; clone 41D10, no. 9516, Cell Signaling Technology), rabbit polyclonal anti-Smad1 (1:1000 for WB; Cell Signaling Technology, no. 9743), goat polyclonal anti-mouse PDGFRA (1:50 for IF; AF1062, R&D Systems), rabbit polyclonal anti-HA antibody (1:7500 for WB; ab9110, Abcam), rabbit polyclonal anti-FOXL1 (1:1000 for WB; ABC937, Merck Millipore), mouse monoclonal antiαSMA (1:500 for immunohistochemistry [IHC]; clone 1A4, Dako), rabbit monoclonal anti-Smad4 (1:5000 for WB; ab40759, Abcam), rabbit monoclonal anti-Ki67 (1:200 for IHC; ab16667, Abcam), rabbit polyclonal anti-EpCAM (1:150 for IHC; abcam71916, Abcam), sheep polyclonal anti-FAP (1:100 for IF; AF3715, R&D Systems), and rabbit polyclonal anti-phospho-Smad1/5/8 (1:100 for IHC; AB3848-I, Merck Millipore). Mouse monoclonal GREM1-neutralizing antibody (Ab7326) and mouse monoclonal IgG1 isotype (Ab101.4) were provided by UCB Pharma.

RNA In Situ Hybridization

All ISH analyses were performed on formalin-fixed and paraffin-embedded human and mouse tissue samples by using RNAscope technology (RNAscope 2.5 HD Detection Kit or RNAscope Multiplex Fluorescent Reagent Kit v2, Advanced Cell Diagnostics [ACD]) following the manufacturer's instructions. Briefly, tissue sections were baked in a dry oven (HybEZ II Hybridization System, ACD) at 60°C for 1 hour and deparaffinized, followed by incubation with Pretreat 1, 2, and 3 (ACD). Slides were incubated with relevant probes for 1 hour at 40°C, followed by successive incubations with Amp1–6 reagents. Staining was visualized with diaminobenzidine (DAB) or TSA Plus Cyanine 5 (NEL745001KT, PerkinElmer). For survival analysis with human CRC samples, ISH samples were quantified by a clinical pathologist in a blinded manner, following a semi-quantitative scoring method recommended by the manufacturer (ACD), where each signal was evaluated as 0 (no staining or <1 dot/cell), 1 (1–3 dots/cell), 2 (4–9 dots/cell with no or very few dot clusters), 3 (10–15 dots/cell and <10% of dots in clusters), or 4 (>15 dots/cell and >10% of dots in clusters). For other ISH analyses, cells with a score of 0 were defined as negative, and cells with a score of 1 or more were defined as positive. In normal colorectal samples, only lamina propria areas were quantified (Figures 2*B* and 3*B* and *D*).

Probes Used for In Situ Hybridization

ISH (RNAscope) probes used in the study were human *PPIB* (a positive control probe, NM_000942.4, regions 139–989; catalogue number 313901), bacterial *DapB* (a negative control probe, EF191515, regions 414–862; catalogue number 310043), human *ISLR* (NM_005545.3, region 275–1322; catalogue number 455481 or 455481-C2), mouse *Islr* (NM_012043.4, regions 763–1690; catalogue number 450041 or NM_012043.4, regions 277–2225; catalogue number 453321-C2), human *GREM1* (NM_013372.6, regions 175–1472; catalogue number 312831 or 312831-C2), mouse *Grem1* (NM_011824.4, regions 398–1359; catalogue number 314741), human *FOXL1* (NM_005250.2, regions 1610–2981; catalogue number 558081), and mouse *Foxl1* (NM_008024.2, regions 954–1931; catalogue number 407401).

Luciferase Reporter Assay

DNA fragments of 4.3 kilo base pairs (–1524 to +2841; +1 is the transcriptional start site (TSS) of exon 1 of human *GREM1*) spanning human *GREM1* promoter and intron regions were generated by custom gene synthesis by Gene Universal and inserted into the upstream of the luciferase (NanoLuc) reporter gene in the pNL2.1 vector (Promega). This region contained a putative hFOXL1-binding site at TSS +2103 to +2109 (corresponding to TSS +2008 to +2014 in mouse *Grem1*), as analyzed by the JASPAR database.[1] Sequences from TSS +2103 to +2841, which contains the putative hFOXL1-binding site, were deleted from the *GREM1* promoter (4.3 kbps)–NanoLuc plasmid, resulting in the generation of the truncated *GREM1* promoter (3.6 kbps)–NanoLuc vector.

The 3.5–kilo base pair human *ISLR* promoter and intron regions (–2321 to +1179, +1 is the TSS of exon 1 of human *ISLR*) were inserted upstream of the luciferase (NanoLuc) reporter gene in the vector pNL2.1 (Promega), as described previously.[2]

The pNL2.1 vectors containing the *GREM1* or *ISLR* promoter regions were cotransfected with pEF1a (human elongation factor 1 alpha promoter)–firefly into YH2 cells by using Lipofectamine 2000 (11668019, Thermo Fisher Scientific). Cell lysates were collected 48 hours after transfection by using a passive lysis buffer (E1941, Promega). The luminescence values of cell lysates were measured by using a Nano-Glo Dual-Luciferase Reporter Assay System (N1610, Promega) and a GloMax microplate reader (GM3000, Promega) following the manufacturer's instructions. The NanoLuc luminescence levels were normalized to the firefly luminescence levels.

To evaluate BMP signaling activation in GFP-, *Islr*-, and *Grem1*-overexpressing YH2 cells, YH2 cells were cotransfected with pGL3 BMP-responsive element–firefly (Addgene; plasmid no. 45126) and pTK-*Renilla* (E2241, Promega). Cells were grown in Dulbecco's modified Eagle medium (DMEM) containing 1% fetal bovine serum (FBS), and cell lysates were collected 48 hours after transfection by using passive lysis buffer (E1941, Promega). Luminescence values of cell lysates were measured by using a Dual-Luciferase Reporter Assay System (E1960, Promega) and a GloMax microplate reader (GM3000, Promega). The firefly luminescence levels were normalized to the *Renilla* luminescence levels. Each assay was performed with at least 2 technical replicates.

Promoter Analyses and Chromatin Immunoprecipitation

Promoter analyses were performed by using the JASPAR database.[1] Promoter analyses of the genomic area around the *Grem1* TSS for putative FOXL1-binding sites identified a binding motif for FOXL1 at TSS +2008 to +2014 bps, an intronic region of the mouse *Grem1* gene, which was conserved in humans.

Chromatin was prepared from 3 independent replicates of *FOXL1*-HA–overexpressing YH2 cells by using a truChIP Chromatin Shearing Kit with Formaldehyde (520154, Covaris) according to the manufacturer's instructions. Chromatin shearing was performed by using a Covaris M220 focused ultrasonicator. ChIP assay was performed by using the EpiQuik Chromatin Immunoprecipitation Kit (P-2002, Epigentek) and IgG isotype (ab171870, Abcam), anti-HA antibody (ab9110, Abcam), and anti-Histone H3 antibody (ab1791, Abcam) as per the manufacturer's protocol. Prepared DNA samples were analyzed by quantitative PCR with the following primers: mouse *Grem1* promoter (forward: 5′-gcaccgttggattaaggctc-3′; reverse: 5′-tgaagatcataggaaagctgtgaag-3′).

Histopathologic Examination

All histopathologic examinations were performed by using formalin-fixed and paraffin-embedded tissues. Paraffin blocks were cut into 4–5-μm sections. H&E staining and Picro-Sirius red staining were performed by Histology Services (University of Adelaide) or Sept Sapie (Tokyo, Japan).

Immunohistochemistry and Immunofluorescence

Formalin-fixed and paraffin-embedded tissue sections were deparaffinized with xylene and rehydrated with phosphate-buffered saline (PBS), followed by antigen retrieval by boiling samples in antigen retrieval buffer (pH 6 or pH 9; H-3300, Vector Laboratories; S1699, Dako; or S2367, Dako) for 30 minutes. Inactivation of endogenous peroxidase was performed with 0.5% H_2O_2 in methanol for 15 minutes, followed by washing with PBS. Then, sections were treated with blocking buffer (X0909, Dako) for 30 minutes, incubated with the indicated primary antibodies overnight at 4°C, and washed with PBS. Sections were incubated with horseradish peroxidase (HRP)–polymer secondary antibody (ab214879 or ab214880, both from Abcam) for 30 minutes, followed by signal detection with DAB solution (K3468, Dako).

For IF studies, deparaffinized sections were treated with blocking buffer (X0909, Dako) for 30 minutes, incubated with the indicated primary antibodies overnight at 4°C, and washed with PBS. Sections were then incubated with Alexa Fluor 488/594/647–conjugated secondary antibodies (all from Thermo Fisher Scientific) for 1 hour at room temperature. The sections were then mounted with ProLong Gold antifade reagent containing 4′6-diamidino-2-phenylindole (DAPI) (Thermo Fisher Scientific), and fluorescence was examined by using a confocal laser-scanning microscope (TCS SP8 MP, Leica) or an inverse IF microscope BZ-X710 (Keyence) with optical sectioning.

Combined Single-Molecule Fluorescence In Situ Hybridization and Immunofluorescence

Combined smFISH and IF were implemented by first performing smFISH, followed by IF. After smFISH, the sections were blocked with blocking buffer (X0909, Dako) and then incubated with a primary antibody overnight at 4°C. The sections were washed in 1× Phosphate-buffered saline containing 0.05% TWEEN 20 (P1379, Sigma) 3 times and then incubated with Alexa Fluor 555–conjugated secondary antibody (Thermo Fisher Scientific) for 60 minutes at room temperature. The sections were then mounted with ProLong Gold antifade reagent containing DAPI (Thermo Fisher Scientific). Fluorescence was examined by using an inverse IF microscope BZ-X710 (Keyence) with optical sectioning.

Western Blot Analysis

Cells were lysed in a lysis buffer (78501, Thermo Fisher Scientific) supplemented with Complete Protease Inhibitor (Roche) and PhosSTOP Phosphatase Inhibitor cocktails (4906845001, Roche). Lysates were clarified by centrifugation at 12,000*g* for 10 minutes at 4°C. Then, sodium dodecyl sulfate (SDS) sample buffer (10 mmol/L Tris-HCl, 2% SDS, 2 mmol/L EDTA, 0.02% bromophenol blue, 6% glycerol; pH 6.8) was added. To prepare CM and total cell lysates for WB, YH2 cells were incubated for 72 hours in protein-free and FBS-free medium (Freestyle 293 expression medium; 12338018, Gibco). The medium was collected, centrifuged at 400*g* for 5 minutes, and filtered through a 0.45-μm filter (16533, Sartorius) to remove cell debris. The medium was concentrated 10-fold by using Amicon Ultra 3-kDa centrifugal filters (UFC500396, Millipore).

The separation was performed by SDS–polyacrylamide gel electrophoresis using a precast gel (4568094, Bio-Rad). Proteins were then transferred to polyvinylidene difluoride membranes (1620177, Bio-Rad) by using a semidry transfer system (1703940, Bio-Rad). The membranes were blocked in 5% milk in PBS containing 0.05% Tween 20 and then incubated with primary antibodies. Proteins were detected by HRP–conjugated secondary antibodies (NA9310V, GE Healthcare; N934VS, GE Healthcare; ab97120, Abcam), followed by signal development using an

HRP substrate (WBLUR0500, Millipore). The blots were imaged by using ChemiDoc MP (Bio-Rad) and quantified with Image Lab software, version 6.0.1 (Bio-Rad).

Plasmids

The cloning of mouse *Islr* complementary DNA (cDNA) was described previously.[3] A DNA fragment for the codon-optimized mouse *Grem1* gene was generated by using a custom gene synthesis service by Gene Universal. These genes were subcloned into bidirectional pLenti-EF1a–multiple cloning site–PGK-Puro vectors, resulting in the generation of pLenti-EF1a-*Islr*-PGK-Puro and pLenti-EF1a-*Grem1*-PGK-Puro.

To overexpress FOXL1 using a lentiviral expression system, DNA fragments for mouse codon–optimized human *FOXL1* gene with either no epitope tags or 3× HA tags at the amino terminus were generated by using a custom gene synthesis service by Gene Universal. These genes were subcloned into unidirectional pLenti-EF1a-GFP-p2A-Puro-EF1a–multiple cloning site vectors, resulting in the generation of pLenti-EF1a-GFP-p2A-Puro-EF1a-*FOXL1* and pLenti-EF1a-GFP-p2A-Puro-EF1a-*FOXL1*–3× HA.

To enable CRISPR/Cas9–mediated mouse *Foxl1* knockdown, the following mouse nontargeting gRNA (guide RNA) sequence or gRNA sequence targeting mouse *Foxl1* was subcloned into pLentiCRISPR v2 eSpCas9 plasmid by GenScript:

Nontargeting gRNA: 5′- AAAAAGTCCGCGATTACGTC-3′
g*Foxl1*: 5′-GGGCTGTACACGTACAACAG-3′

To overexpress *Islr* using the AAV expression system, mouse *Islr* cDNA was subcloned into a self-complementary AAV plasmid (VPK-430, Cell Biolabs), and a pAAV-CMV-*Islr* vector was generated. As a control, a self-complementary AAV plasmid expressing a red fluorescent protein, mRuby2 (pAAV-CAG-mRuby2, Addgene plasmid no. 99123), was used.

Lentivirus Production and Transduction

293T cells were cotransfected with psPAX2 (Addgene, plasmid no. 12260), pMD2.G (Addgene; plasmid no. 12259), and a lentivirus vector plasmid. At 48 and 72 hours after transfection, viral supernatants were harvested, filtered through a 0.45-μm filter, and concentrated by using Amicon Ultra Centrifugal Filters (Merck Millipore, UFC910024). Concentrated lentivirus particles were used for transduction. At 48 hours after transduction, positively transduced cells were selected with 2–4 μg/mL puromycin if the lentivirus vectors contain a puromycin resistance gene.

Retrovirus Production and Transduction

The 293T cells were cotransfected with the pEQ ecotropic-packaging vector[4] and a retrovirus vector plasmid. At 48 and 72 hours after transfection, viral supernatants were harvested, filtered through a 0.45-μm filter, and concentrated by using an ultracentrifuge (Optima XPN, Beckman Coulter). Concentrated retrovirus particles were used for transduction.

Cell Culture

A mouse colonic fibroblast cell line, YH2 cells,[5] was generously provided by Professor Tony Burgess (The Walter and Eliza Hall Institute of Medical Research). Normal mouse colonic organoids were isolated from a Rosa26-Cas9 mouse (JAX stock no. 024858; C57BL/6 × 129 genetic background) housed under pathogen-free conditions in the SAHMRI Bioresources facility. YH2 cells were cultured in DMEM (Sigma) supplemented with 10% FBS (Gibco).

All cell lines used were routinely screened for *Mycoplasma* contamination by MycoAlert Mycoplasma Detection Kit (LT07-118, Lonza).

In Figure 1*F* and *G* and Supplementary Figures 5 and 6, YH2 cells were serum-starved in DMEM containing 1% FBS for 48 hours before stimulation with recombinant human BMP7 (PHC9541, Thermo Fisher) or BMP2 (cyt-261, Prospec) at the concentration indicated in figure legends. Proteins and RNAs were collected 30 minutes and 24 hours, respectively, after the addition of the recombinant BMP7 or BMP2. In Figure 4*J*, YH2 cells were stimulated with a vehicle, 10 ng/mL of recombinant human TGF-β1 (240-B-002, R&D systems) + DMSO, or 10 ng/mL of recombinant human TGF-β1 + 10 μmol/L of galunisertib (ADV465749242; AChemBlock) in DMEM containing 1% FBS for 24 hours. In Supplementary Figure 18*A* and *B*, recombinant human BMP2 (cyt-261, Prospec), recombinant human GREM1 (provided by UCB Pharma), GREM1-neutralizing antibody (Ab7326), and mouse monoclonal IgG1 isotype (Ab101.4) were used for the experiment.

Isolation of Primary Mouse Colonic Fibroblasts

The isolation and culture of primary mouse colonic fibroblasts were performed by using the protocol described elsewhere[6] with modifications. Briefly, colons were harvested from 8–16-week-old C57BL/6J mice and thoroughly washed with cold PBS. The colon tissues were incubated in PBS supplemented with 3 mmol/L EDTA and 0.05 mmol/L dithiothreitol for 60 minutes at room temperature to remove epithelial cells. After washing the colon 3 times with PBS, the colons were digested in 1 mg of collagenase type IV (17104-019, Gibco), 1 mg of dispase (17105-041, Gibco), and 2000 units of DNase I (D4527, Sigma) in 15 mL of RPMI 5 media (described as follows) at 37°C for 30 minutes. The supernatant was filtered through a 70-μm cell strainer, and the cell suspension was plated on T75 flasks. Cells were cultured in RPMI 5 media: RPMI media (R8756, Sigma) supplemented with 5% FBS (Gibco), 2 mmol/L L-glutamine, 1% penicillin/streptomycin, 10 mmol/L HEPES, 1 mmol/L sodium pyruvate, and 0.05 mmol/L 2-mercaptoethanol.[6] At passage 2, cells were transduced with lentivirus encoding Cas9 protein and nontargeting gRNA or gRNA targeting *Foxl1*. RNA and protein collections were performed at passage 4.

Organoid Culture and Organoid Genome Editing

The basal culture medium for mouse colon organoids was Advanced DMEM (ADMEM)/F12 (Life Technologies) supplemented with 1× gentamicin/antimycotic/antibiotic

(Life Technologies), 10 mmol/L HEPES, 2 mmol/L GlutaMAX, 1× B27 (Life Technologies), and 1× N2 (Life Technologies). The following niche factors were used: 50 ng/mL mouse recombinant epidermal growth factor (Peprotech), 100 ng/mL mouse recombinant noggin (Peprotech), 20% R-spondin-2 CM, and 50% Wnt-3A CM. Organoids were plated in 50 μL growth factor-reduced Matrigel (356231, Corning) on a 24-well dish and 500 μL of the medium was added to organoids

gRNAs specific for each target gene were either previously published or designed de novo by using the CRISPR design tool.[7] *Apc* and *p53* gRNA oligos were cloned into pLentiGuide-CMV-dtomato (modified from Addgene plasmid no. 17452). *Smad4* gRNA oligos were cloned into plentiGuide-Puro (Addgene Plasmid no. 52963).

Guide RNA Sequences. *Apc*: 5′-GGAAGCCTTGTGGGACATGG-3′

Trp53: 5′-GTGTAATAGCTCCTGCATGG-3′[8]

Smad4: 5′-CAAAAGCGATCTCCTCCCGA-3′[9]

Normal colon organoids isolated from a Rosa26-Cas9 mouse (JAX stock no. 024858)[8] were transduced with lentivirus expressing sg*Apc* and sg*Trp53*. Three days later, media was changed to ADMEM (no Wnt, no Rspo, +Nutlin-3a; SML0580, Sigma) to enrich for correctly targeted clones (Supplementary Figure 16). After confirmation of correctly targeted monoclonal lines by amplicon sequence, retrovirus expressing firefly (Retro-SFG-NES-HSV1-tk-GFP-Luciferase)[10] was transduced to the AP ($Apc^{\Delta/\Delta}$, $Trp53^{\Delta/\Delta}$) tumoroid line. Then, a monoclonal line that showed high luciferase signals was selected by evaluating the luciferase signals by a luminometer. The luciferase-expressing AP monoclonal tumoroid was further transduced with lentivirus expressing sg*Smad4* and a puromycin resistance gene. Three days later, media was changed to ADMEM (no Wnt, no Rspo, no Noggin, +puromycin, +Nutlin-3a; SML0580, Sigma) to enrich for correctly targeted clones, followed by handpicking of monoclonal lines.

Handpicked monoclonal organoid lines were screened for loss of function insertions/deletions by using amplicon sequences using the following primers, as described.[11] Overhang sequences are underlined.

Apc forward: 5′-CTGAGACTTGCACATCGCAGCTTAATTCAGGCAAATCCTAAGAGAG-3′

Apc reverse: 5′-GTGACCTATGAACTCAGGAGTCGGTCTGTTTGCCATGAGATTCC-3′

Trp53 forward: 5′-GTGACCTATGAACTCAGGAGTCTAGTGAGGTAGGGAGCGACTTC-3′

Trp53 reverse: 5′-CTGAGACTTGCACATCGCAGCCCAAAGAGCGTTGGGCATGTG-3′

Smad4 forward: 5′-CTGAGACTTGCACATCGCAGCCTGGTGCTCCATTGCTTACT-3′

Smad4 reverse: 5′-GTGACCTATGAACTCAGGAGTCACTTAATTCCTCGATATTTAAGCTC-3′

To detect a large deletion in *Smad4* in APS tumoroids, PCR was performed by using the following primers: *Smad4* forward: 5′-TTGTGTCAGCTCAGAGTGGGTC-3′ and *Smad4* reverse: 5′-GCAAACCACACGACGATGC-3′. After gel electrophoresis and purification of PCR products, Sanger sequencing for the PCR products was performed to examine mutations.

Conditioned Medium Experiments

In Figure 5*A–E* and Supplementary Figure 17*A–D*, monolayer GFP-overexpressing or *Grem1*-overexpressing YH2 cells were cultured for 72 hours in DMEM (Gibco) containing 1% FBS (Gibco), 1% L-glutamine, and 1% penicillin/streptomycin. In Figure 5*F–J*, monolayer GFP-overexpressing or *Islr*-overexpressing YH2 cells were cultured for 72 hours in DMEM containing 1% FBS, 1% L-glutamine, and 1% penicillin/streptomycin with 10 ng/mL recombinant human BMP7 (PHC9541, Thermo Fisher Scientific). The medium was collected and filtered through a 0.45-μm filter (16533, Sartorius) to remove cell debris. Trypsinized tumoroid fragments equivalent to 2500 cells were plated in 50 ul growth factor-reduced Matrigel (356231, Corning) on a 24-well dish. At 30 minutes later, 500 mL of the CM was added to tumoroids. In Figure 5*A–E*, a mouse GREM1-neutralizing antibody (Ab7326, 100 μg/mL) or a mouse IgG1 isotype (Ab101.4, 100 μg/mL) was added to each well every time CM was added to the tumoroids throughout the course of the experiments.

Organoid area measurements, luciferase assays, and qRT-PCR were performed 8 days and 12 days after plating AP tumoroids and APS tumoroids, respectively, in experiments using *Grem1*-overexpressing YH2 cells. In experiments using *Islr*-overexpressing YH2 cells, organoid area measurements, luciferase assays, and qRT-PCR were performed 7 days after plating AP tumoroids.

Luciferase Assays for Firefly-Expressing Tumoroids

For luciferase assays using firefly-expressing tumoroids (AP tumoroids and APS tumoroids), after the removal of medium, tumoroids and Matrigel were lysed with 200 μL of 2× passive lysis buffer (E1941, Promega). Firefly luminescence values of cell lysates were measured with Luciferase Assay Reagent II (E1960, Promega) and a GloMax microplate reader (GM3000, Promega) following the manufacturer's instructions.

Tumoroid Area Quantification

The images of each well (1 picture/independent replicate) were captured by using an inverted microscope (IX53, Olympus; a 2× objective lens). Each organoid was quantified by using ImageJ software (National Institutes of Health)[12] to outline an organoid shape and measure an organoid area.

Quantitative Polymerase Chain Reaction

Total RNA was extracted by using TRIzol reagent (Invitrogen) and an RNeasy Mini Kit (Qiagen). Purified RNA samples were reverse-transcribed by using Transcriptor Universal cDNA Master (Roche) according to the manufacturer's instructions, followed by the dilution of cDNA at 1:5.

qRT-PCR of the cDNAs was performed with KAPA PROBE or SYBER FAST qPCR Master Mix (KAPABiosystems) and was run on a QuantStudio 7 Flex Real-Time PCR System (Thermo Fisher Scientific). The data were analyzed using the 2-ΔΔCt method and normalized to *Gapdh* expression levels.

Probes or primers (generated by IDT) used in this study are as follows. Primers were designed with the Primer-BLAST online program.[13]

Probes. Mouse *Grem1* (Mm.PT.58.11631114), mouse *Islr* (Mm.PT.58.12037488), mouse *Id1* (Mm.PT.58.6622645.g), mouse *Id2* (Mm.PT.58.13116812.g), mouse *Id3* (Mm.PT.58.29482466.g), mouse *Id4* (Mm.PT.58.6851535), mouse *Acta2* (Mm.PT.58.16320644), and mouse *Serpine1* (Mm.PT.58.6413525), mouse *Gapdh* (Mm.PT.39a.1), mouse *Lgr5* (Mm.PT.58.12492947), and mouse *Krt20* (Mm.PT.58.43092140).

Primers. Mouse *Grem1* (forward: 5′-GCTCTCCTTCGTCTTCCTC-3′, reverse: 5′-AGTGTATGCGGTGCGATTC-3′), mouse *Islr* (forward: 5′-TGCGAGCAATCCAGTCCTTAGATG-3′, reverse: 5′-AGCCCAACAAAGCAGGCACAG-3′), mouse *Foxl1* (forward: 5′-GTCGCTCAACGAGTGCTTCG-3′, reverse: 5′-TGCGCCGATAATTGCCGTTC-3′), and mouse *Gapdh* (forward: 5′-CCTCGTCCCGTAGACAAAATG-3′, reverse: 5′- TGTAGTTGAGGTCAATGAAGGG-3′)

Analyses of Publicly Available Complementary DNA Gene Expression Microarray and RNA-Sequencing Data Sets

For GSE39396,[14] microarray data were downloaded by using the GEOquery R package, version 2.54.1. Differential gene expression was analyzed as described in the limma software manual.[15] Briefly, a linear modeling approach was used by the function lmFit, followed by the empirical Bayes estimation through functions eBayes and topTable. Differentially up-regulated genes were defined as genes that showed a $\log_2$ fold change of greater than or equal to 2 and false discovery rate of less than 0.05. The top 150 differentially up-regulated gene probes in FAP^+ CAFs in each group (FAP^+ CAFs vs $EpCAM^+$ cancer cells, FAP^+ CAFs vs $CD31^+$ endothelial cells, and FAP^+ CAFs vs $CD45^+$ immune cells) were selected based on the $\log_2$ fold change. Volcano plots were generated by using the ggplot2 package, version 3.2.1. A human gene list in a Gene Ontology (BMP signaling pathway; GO0030509) was downloaded from AmiGO 2 (http://amigo.geneontology.org/amigo/search/ontology).

Expression microarray data (GSE17538, 39582, and 41258) and RNA-seq data (GSE94072) were obtained from the Gene Expression Omnibus (GEO) (https://www.ncbi.nlm.nih.gov/geo/). For survival analyses, gene expression levels for the following probes were used: 218469_PM_s_at for *GREM1* and 207191_PM_s_at for *ISLR*. Patient information in GSE41258,[16] GSE39582,[17] and GSE17538[18] were also obtained from the GEO. False delivery rate (FDR)–adjusted *P* values were calculated by GEO2R (https://www.ncbi.nlm.nih.gov/geo/geo2r). In GSE41258, normal colon samples, polyp (adenoma) samples, primary colon tumor samples, and liver metastasis samples included in a previous analysis[16] were analyzed by choosing samples with a label of "Included in analysis; Yes" on the GEO website. Optimal cutoff levels for survival analyses were determined by using the X-tile software.[19] Patients with survival time 0 (events at the time of patient enrolment) were not included in survival analysis by Prism 8 software (GraphPad; https://www.graphpad.com/guides/prism/8/statistics/stat_qa_survival_analysis.htm). In GSE94072, fragments per kilobase of exon per million fragments (FPKM) data were directly downloaded from the GEO website (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE94072).

Analyses of GREM1 and ISLR Expression Levels in the Consensus Molecular Subtypes of Colorectal Cancer

For The Cancer Genome Atlas data, the consensus molecular subtype (CMS) labels were downloaded from the supplementary materials of a previous publication.[20,21] Expression values and sample identifiers were downloaded from the Genomic Data Commons data portal (https://portal.gdc.cancer.gov/). Expression *Z*-scores for tumor samples were calculated based on the standard deviation and mean value for the normal colorectal tissues. Violin plots were then generated by using the ggplot2 package, version 3.2.1.

The CMS classifier[20] was applied to the downloaded GSE39582 expression microarray data to classify patients with colon cancer into CMSs. This allowed us to classify 566 colon cancer samples into 70 CMS1 tumors, 177 CMS2 tumors, 97 CMS3 tumors, 107 CMS4 tumors, and 115 unclassified tumors.

Analyses of Publicly Available Single-Cell RNA-Sequencing Data

scRNA-seq data (GSE132465)[22] were analyzed by using BBrowser2, version 2.6.4 (BioTuring). Unique molecular identifier counts available on the GEO website (https://www.ncbi.nlm.nih.gov/geo/) were used for normalization by the software. Processed scRNA-seq data (GSE81861)[23] were directly downloaded from the GEO website, and $\log_2$ (FPKM + 1) values were used to plot *GREM1* and *ISLR* expression.

In Supplementary Figure 4*A*, fibroblasts include myofibroblasts and stromal 1, stromal 2, and stromal 3 fibroblasts.[22] Endothelial cells include tip-like endothelial cells, stalk-like endothelial cells, proliferative endothelial cells, and lymphatic endothelial cells.[22] In Supplementary Figure 4*A* and *B*, immune cells comprise T cells, B cells, myeloid cells, and mast cells.[22,23] In Spearman correlation analyses, given our focus on *GREM1*- or *ISLR*-expressing CAFs, 1156 CRC CAFs that show detectable *GREM1* or *ISLR* transcripts (log-normalized *GREM1* expression of >0 or log-normalized *ISLR* expression of >0) were analyzed among all CRC CAFs (1501 cells; myofibroblasts and stromal 1, stromal 2, and stromal 3 fibroblasts from CRC tissues). Log-normalized expression values in the 1156 CRC CAFs were

transformed to *Z*-scores in the Spearman correlation analyses.

Tamoxifen Administration to Mice

Grem1-CreERT2 (JAX stock no. 027039);Rosa26-LSL (LoxP-stop-LoxP)–tdtomato (JAX stock no. 007909) and *Islr*-CreERT2[2];Rosa26-LSL-tdtomato were subjected to tamoxifen administration. First, 6 mg of tamoxifen (T5648, Sigma) dissolved in peanut oil was administered by oral gavage 4 times every other day to 6-week-old adult mice to sufficiently induce CreERT2-mediated recombination, followed by harvest of the 8-week-old mice. The colons and small intestines were removed, opened longitudinally, and fixed in 10% formalin overnight at room temperature. Tissues were subsequently dehydrated and paraffin embedded, and 4-μm sections were used for histologic analysis (ISH/IF and IF).

Mouse Genotyping

Genomic DNA extracted from mouse ear or tails was used for PCR genotyping. The sequences of the primers were as follows:

Islr–wild type (WT) forward: 5′-ACACGACCTTGGCAAGTCCCAGC-3′, *Islr*-WT reverse: 5′-GTCTGCAATCTGGAAGCCATACTTCTCC-3′;

Islr-CreERT2 forward: 5′-ACACGACCTTGGCAAGTCCCAGC-3′, *Islr*-CreERT2 reverse: 5′-CGATCCCTGAACATGTCCATCAGG-3′;

Grem1-CreERT2 forward: 5′-TTAATCCATATTGGCAGAACGAAAACG-3′, *Grem1*-CreERT2 reverse: 5′-CAGGCTAAGTGCCTTCTCTACA-3′.

Rosa26-LSL-tdtomato forward: 5′-AGATCCACCAGGCCCTGAA-3′, Rosa26-LSL-tdtomato reverse: 5′-GTCTTGAACTCCACCAGGTAG-3′;

Rosa26-Cas9 forward: 5′-CGTCGTCCTTGAAGAAGATGGT-3′, Rosa26-Cas9 reverse: 5′-CACATGAAGCAGCACGACTT-3′; and

Rosa26-WT forward: 5′-TTCCCTCGTGATCTGCAACTC-3′, Rosa26-WT reverse: 5′-CTTTAAGCCTGCCCAGAAGACT-3′.

An Azoxymethane/Dextran Sulfate Sodium Mouse Model of Colorectal Cancer

A colitis-associated AOM/DSS mouse model of CRC was generated as described.[24] Briefly, AOM (Sigma, A5486; 10 mg/kg) was administered by intraperitoneal injection to C57BL/6J mice obtained from Charles River Laboratories Japan. One week later, the mice commenced the first of 3 cycles of DSS to induce inflammation (MP Biomedicals, 160110; molecular weight, 36,000–50,000 Da). Each cycle comprised 5 days of 3% DSS (weight/volume), then 16 days of normal drinking water. At 10 weeks after AOM injection, mice were killed, and colons were removed, opened longitudinally, and fixed in 10% formalin overnight at room temperature. Tissues were subsequently dehydrated and paraffin embedded, and 4-μm sections were used for histologic analysis.

Mouse Models of Colorectal Cancer Liver Metastasis

Tumoroids were isolated from Matrigel and dissociated to single cells by using TrypLE. The cells were filtered through 40-μm meshes to remove cell clumps and suspended in cold PBS containing 10 μmol/L Y-27632 (In Vitro Technologies); 5.0×10^5 cells in 100 μL were injected directly into the portal vein using a 33G syringe.

Male and female Rosa26-Cas9 mice (JAX stock no. 024858; C57BL/6 × 129 genetic background; 6 to 24 weeks old), housed under pathogen-free conditions in the SAHMRI Bioresources facility, were used for portal vein injection.

In all animal experiments, mice were allocated randomly to different treatment groups. Sample sizes were determined based on pilot experiments and sample availability.

Adeno-Associated Virus Packaging

AAV8 preparation was performed by the Vector and Genome Engineering Facility, Children's Medical Research Institute. The AAV vectors used in this study were packaged by using a standard transient transfection as described previously.[25] In short, pAAV transfer vector, pAd5 helper plasmid,[26] and AAV helper plasmid encoding rep2 and cap8 (pR2C8) were transfected by using PEI (Polysciences; catalogue no. 239662) into HEK293 cells. Assembled vector particles were purified by using iodixanol density gradients as described.[27] Vector genomes were quantified by using qRT-PCR as previously described.[28]

Intravenous Adeno-Associated Virus 8 Injection

Male and female Rosa26-Cas9 mice (JAX stock no. 024858, C57BL/6 × 129 genetic background, 6 to 24 weeks old) were administered intravenously via the tail vein, with a 150-μL injection of 1.0×10^{11} viral genomes of AAV8-mRuby2 or AAV8-*Islr* per mouse. A 2-week interval after AAV injection was allowed for protein expression to occur before portal vein injection of tumoroids. In Figure 6*D*, *E*, and *I*–*N*, AAV8-treated mice were harvested for histopathologic analyses 3–4 weeks after tumor injection, before prominent necrosis complicated histologic assessment.

Subcutaneous Injection of the GREM1-Neutralizing Antibody

Subcutaneous injection of the GREM1-neutralizing antibody or IgG isotype was commenced 1 week after tumoroid injection. A 30-mg/kg dose of the GREM1-neutralizing antibody (Ab7326) or IgG isotype (Ab101.4) was subcutaneously administered to each mouse twice a week until endpoint criteria were met.

In Vivo Imaging System

Liver metastasis tumor growth kinetics were assessed with an in vivo imaging system (IVIS) by using a Xenogen IVIS Spectrum Imaging System (PerkinElmer) 10 minutes after intraperitoneal injection of 150 mg/kg of D-luciferin

(L-8220, Biosynth Carbosynth). Luciferase activity was quantified by using Living Image software (PerkinElmer).

A Humane Endpoint in Animal Experiments

In survival analyses, to determine the humane endpoint in animal experiments, a clinical record score was used. Scores were obtained by giving 1 point for the presence of each of the following observations: weight loss of >15%, hunched/ruffled coat, dehydration, absence of movement, or facial grimace. Once a score of 3 was reached, the mice were killed.

Collagen Gel Contraction Assay

Collagen gels were prepared by mixing YH2 cells with PureCol bovine type I collagen (Advanced Biomatrix), 1× PBS, and 1 mol/L NaOH (final concentration: 1.0 mg/mL PureCol). The mixture containing 3.0×10^5 cells/mL was seeded in 24-well cell culture plates and allowed to polymerize for 20 minutes. DMEM supplemented with 10% FBS and 1% penicillin/streptomycin was added, and gels were released from the wells. The gel area was measured with ImageJ. Representative images were processed using Adobe Photoshop CC (Adobe).

Human Colorectal Samples

All ISH was performed by using surgery samples from patients with CRC who were diagnosed with adenocarcinoma of the colon or rectum at Nagoya University Hospital, Japan. This study was conducted in accordance with the Helsinki Declaration for Human Research and approved by the Ethics Committee of Nagoya University Graduate School of Medicine (approval no. 2017-0127).

Quantitative Image Analysis

IHC or ISH images were processed into separate channels representing nuclei staining (hematoxylin) and IHC or ISH staining (DAB) by using a color deconvolution function in Fiji software (ImageJ). Then, binary images were generated by intensity thresholds, and DAB^+ areas were calculated by ImageJ. To evaluate pSmad1/5/8 staining intensity, after an application of the lower threshold, the DAB staining intensity per high-power field was calculated by ImageJ.

In Figure 2*B*, (1) normal rectal mucosa adjacent to adenocarcinoma, (2) adenoma in carcinoma, and (3) adenocarcinoma were randomly selected from ISH samples and were used for quantification. Stromal areas (total areas excluding epithelial cells) visualized by hematoxylin counterstaining were outlined and measured by a clinical pathologist by using Adobe Photoshop CC (Adobe). In the normal mucosa, only lamina propria regions were quantified.

The Ki-67 labeling index was calculated as the percentage of $Ki\text{-}67^+$ cells in the total epithelial cells as visualized by hematoxylin counterstaining.

The MRI fibrosis tool (https://github.com/MontpellierRessourcesImagerie/imagej_macros_and_scripts/wiki/MRI_Fibrosis_Tool) was used to quantify the percentage of fibrosis areas (Picro-Sirius red^+ areas) at default settings.

Distance Analysis Using In Situ Hybridization Samples

In Figure 4*H* and *I*, serial sections of human desmoplastic rectal cancer samples, in which the stroma accounted for more than 40% of the total tumor area, were evaluated. The minimum edge-to-edge distance between ISH signals and the closest tumor glands was calculated by the DiAna plugin[29] for ImageJ. Briefly, tumor glands in ISH images were outlined by a clinical pathologist. Then, to identify ISH signals (DAB^+ signals), the ISH images were subjected to color deconvolution before application of intensity thresholding and a watershed algorithm in Fiji software. ISH signals were segmented by using a minimum size threshold by the DiAna plugin. Then, the smallest edge-to-edge distance between each ISH signal and the closest tumor gland was computed by the DiAna plugin.

Evaluation of Tumor Differentiation and Tumor Budding

Evaluation of tumor differentiation and tumor budding was performed by a clinical pathologist. Poorly differentiated adenocarcinomas were defined as cancers that exhibit glandular differentiation in 5%–50% of the tumor, as described.[30]

Tumor budding was defined as a single tumor cell or a cell cluster of up to 4 tumor cells, as described.[31]

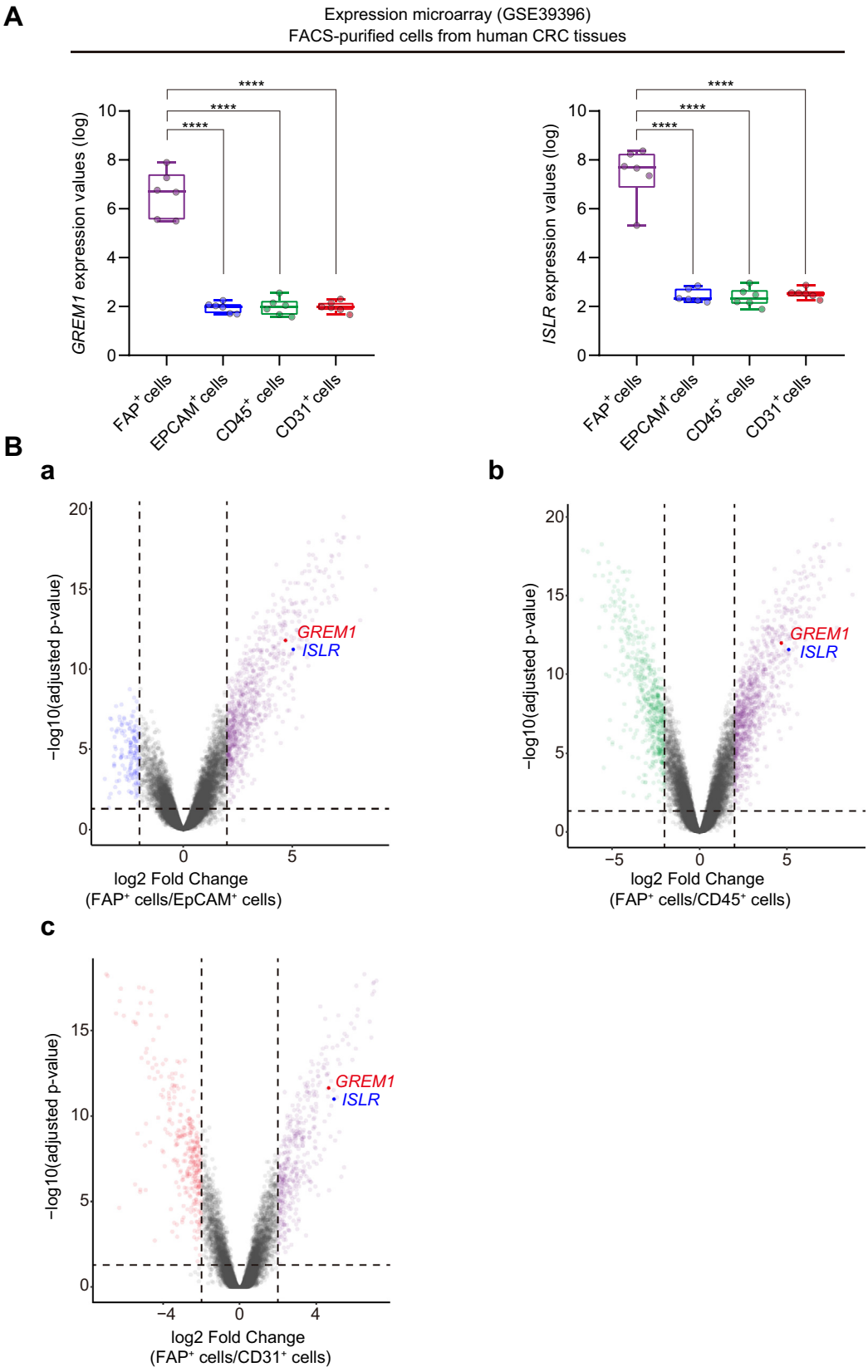
Data and Materials Availability

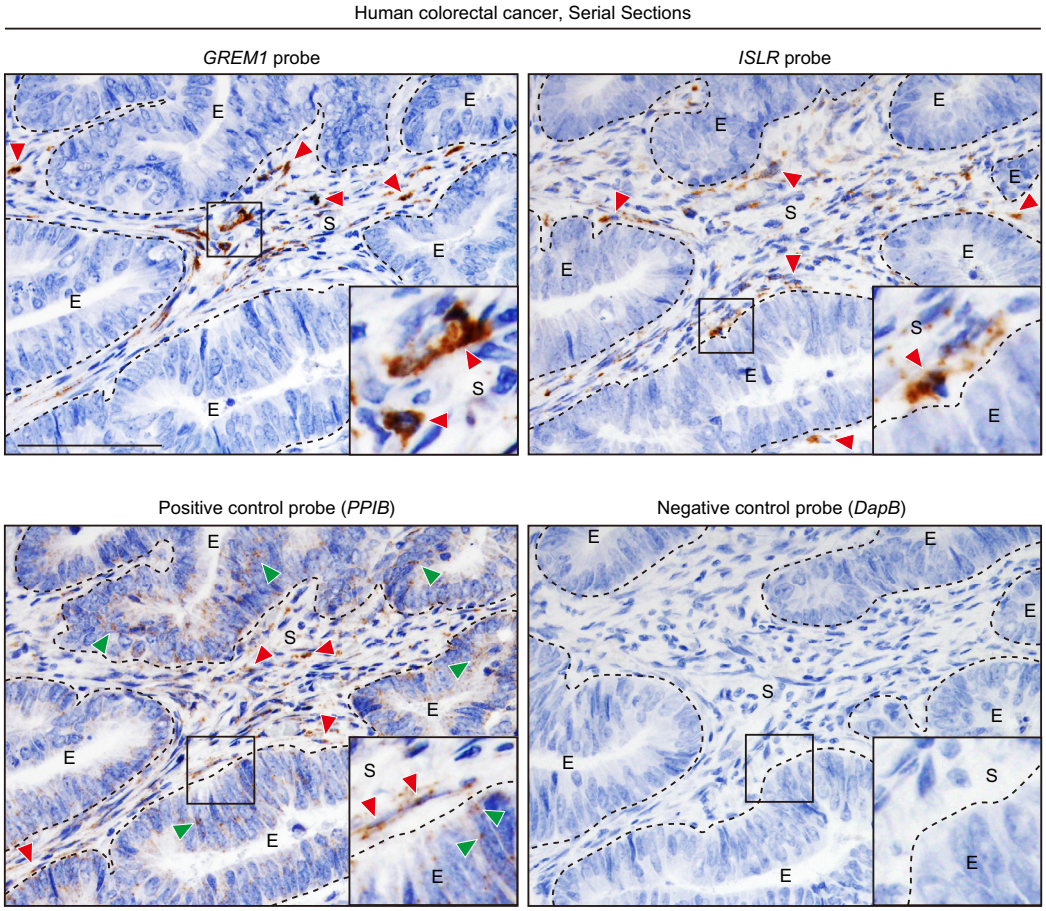
Data that support the findings of this study are available from the corresponding author upon reasonable request. Noncommercially available materials described in this study may be obtained with a material transfer agreement. Requests for materials should be addressed to Susan L. Woods.

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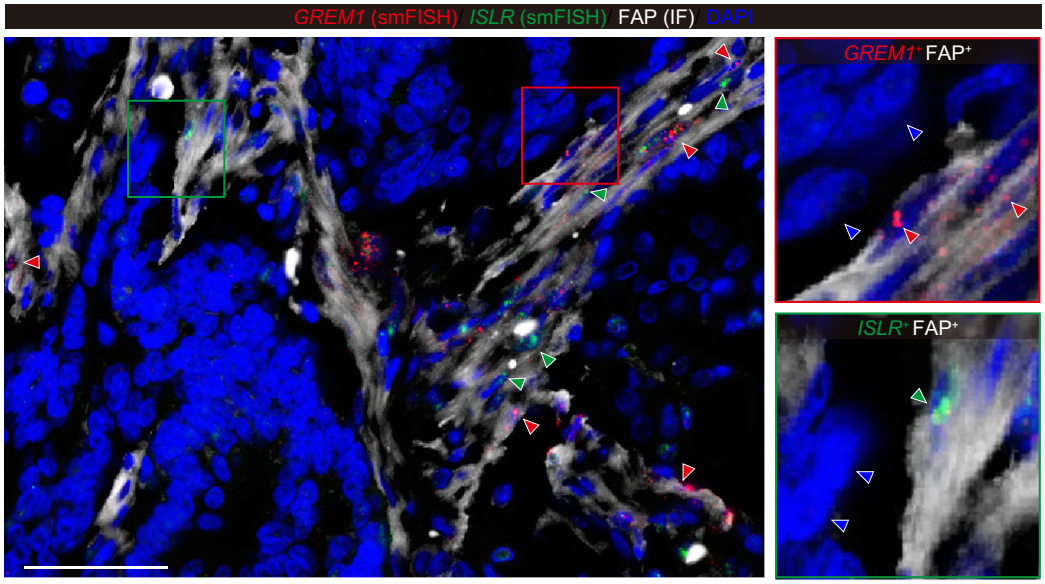
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A
Expression microarray (GSE39396)
FACS-purified cells from human CRC tissues
GREM1 expression values (log)
ISLR expression values (log)
****
FAP+ cells
EPCAM+ cells
CD45+ cells
CD31+ cells
B
a
b
c
−log10(adjusted p-value)
GREM1
ISLR
log2 Fold Change
(FAP+ cells/EpCAM+ cells)
(FAP+ cells/CD45+ cells)
(FAP+ cells/CD31+ cells)

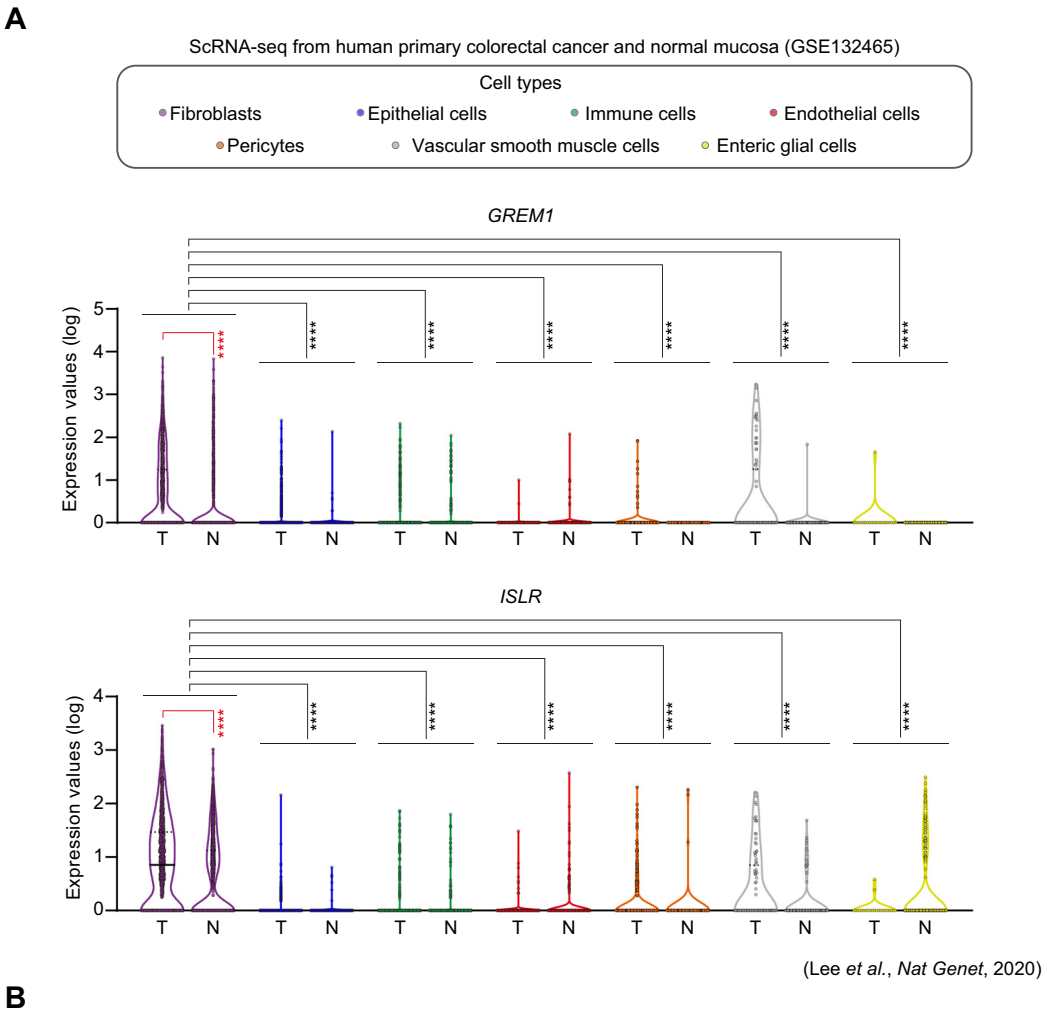


Supplementary Figure 2. Validation of ISH staining using positive and negative control probes. Representative ISH pictures using serial sections from human colorectal cancer. ISH for *GREM1* and *ISLR* shows 3,3′-DAB$^+$ staining specifically in the stroma, whereas positive control probe (*PPIB*, a ubiquitously expressed gene) staining is observed both in the epithelial cells and stroma. No apparent background staining is seen in ISH for a negative control probe that targets the bacterial *DapB* gene. Dotted lines indicate the borders between epithelial cells (E) and the stroma (S). Red and green arrowheads denote DAB$^+$ staining in the stroma and epithelial cells, respectively. The boxed areas are magnified in the insets. Scale bar, 100 μm.

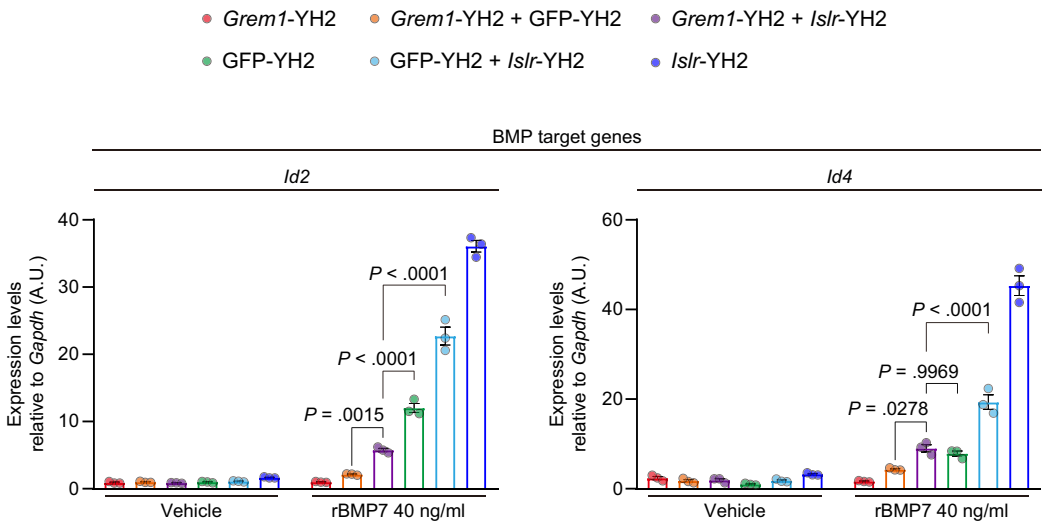
Supplementary Figure 1. Analyses of a human CRC expression microarray data set reveal that *GREM1* and *ISLR* are highly expressed in FAP$^+$ CAFs compared to EpCAM$^+$ epithelial cells, CD45$^+$ immune cells, and CD31$^+$ endothelial cells. (*A*) *GREM1* and *ISLR* expression levels in a cDNA expression microarray from FACS-purified cells from human primary CRC tissues (GSE39396); n = 6 patients each. One-way ANOVA followed by Tukey post hoc multiple comparisons. ****$P < .0001$. Box plots have whiskers of maximum and minimum values; the boxes represent first, second (median), and third quartiles. (*B*) Volcano plots showing differentially expressed transcripts between FAP$^+$ CAFs and EpCAM$^+$ epithelial cells (*a*), between FAP$^+$ CAFs and CD45$^+$ immune cells (*b*), and between FAP$^+$CAFs and CD31$^+$ endothelial cells (*c*) (GSE39396); n = 6 patients each. The adjusted *P* value cutoffs and the $\log_2$ fold change cutoffs for differentially expressed genes were 0.05 and 2, respectively. Semitransparent purple, blue, green, and red dots denote differentially expressed transcripts, which are up-regulated in FAP$^+$ CAFs, EpCAM$^+$ epithelial cells, CD45$^+$ immune cells, and CD31$^+$ endothelial cells, respectively. Gray dots represent transcripts that are not differentially expressed. *GREM1* and *ISLR* gene probes used for the analyses are as follows: 218468_PM_s_at (*GREM1*) and 207191_PM_s_at (*ISLR*).



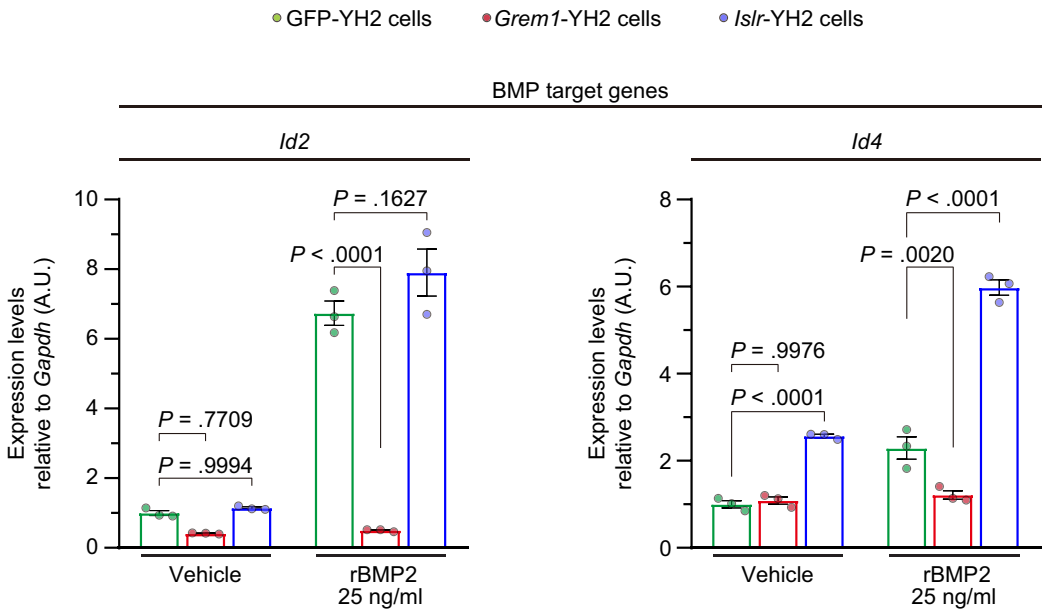
Supplementary Figure 3. *GREM1* and *ISLR* transcripts are detected in FAP^{+} CAFs in single-molecule FISH on human CRC sections. Representative picture of *GREM1* and *ISLR* dual smFISH followed by FAP IF, using human CRC tissue sections. Red and green arrowheads denote *GREM1*$^{+}$FAP^{+} cells and *ISLR*$^{+}$FAP^{+} cells, respectively. Blue arrowheads indicate FAP^{-} cells that do not express *GREM1* or *ISLR*. N = 3 patients. Scale bar, 50 μm.



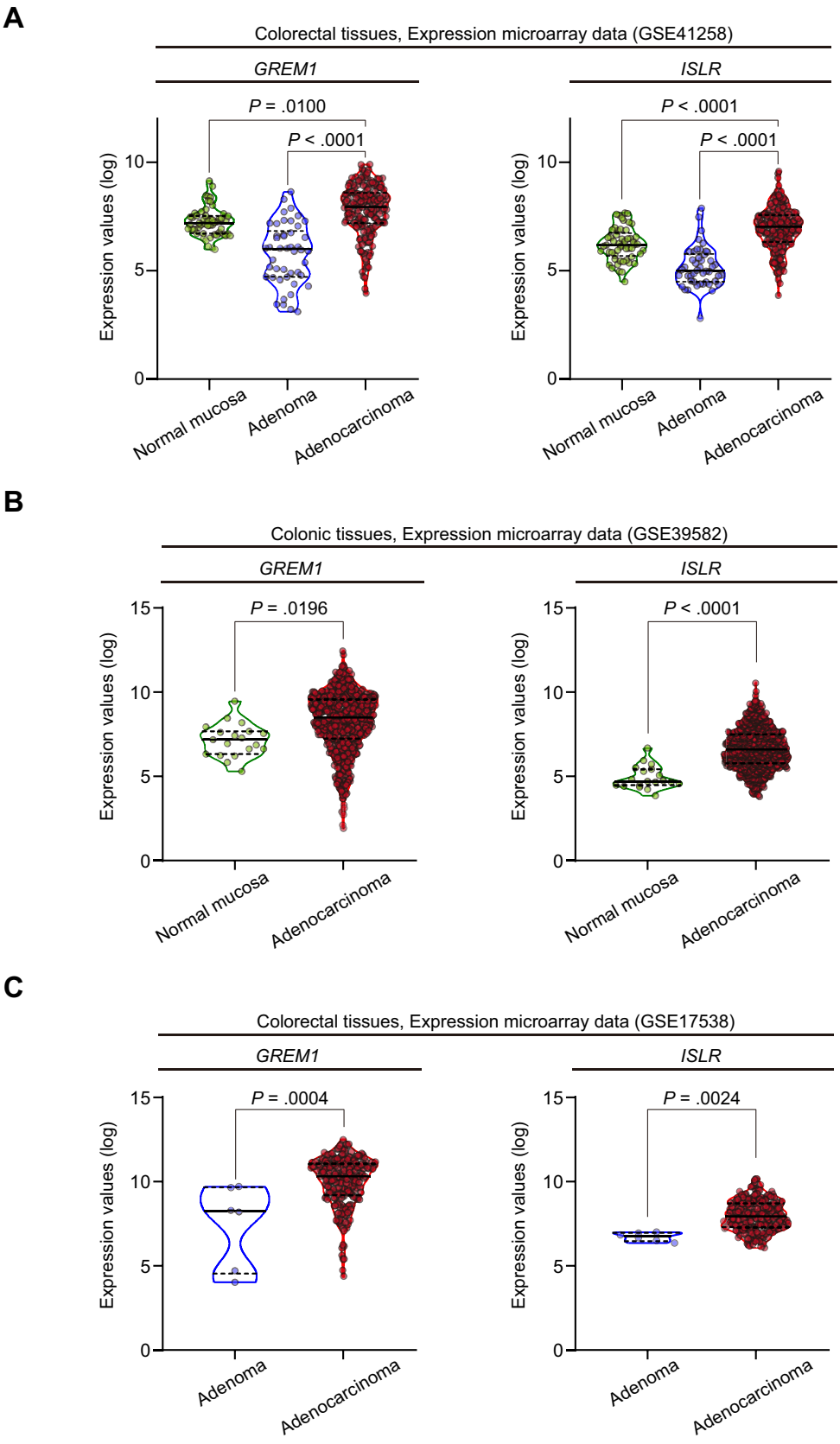
Supplementary Figure 4. *GREM1* and *ISLR* are highly expressed in fibroblasts in single-cell RNA-seq data from human normal colorectal mucosa and primary CRC. (*A*) Violin plots depicting *GREM1* and *ISLR* expression levels in scRNA-seq of unsorted cells from human primary CRC tissues and normal colorectal tissues (GSE132465).[22] In fibroblasts, *GREM1* and *ISLR* transcripts are differentially up-regulated in tumors compared with normal samples (Wilcoxon tests, red asterisks); n = 1501; 1961; 17,469; 1070; 27,080; 12,137; 768; 739; 353; 92; 91; 123; 23; and 282 cells (*left to right*). Tumor samples from 23 patients. Normal samples from 10 of the patients. (*B*) Violin plots showing *GREM1* and *ISLR* expression levels in scRNA-seq of unsorted cells from human primary CRC tissues and normal colorectal tissues (GSE81861)[23]; n = 17, 9, 272, 160, 71, 42, 4, and 2 cells (*left to right*). Tumor samples from 11 patients. Normal samples from 7 of the patients. Kruskal-Wallis tests followed by Dunn post hoc multiple comparisons were used to compare *GREM1* or *ISLR* expression levels in fibroblasts from tumor and normal samples with those in other cell populations (*black asterisks*). ****$P < .0001$, *$P = .0132$. Solid black lines, median; dotted black lines, quartiles. N, normal samples; T, tumor samples.



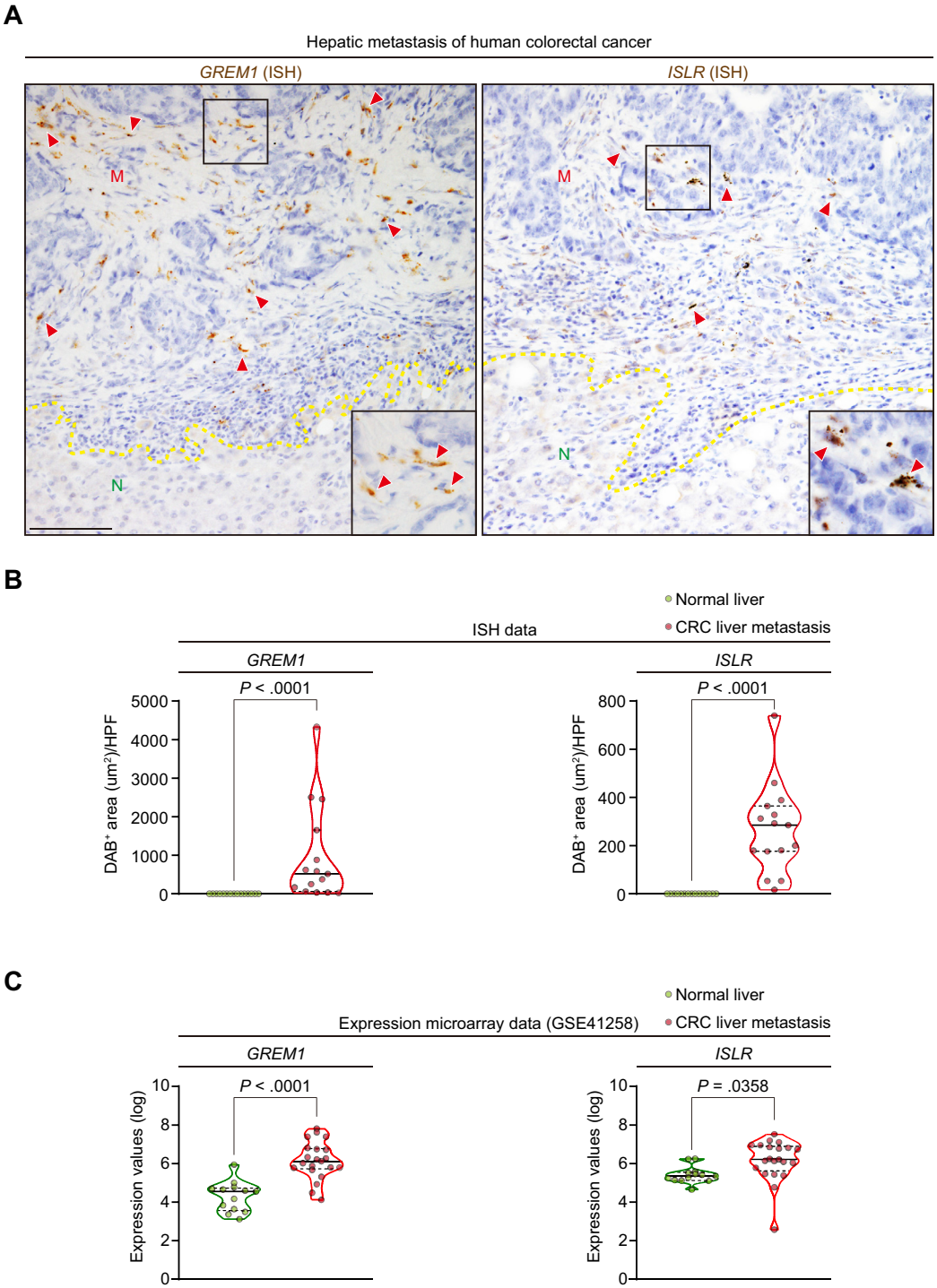
Supplementary Figure 5. GREM1 and ISLR overexpression counteract each other's effects on BMP7-*Id2/4* signaling in vitro. qRT-PCR for BMP target genes *Id2* and *Id4*. YH2 cells that overexpress GREM1, GFP, or ISLR were admixed at a 1:1 ratio and stimulated with 40 ng/mL of recombinant BMP7 for 24 hours (*orange, purple, and aqua*). *Grem1*-YH2 cells, GFP-YH2 cells, or *Islr*-YH2 cells alone were used as controls (*red, green, and blue, respectively*). The same number of cells (a total of 5.0×10^4 cells/well) were seeded in a 6-well plate in each condition. n = 3 each; mean ± SEM. Two-way ANOVA with post hoc Tukey multiple comparisons. A.U., arbitrary unit.



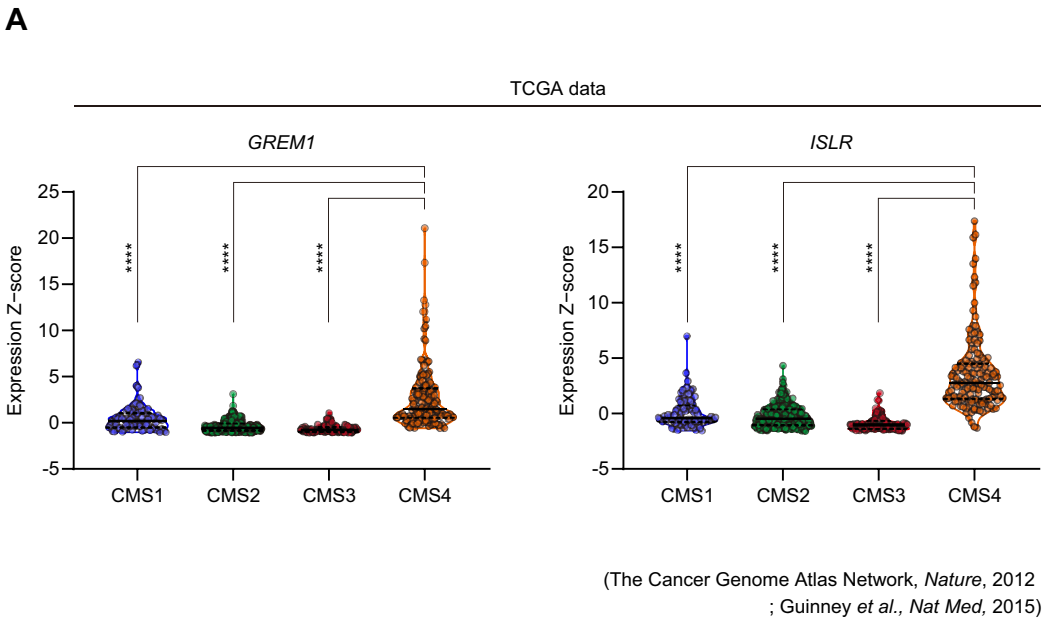
Supplementary Figure 6. BMP2-induced increase in *Id4* is prevented by GREM1 and augmented by ISLR in YH2 cells. qRT-PCR for *Id2* and *Id4* in YH2 cells. GFP-, *Grem1*-, or *Islr*-overexpressing YH2 cells were treated with 25 ng/mL of recombinant BMP2 for 24 hours. Note that GREM1 prevented the BMP2-induced increase in *Id2*, whereas ISLR did not significantly promote the BMP2-mediated increase in *Id2* in YH2 cells. n = 3 each. Mean ± SEM. Two-way ANOVA with post hoc Tukey multiple comparisons.



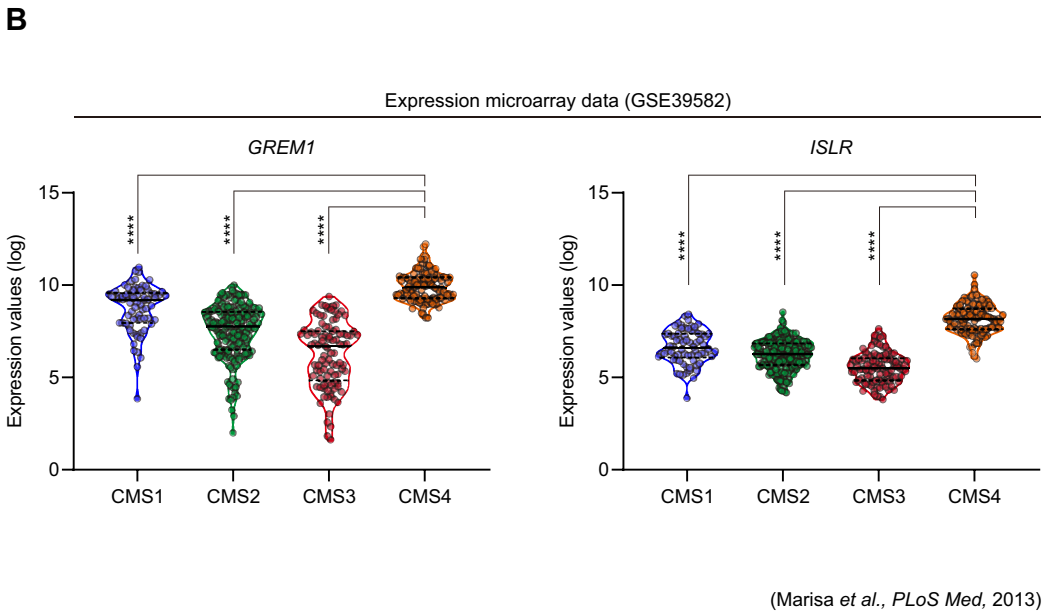
Supplementary Figure 7. *GREM1* and *ISLR* expression levels are up-regulated during human CRC carcinogenesis. (*A*) Violin plots depicting *GREM1* and *ISLR* expression in cDNA expression microarray data from human normal colorectal mucosa, colorectal adenoma, and primary colorectal adenocarcinoma (GSE41258).[16] n = 53 (normal mucosa), 46 (adenoma), and 182 patients (adenocarcinoma). (*B*) Violin plots depicting *GREM1* and *ISLR* expression in cDNA expression microarray data from human normal colon mucosa and primary colon adenocarcinoma (GSE39582).[17] n = 19 (normal colon mucosa) and 566 patients (primary colon adenocarcinoma). (*C*) Violin plots depicting *GREM1* and *ISLR* expression in cDNA expression microarray data from human colorectal adenoma and primary colorectal adenocarcinoma (GSE17538).[18] n = 6 (colorectal adenoma) and 232 patients (primary colorectal adenocarcinoma). One-way ANOVA followed by Tukey post hoc multiple comparisons (*A*). FDR-adjusted *P*- values are shown (*B* and *C*). Solid black lines, median; dotted black lines, quartiles.



Supplementary Figure 8. *GREM1* and *ISLR* are expressed in the stroma of human CRC hepatic metastases. (*A*, *B*) ISH for *GREM1* and *ISLR* in the hepatic metastases of human CRC. (*A*) Representative images. *GREM1* and *ISLR* are expressed by fibroblastic cells in the stroma of the liver metastases. Dotted yellow lines indicate the borders between the normal liver (N) and CRC liver metastases (M). Red arrowheads denote *GREM1* or *ISLR* expression. The boxed areas are magnified in the insets. Scale bars, 100 μm. (*B*) Violin plots showing quantification of 3,3′-DAB$^+$ areas by ImageJ; 5 HPFs/patient, 3 patients each. Solid black lines, median; dotted black lines, quartiles. (*C*) Violin plots depicting *GREM1* and *ISLR* expression in cDNA expression microarray data from human normal liver and liver metastasis of CRC (GSE41258). n = 13 (normal liver) and 21 patients (CRC liver metastasis). Solid black lines, median; dotted black lines, quartiles. Mann-Whitney *U* test (B). FDR-adjusted *P* values are shown in (*C*). HPF, high-power field.

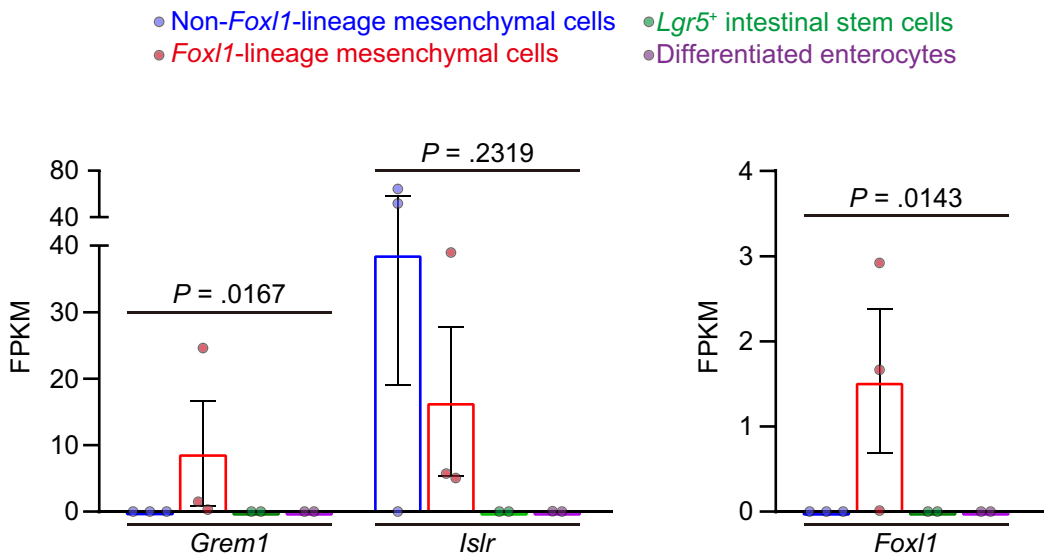


(The Cancer Genome Atlas Network, *Nature*, 2012 ; Guinney *et al., Nat Med,* 2015)

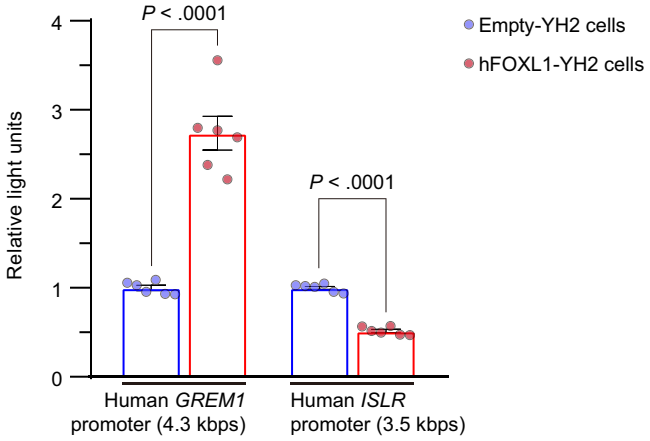


(Marisa *et al., PLoS Med,* 2013)

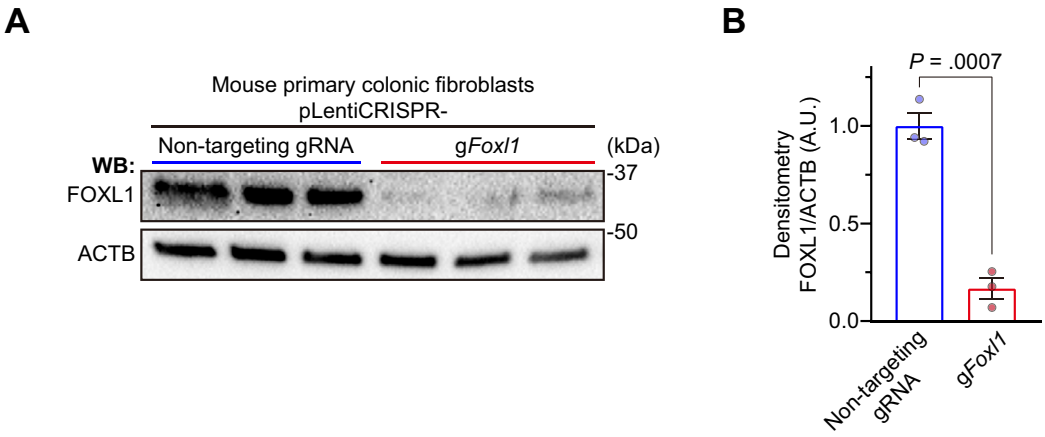
Supplementary Figure 9. *GREM1* and *ISLR* are highly expressed in CMS 4 CRC. (*A*) Violin plots showing expression levels of *GREM1* and *ISLR* in 4 CMSs.[20] Primary colon and rectal adenocarcinoma samples in TCGA data were analyzed; n = 76 (CMS1), 220 (CMS2), 72 (CMS3), and 143 patients (CMS4). (*B*) Violin plots depicting *GREM1* and *ISLR* expression levels in 4 CMSs. Primary colon adenocarcinoma samples were analyzed by using cDNA expression microarray data (GSE39582).[17] n = 70 (CMS1), 177 (CMS2), 97 (CMS3), and 107 patients (CMS4). Kruskal-Wallis test followed by Dunn post hoc multiple comparisons. Solid black lines, median; dotted black lines, quartiles. ****$P < .0001$. TCGA, The Cancer Genome Atlas.



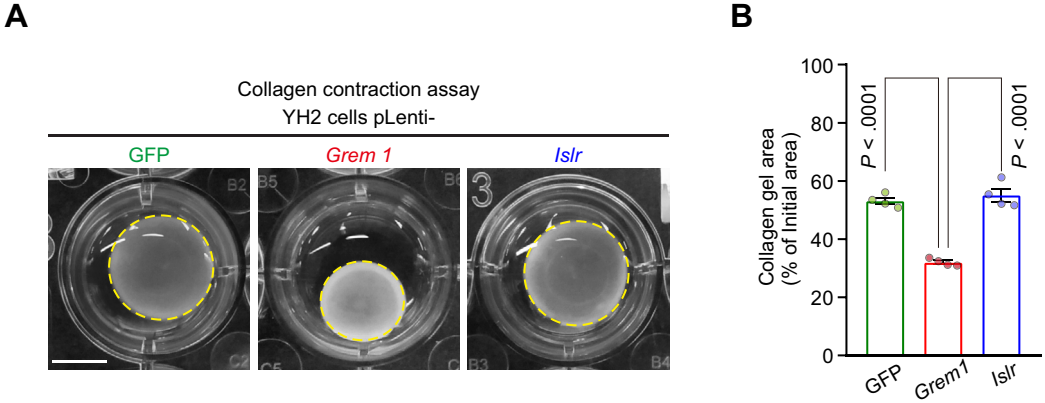
Supplementary Figure 10. *Grem1* is highly expressed in *Foxl1*-lineage telocytes in the mouse small intestine. Expression levels of *Grem1*, *Islr*, and *Foxl1* in RNA-seq data from FACS-purified non–*Foxl1*-lineage mesenchymal cells, *Foxl1*-lineage mesenchymal cells (sorted from *Foxl1*-Cre;Rosa26-YFP or *Foxl1*-Cre;Rosa26-mT/mG), *Lgr5*$^{+}$ intestinal stem cells (sorted by *Lgr5*-eGFP^{high} expression), and differentiated enterocytes (GSE94072). Note that FPKM values for *Grem1* and *Foxl1* were 0 in all 3 biological replicates of non-*Foxl1* lineage mesenchymal cells. n = 3 (non–*Foxl1*-lineage mesenchymal cells), 3 (*Foxl1*-lineage mesenchymal cells), 2 (*Lgr5*$^{+}$ intestinal stem cells), and 2 (differentiated enterocytes) mice. All data are represented as mean ± SEM. Statistical analyses were performed using the Kruskal-Wallis test.



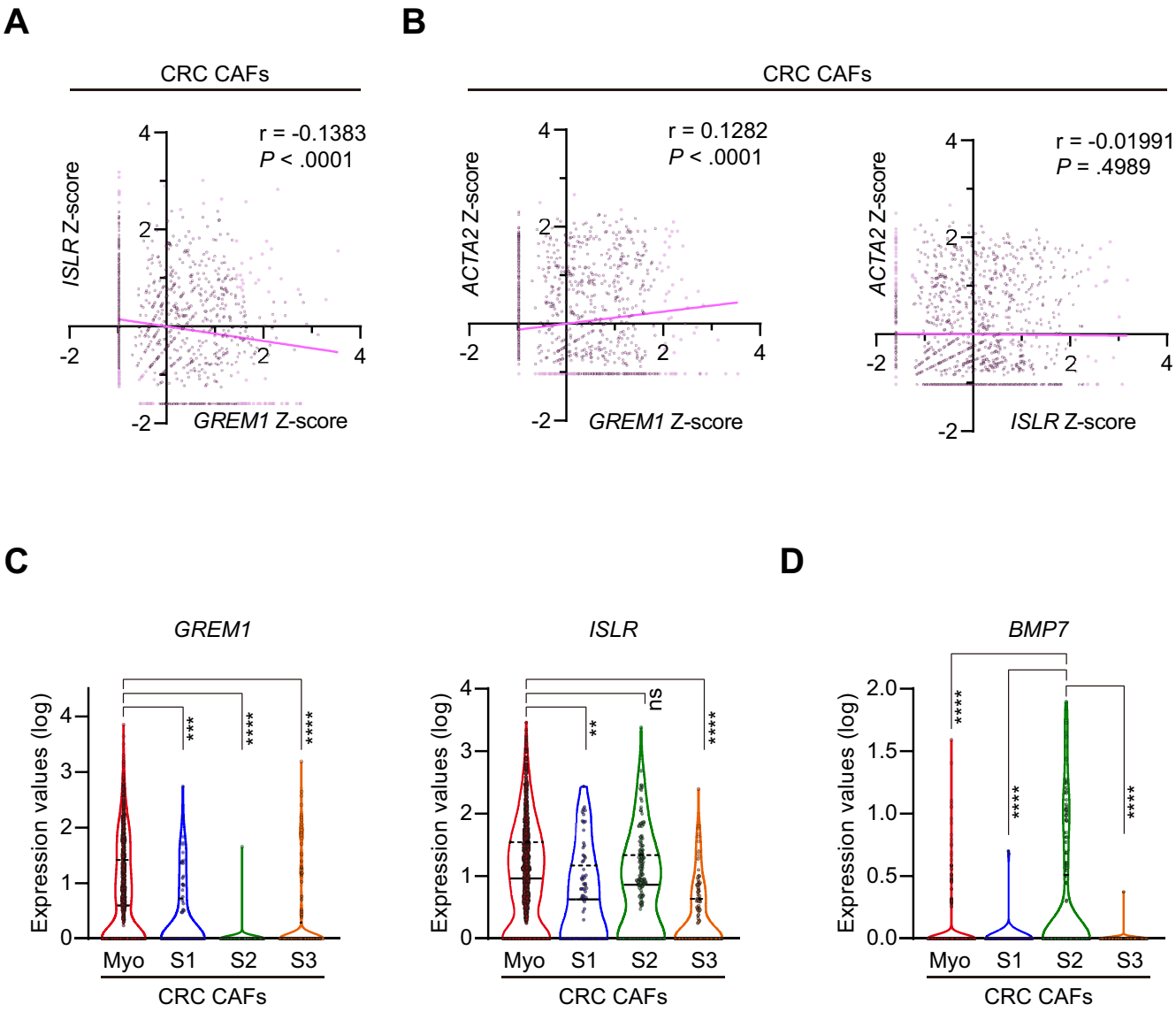
Supplementary Figure 11. Lentivirus-mediated FOXL1 overexpression increases *GREM1*-promoter activity and decreases *ISLR*-promoter activity. Luciferase assays of human *GREM1*- or *ISLR*-promoter regions using human FOXL1-overexpressing YH2 cells and control empty YH2 cells. n = 6 each. Data are represented as mean ± SEM. Statistical analysis was performed by using the 2-tailed unpaired Student *t* test.



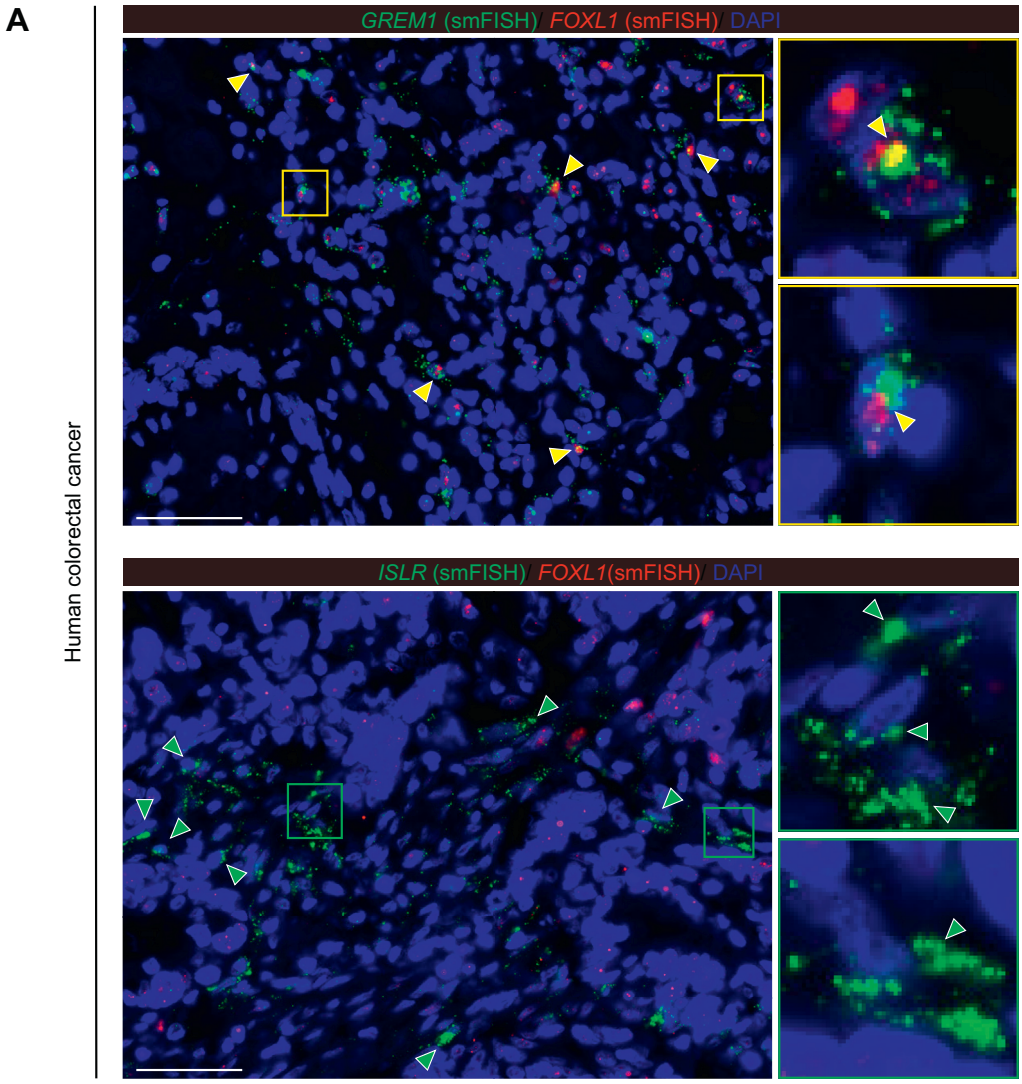
Supplementary Figure 12. Validation of CRISPR/Cas9–mediated FOXL1 knockdown in mouse primary colonic fibroblasts by WB. (*A*) WB showing FOXL1 knockdown by lentiviral transduction. (*B*) Densitometry of the WB in *A* showing decreased FOXL1 protein expression by lentiviral transduction. n = 3 mice each. Data are represented as mean ± SEM. Statistical analysis was performed by using the 2-tailed unpaired Student *t* test.



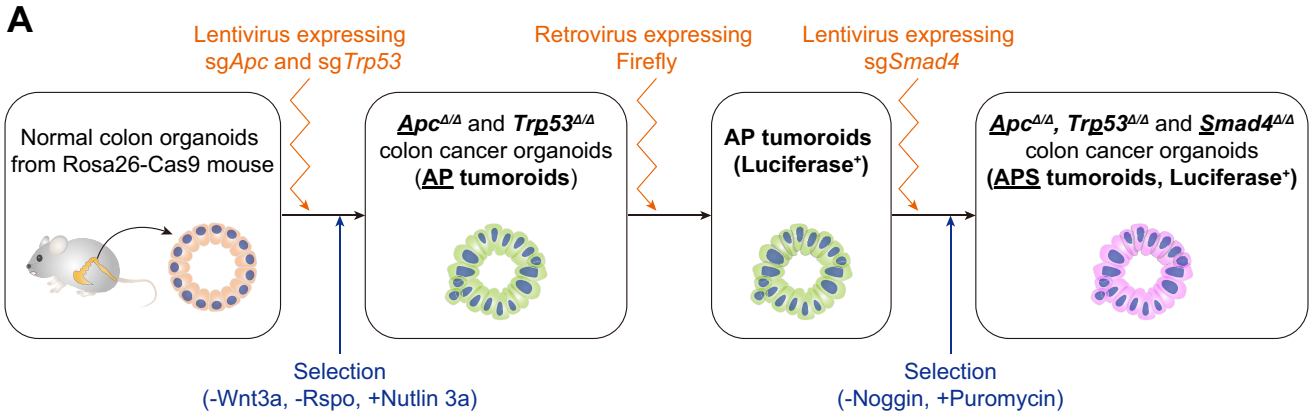
Supplementary Figure 13. Increased collagen gel contraction in *Grem1*-overexpressing YH2 cells. (*A*, *B*) Collagen gel contraction assay using GFP-, *Grem1*- and *Islr*-overexpressing YH2 cells. (*A*) Representative pictures. Collagen gel areas are outlined by dotted yellow lines. Scale bars, 5 mm. (*B*) Quantification of collagen gel areas. Collagen gel areas were evaluated 12 hours after seeding cells; n = 4 each. Data are represented as mean ± SEM. Statistical analysis was performed by using 1-way ANOVA with Tukey post hoc multiple comparisons.



Supplementary Figure 14. ScRNA-seq from human CRC tissues shows that *GREM1* identifies a myofibroblastic CAF subpopulation that is distinct from *ISLR*$^{+}$ CAFs. (*A*, *B*) Spearman correlation analyses of gene expression levels in scRNA-seq data from human CRC tissues (GSE132465).[22] (*A*) *GREM1* transcript levels are inversely correlated with *ISLR* expression in human CRC CAFs. (*B*) *GREM1*, but not *ISLR*, transcripts are positively correlated with *ACTA2* transcripts. In a total of 1501 CAFs, 1156 human CRC CAFs that express *GREM1* or *ISLR* transcripts were analyzed. Spearman *r* and *P* values are reported in the figure. Solid pink line, linear regression. (*C*, *D*) Violin plots depicting *GREM1* and *ISLR* expression (*C*) and BMP7 expression (*D*) in 4 different CRC CAF subclusters[22]; n = 1146 (myofibroblasts), 73 (S1 fibroblasts), 158 (S2 fibroblasts), and 124 cells (S3 fibroblasts). Kruskal-Wallis test followed by Dunn post hoc multiple comparisons. ****$P < .0001$; ***$P = .0005$; **$P = .0098$; ns, $P = .1375$. Solid black lines, median; dotted black lines, quartiles. Myo, myofibroblasts; S1, stromal 1 fibroblasts; S2, stromal 2 fibroblasts; S3, stromal 3 fibroblasts.



Supplementary Figure 15. *GREM1*$^{+}$ CAFs shows higher positivity for *FOXL1* in human CRC than *ISLR*$^{+}$ CAFs. (*A*, *B*) Dual smFISH for *GREM1*/*FOXL1* (*upper panel in A*) and *ISLR*/*FOXL1* (*lower panel in A*) in human CRC samples. (*A*) Representative pictures. Yellow and green arrowheads denote *GREM1*$^{+}$*FOXL1*$^{+}$ cells and *ISLR*$^{+}$*FOXL1*$^{-}$ cells, respectively. The boxed areas are magnified in the adjacent panels. (*B*) Semiquantification of *FOXL1* positivity in *GREM1*$^{+}$ cells and *ISLR*$^{+}$ cells. Three HPFs/patient, 3 patients. Mean ± SEM. Mann-Whitney *U* test. Scale bars, 50 μm.



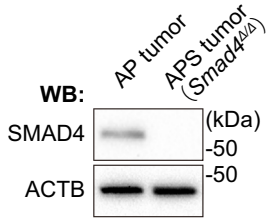
B

AP tumoroids

Apc		
Reference sequence	sgApc / PAM: GACCAGGAAGCCTTGTGGGACA TGGGGGCA	
		Change to Protein
Apc allele 1	1bp insertion: GACCAGGAAGCCTTGTGGGACAATGGGGGCA	p.Met717AsnfsTer37
Apc allele 2	5bp deletion: GACCAGGAAGCCTTGT-----A TGGGGGCA	p.Trp715TyrfsTer37

Trp53		
Reference sequence	sgTrp53 / PAM: TACATGTGTAATAGCTCCTGCA TGGGGGGC	
		Change to Protein
Trp53 allele 1	2bp deletion: TACATGTGTAATAGCTCCTGCA --GGGGGC	p.Met240ArgfsTer27
Trp53 allele 2	1bp insertion: TACATGTGTAATAGCTCCTGCAATGGGGGGC	p.Met240AsnfsTer28

C



D

APS tumoroids

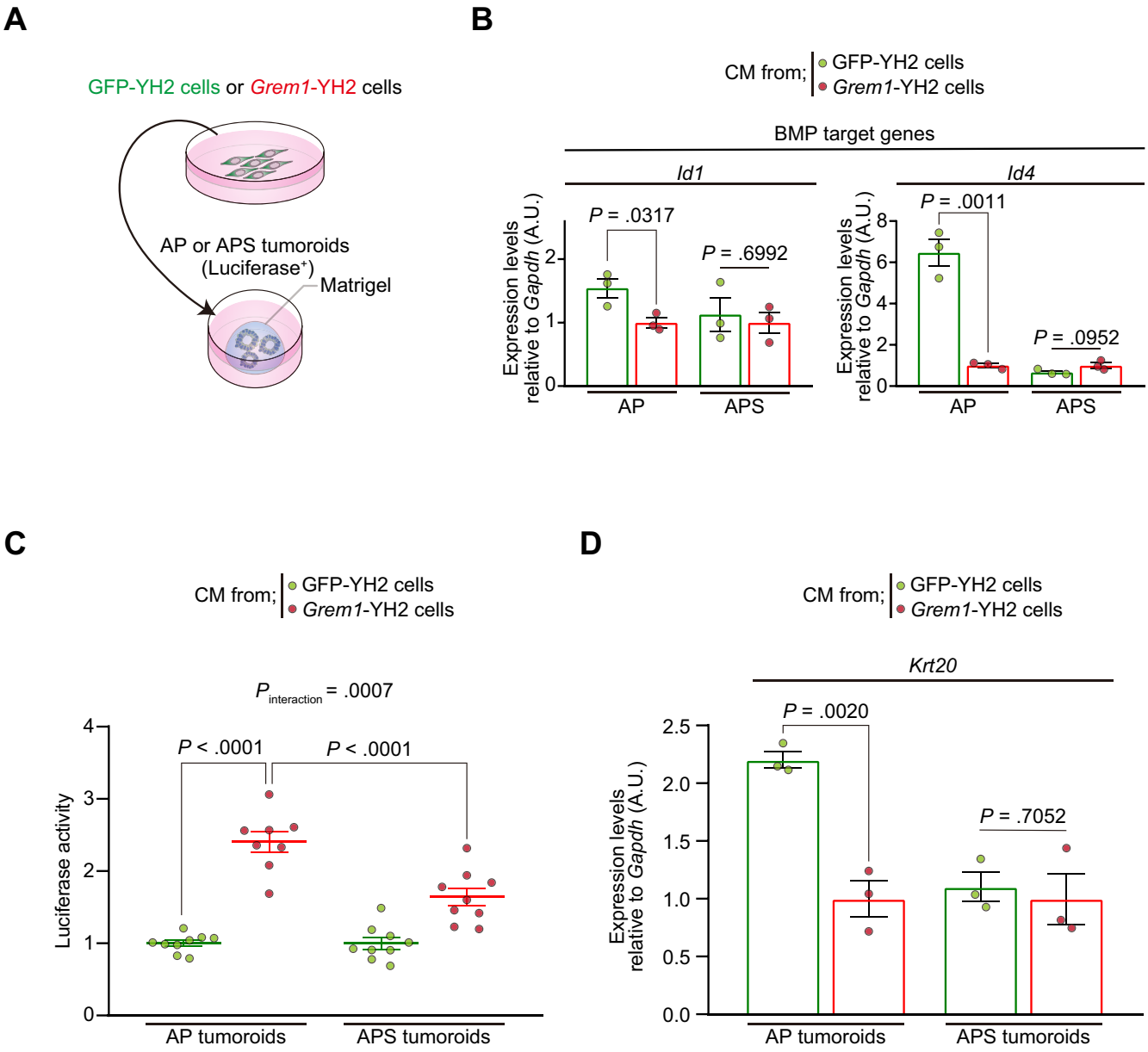
Reference sequence: Mutation hotspot regions, PAM, sgSmad4
GTGGATGGCTATGTGGATCCTTCGGGAGGAGATCGCTTTTGC
L1 loop

Smad4 allele 1: 27 bps deletion
GTGGATGGCT----------TTTGC
Change to Protein: p.Tyr352_Arg360del

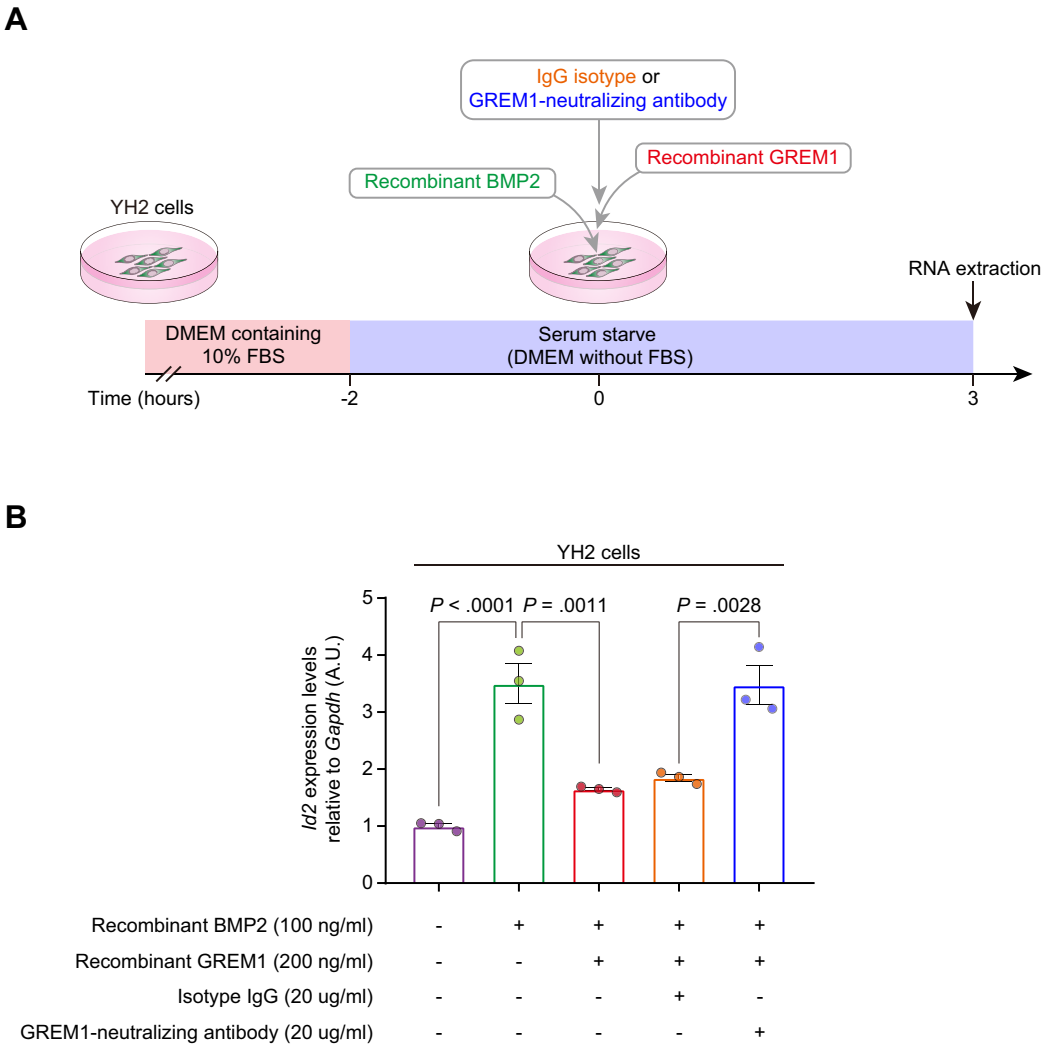
Reference sequence: Smad4 — Exon 9 — Exon 10

Smad4 allele 2: 1909 bps deletion
ACTCTAAGTGTCAGCAGGAAGTAGTTTTGTGGTTACTTTC

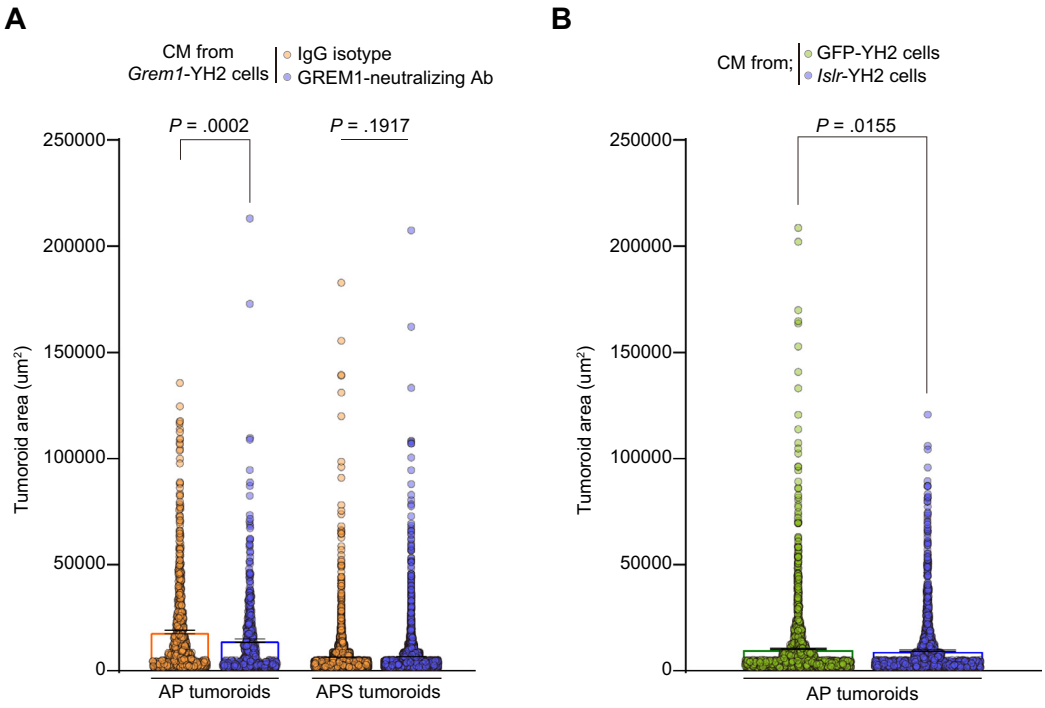
Supplementary Figure 16. Generation of *Apc* $^{\Delta/\Delta}$,*Trp53* $^{\Delta/\Delta}$ organoids (AP tumoroids) and *Smad4*-mutant AP tumoroids (APS tumoroids). (*A*) Schematic illustration of the CRISPR/Cas9–mediated sequential engineering of AP tumoroids and APS tumoroids from normal colon organoids isolated from a Rosa26-Cas9 mouse. Correctly targeted clones were enriched by selection medium as indicated. Monoclonal lines were handpicked after each transduction step. (*B*) *Apc* and *Trp53* DNA sequence verification of biallelic insertion/deletion mutations, which result in prematurely truncated proteins. (*C*) Decreased SMAD4 protein expression in APS tumoroids was validated by WB. (*D*) *Smad4* DNA sequence verification of biallelic mutations, resulting in disruption of the sequence of L1 loop protein, which is involved in the formation of SMAD protein complex formation.[32] sg*Smad4* was designed to target the genomic sequence corresponding to *SMAD4* mutation hotspot regions in human CRC,[32] which were mapped to the L1 loop and its adjacent region.[32] To simplify graphical presentation, the following mutations were not shown in the schematics in *D*: *Smad4* allele 1, c.1005A>G (silent mutation in exon 9) ; *Smad4* allele 2, c.953-831A>T (mutation in an intron; *orange bar*); and c.953-805_953-804insAAAAAATAAATAAAAAAAATAAAT (mutation in an intron; *Magenta bar*). Changes to proteins are reported according to the Human Genome Variation Society nomenclature. Reference sequences are as follows: *Apc*, NM_001360980.1; *Trp53*, NM_011640.3; *Smad4*, NM_008540.3. fs, frameshift mutation; PAM, protospacer adjacent motif; sg, single guide RNA; Ter, translation termination (stop) codon.



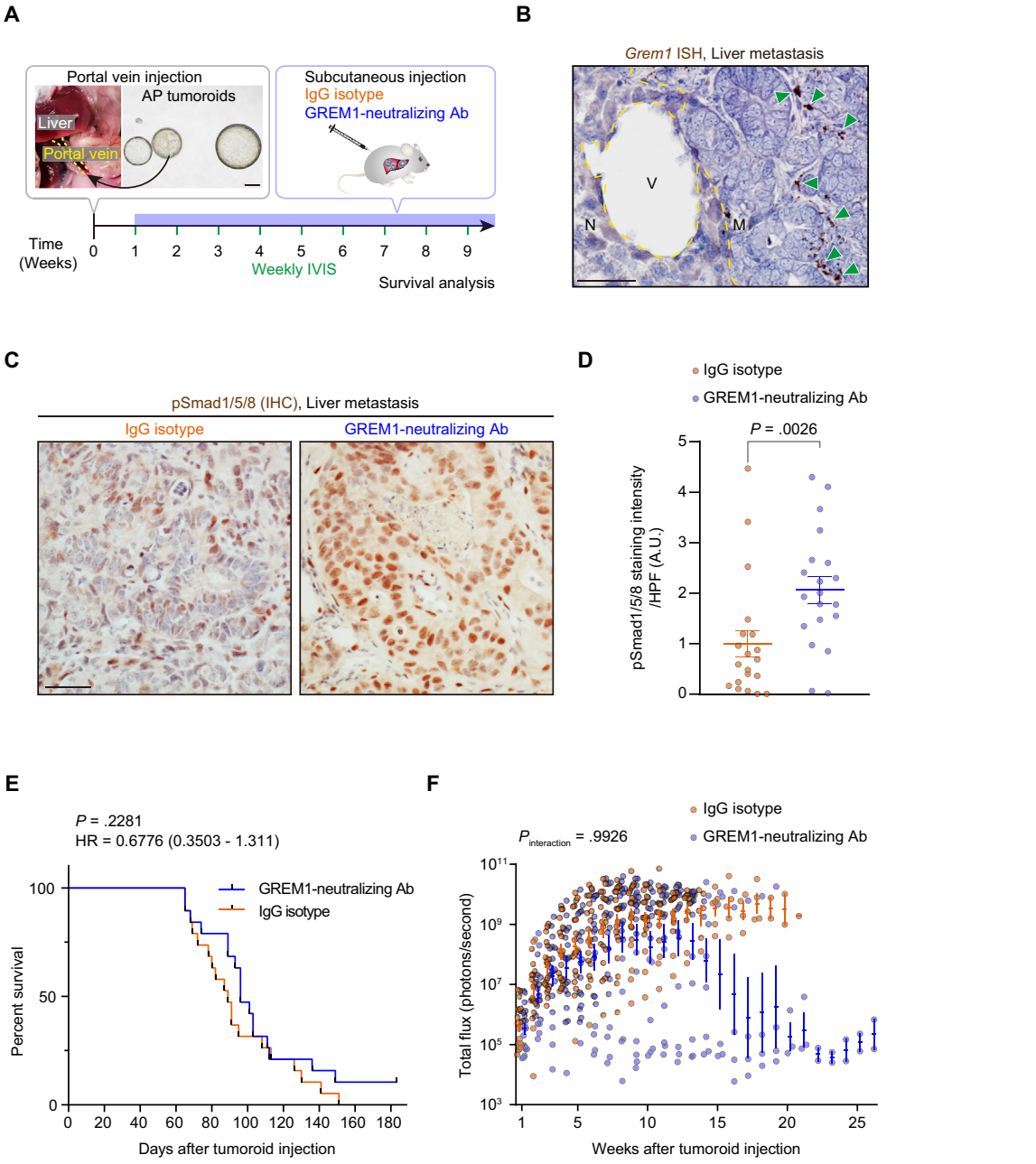
Supplementary Figure 17. CM transfer from *Grem1*-overexpressing YH2 cells decreases BMP signaling and accelerates organoid growth in AP tumoroids, but not in APS tumoroids. (*A*) Experimental schematic depicting CM transfer from *Grem1*-overexpressing YH2 cells or GFP-overexpressing YH2 cells to AP ($Apc^{\Delta/\Delta}$,$Trp53^{\Delta/\Delta}$) tumoroids or APS ($Apc^{\Delta/\Delta}$,$Trp53^{\Delta/\Delta}$,$Smad4^{\Delta/\Delta}$) tumoroids. (*B*) qRT-PCR for BMP target genes in AP and APS tumoroids (n = 3). (*C*) Luciferase signals from AP tumoroids and APS tumoroids (n ≥ 8). (*D*) qRT-PCR for *Krt20* in AP tumoroids and APS tumoroids (n = 3). Note that data normalization was performed within the AP and APS tumoroid groups separately (*B–D*). Statistical analyses were performed by using the 2-tailed unpaired Student *t* test (*B* and *D*) and 2-way ANOVA with Tukey post hoc multiple comparisons (C). Data are represented as mean ± SEM.



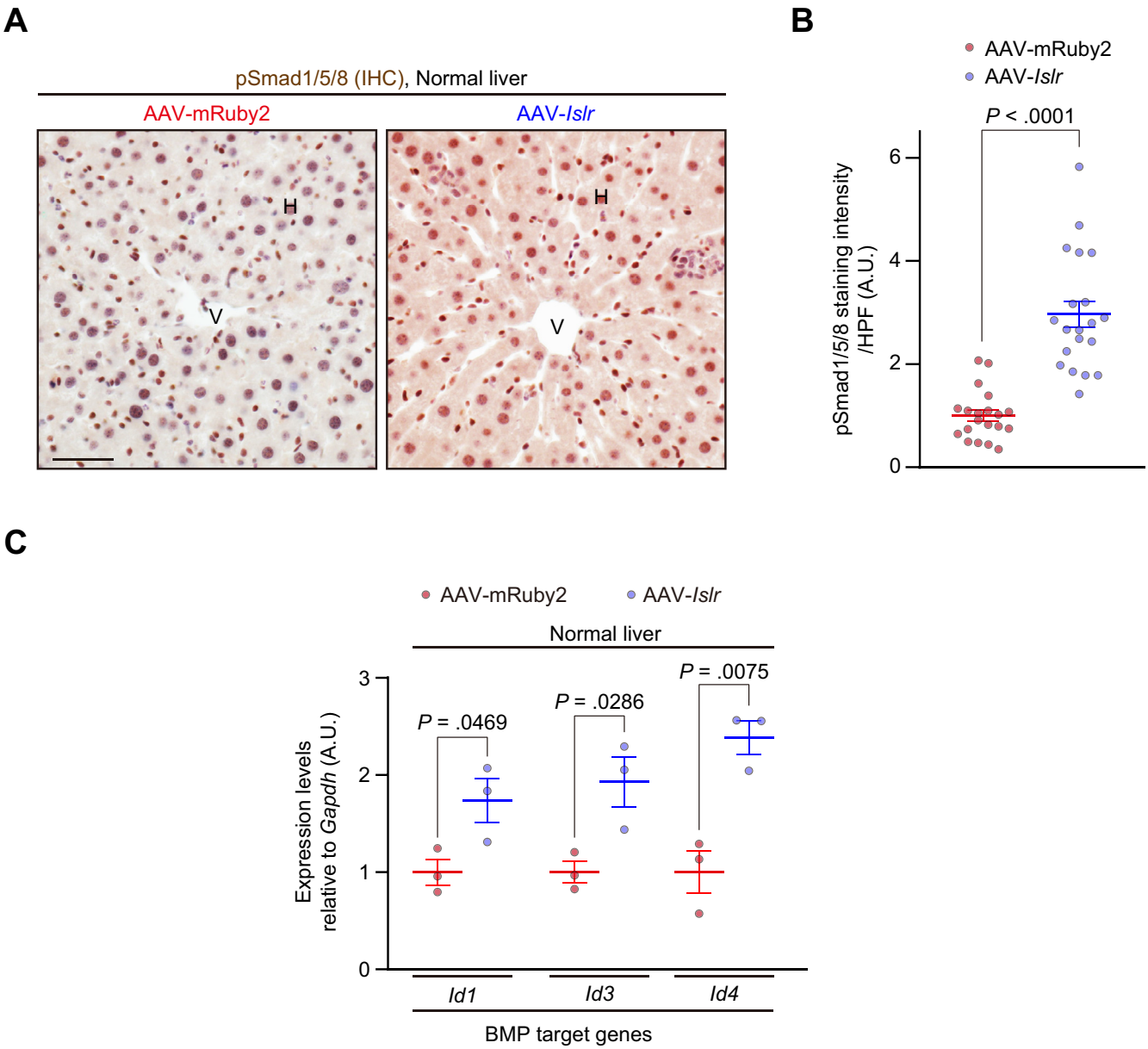
Supplementary Figure 18. Treatment with a GREM1-neutralizing antibody blocks GREM1-mediated inhibition of BMP2 signaling in vitro. (*A*, *B*) Validation of the GREM1-neutralizing antibody using YH2 cells. (*A*) Experimental schematic. YH2 cells were serum-starved for 2 hours before treating with the indicated proteins for 3 hours. (*B*) qRT-PCR of *Id2* in YH2 cells was performed to confirm the effect of the GREM1-neutralizing antibody in neutralizing recombinant GREM1 and restoring BMP signaling. n = 3 each. Data are represented as mean ± SEM. Statistical analyses were performed by using 1-way ANOVA with Tukey post hoc multiple comparisons (*B*).



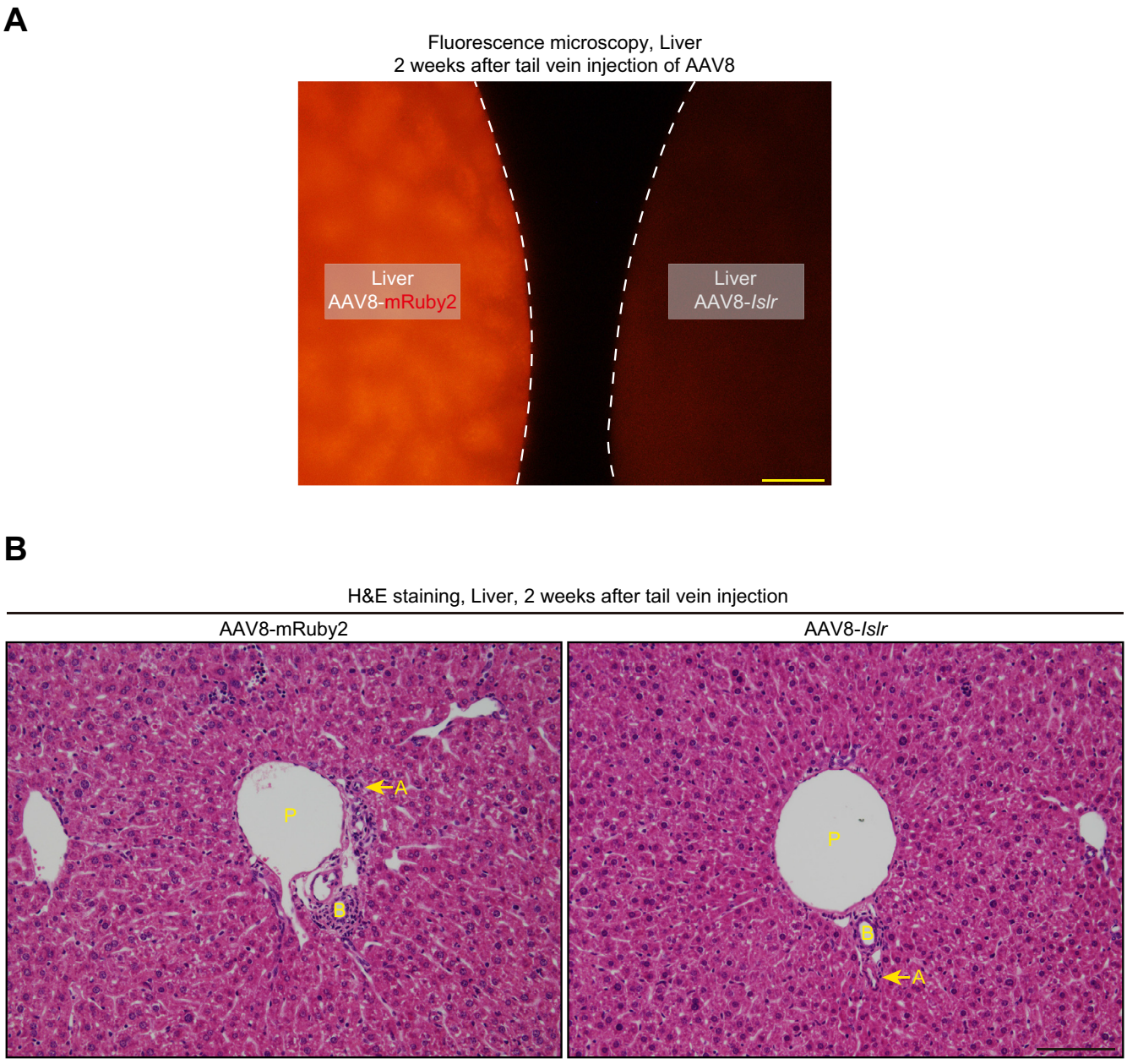
Supplementary Figure 19. Treatment with a GREM1-neutralizing antibody or CM from *Islr*-overexpressing intestinal fibroblasts attenuates tumoroid growth as assessed by tumoroid area. (*A*) Treatment with a GREM1-neutralizing antibody reduces AP tumoroid areas. n = 790, 522, 3571, and 3494 tumoroids (*left to right*) from 6 (AP tumoroids) and 9 independent replicates (APS tumoroids). Note that AP tumoroid and APS tumoroids were evaluated 8 and 12 days after being plated. In this experimental condition, APS tumoroids generated smaller tumoroid areas than AP tumoroids when treated with an IgG isotype ($P < .0001$, Mann-Whitney *U* test). (*B*) CM transfer from *Islr*-overexpressing YH2 cells decreases AP tumoroid areas. n = 1873 (GFP-YH2 cells) and 1713 tumoroids (*Islr*-YH2 cells) from 9 independent replicates. The AP tumoroids were evaluated 7 days after being plated. Tumoroid areas were assessed by Image J. Mean ± SEM, Mann-Whitney *U* test (*A* and *B*). Ab, antibody.



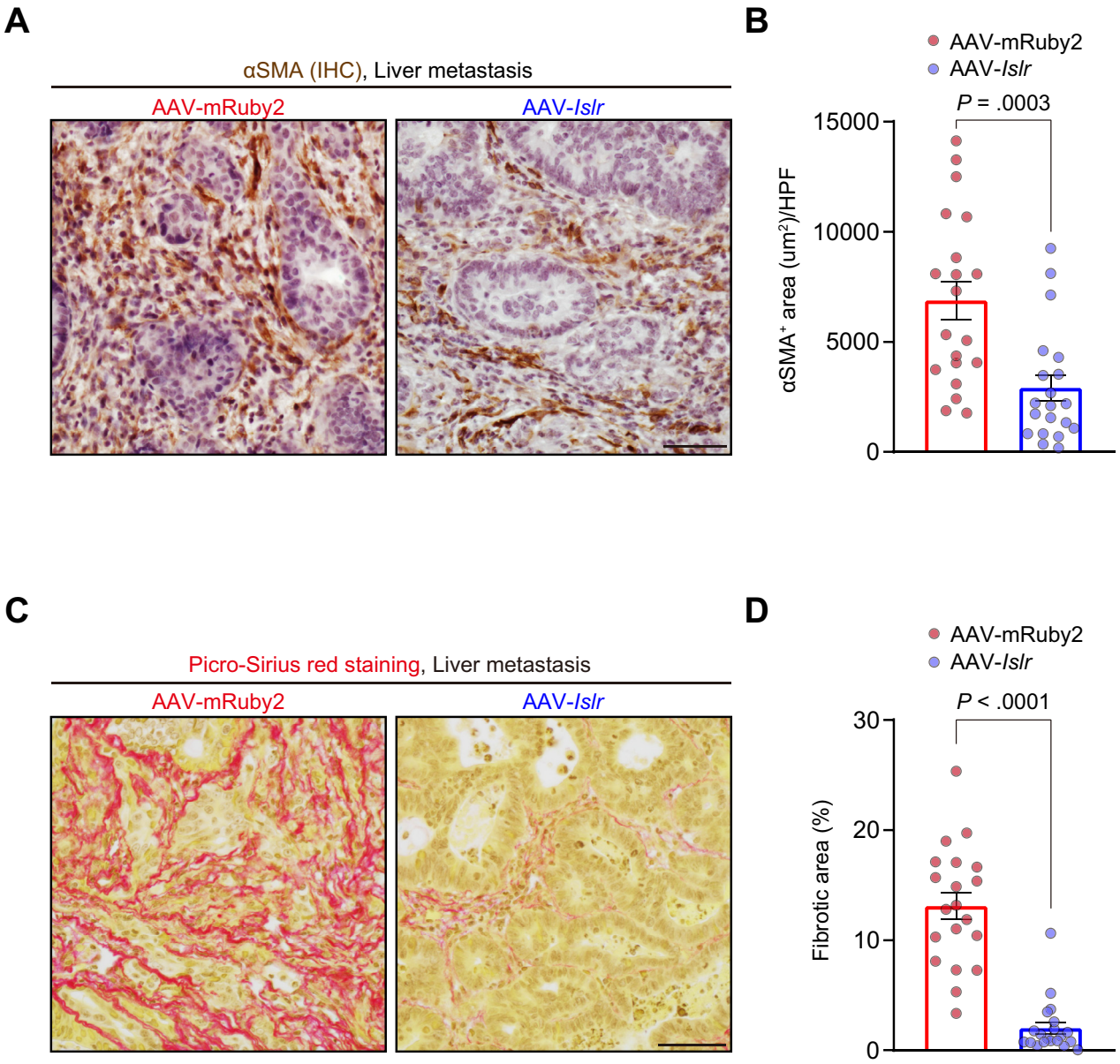
Supplementary Figure 20. A GREM1-neutralizing antibody–treated group showed enhanced BMP signaling and a trend toward prolonged survival in a mouse model of CRC liver metastasis. (*A*) Experimental scheme for portal vein injection of AP tumoroids and subsequent GREM1-neutralizing antibody administration. The GREM1-neutralizing antibody was subcutaneously administered 1 week after portal vein injection of AP tumoroids until mice reached humane endpoints. Yellow dotted lines outline the portal vein. Scale bar, 200 μm. (*B*) ISH for *Grem1* in mouse CRC liver metastasis. *Grem1* was expressed by the fibroblastic cells in the liver metastasis mesenchyme (M) but not in the normal liver (N). Yellow dotted lines demarcate the normal liver (N) and CRC liver metastasis (M). Green arrowheads indicate *Grem1* expression, and V indicates a blood vessel. n = 3 mice. Scale bar, 50 μm. (*C*, *D*) IHC for pSmad1/5/8 in the liver metastases developed in IgG isotype-treated or GREM1-neutralizing antibody–treated mice. (*C*) Representative pictures. Scale bar, 50 μm. (*D*) Quantification of 3,3′-DAB intensity by ImageJ; 4 HPFs/mouse, 5 mice each. (*E*) Kaplan-Meier survival curves. n = 19 (IgG isotype) and 19 mice (GREM1-neutralizing antibody). Hazard ratio (HR) and 95% confidence interval of the HR are shown (log-rank method). (*F*) The growth kinetics of CRC hepatic metastases were assessed during the survival analysis in *E*. AP tumoroid–derived luciferase signals were evaluated by using an IVIS. Mean ± SEM. Mann-Whitney *U* test (*D*), log-rank test (*E*), and 2-way repeated-measures ANOVA (*F*). In *F*, to exclude the effect of loss of samples during the experiment, statistical analysis was performed by using IVIS signal values week before 10, when the first mouse reached a humane endpoint. Note that the same pictures were used in Figure 6*A* and Supplementary Figure 20*A* to show the procedure of portal vein injection.



Supplementary Figure 21. AAV-*Islr* treatment augments BMP signaling in the normal mouse liver. (*A*, *B*) IHC for pSmad1/5/8 in the normal liver from AAV-*Islr*–treated or AAV-mRuby2–treated mice. (*A*) Representative pictures. Scale bar, 50 μm. H, hepatocytes; V, central vein. (*B*) Quantification of 3,3′-DAB intensity by ImageJ; 5 HPFs/mouse, 4 mice each. (*C*) qRT-PCR for BMP target genes using normal liver tissue lysates; n = 3 mice each. Mice were harvested 5–6 weeks after tail vein injection (3–4 weeks after tumor injection) (*A*–*C*). Mean ± SEM. Mann-Whitney *U* test (*B*) and 2-tailed unpaired Student *t* test (*C*).

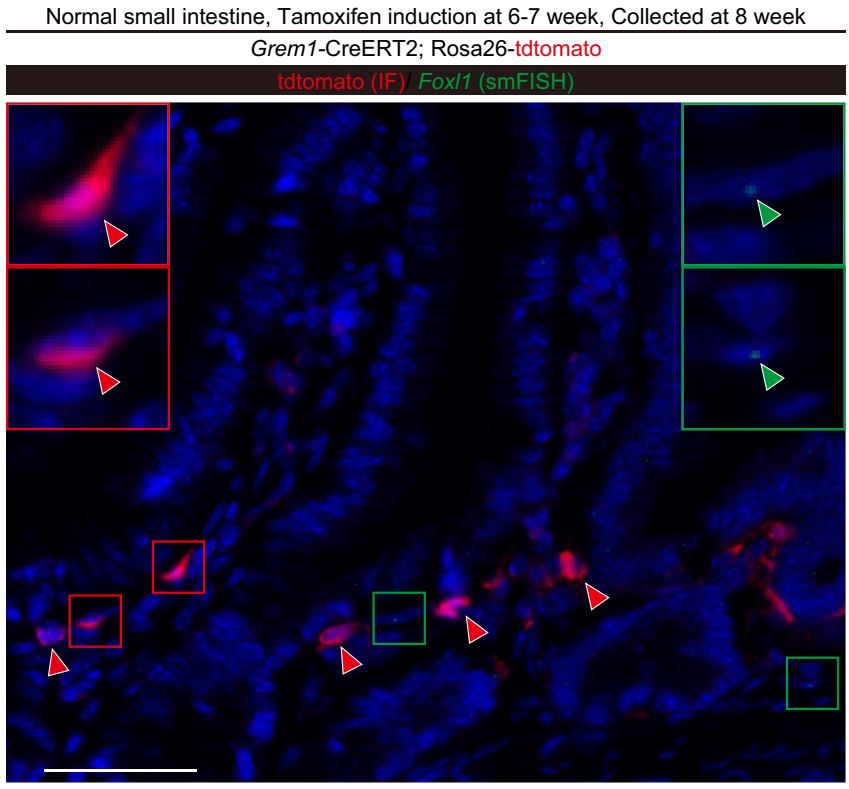


Supplementary Figure 22. No liver injury was apparent in mice administered AAV8-*Islr* via histologic analysis. (*A*) Fluorescent microscopy image showing expression of a red fluorescent protein (mRuby2) in the liver of mice harvested 2 weeks after tail vein injection of AAV8-mRuby2 and AAV8-*Islr*. White dotted lines indicate the edges of the livers. Yellow scale bar, 500 μm. (*B*) H&E staining of the liver section from mice harvested 2 weeks after tail vein injection of AAV8-mRuby2 and AAV8-*Islr*; n = 3 mice each. A, hepatic artery (arrows); B, bile duct; P, portal vein. Scale bar, 100 μm.

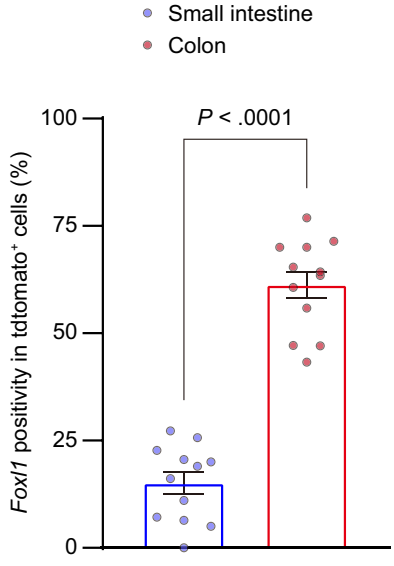


Supplementary Figure 23. Mice administered AAV8-*Islr* show reduced fibrosis in CRC liver metastases. (*A*, *B*) IHC for αSMA using CRC liver metastases in mice administered AAV8-*Islr* or AAV8-mRuby2. (*A*) Representative pictures. (*B*) Quantification of 3,3′-DAB^{+} areas by ImageJ; 5 HPFs/mouse, 4 mice each. (*C*, *D*) Picro-Sirius red staining for collagen using the CRC liver metastasis sections. (*C*) Representative pictures. (*D*) Quantification of Picro Sirius red^{+} areas by ImageJ; 5 HPFs/mouse, 4 mice each. Mean ± SEM. Mann-Whitney *U* test (*B* and *D*). Scale bars, 50 μm.

A



B



Supplementary Figure 24. *Grem1*$^{+}$ cells in the small intestine show lower *Foxl1* positivity than those in the colon. (*A*, *B*) smFISH for *Foxl1* and IF for tdtomato in the small intestine from *Grem1*-CreERT2;Rosa26-tdtomato mice. (*A*) Representative picture. Red and green arrowheads denote *Grem1*$^{+}$*Foxl1*$^{-}$ cells and *Grem1*$^{-}$*Foxl1*$^{+}$ cells, respectively. The boxed areas are magnified in the insets. (*B*) *Foxl1* positivity in the *Grem1*$^{+}$ cells in the small intestine (*blue*). For statistical comparison, the result with the colon from Figure 3*D* is shown on the right (*red*). Four HPFs (400×)/mouse, 3 mice each. Two-tailed unpaired Student *t* test. Scale bars, 50 μm.

Update

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DOI: https://doi.org/10.1053/j.gastro.2021.09.038

Correction

Kobayashi H, Gieniec KA, Wright JA, et al. The Balance of Stromal BMP Signaling Mediated by *GREM1* and *ISLR* Drives Colorectal Carcinogenesis. Gastroenterology 2021;160:1224–1239.

In the above article, there was an error in the spelling of the surname of one of the co-authors. The name in the author list and CRediT authorship contribution should read Tadashi Iida, not Tadashi Lida.

Correction

Pariente B, Torres J, Burisch J, et al. Validation and Update of the Lémann Index to Measure Cumulative Structural Bowel Damage in Crohn's Disease. Gastroenterology 2021;161:853–864.

In the above article, Supplementary Figures 2 and 3 have been modified as requested initially by the editor to obtain higher resolution images (300 × 300 bpi). In addition, in both figures the area of each circle is now proportional to the number of coincident data plotted there, as it is the case for all other figures of the paper (Figures 1, 2, Supplementary Figure 1). The updated figures are shown here.